NEET 2019

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

All questions with detailed solutions of Physics, Chemistry and Biology

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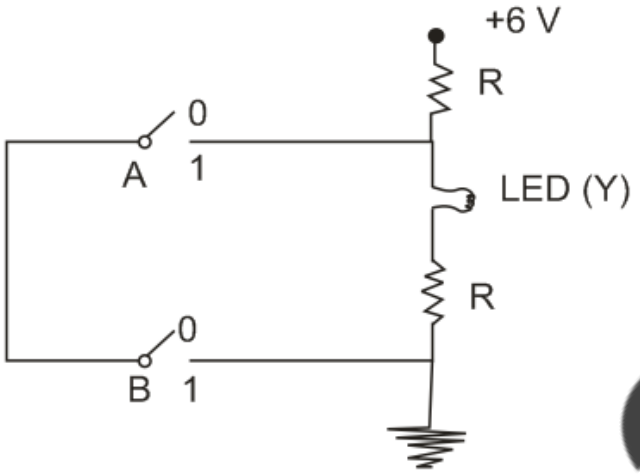
Physics

Q. 1



The correct Boolean operation represented by the circuit diagram drawn is :

Option 1:
AND

Option 2:
OR

Option 3:
NAND

Option 4:
NOR

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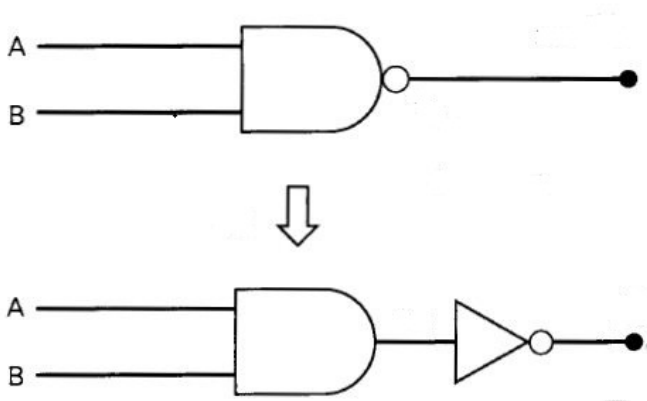
Correct Answer:
NAND

Solution:

NAND gate -

NOT + AND gate



- wherein

$$Y = \overline{A \cdot B}$$

A and B are input

Y is out put

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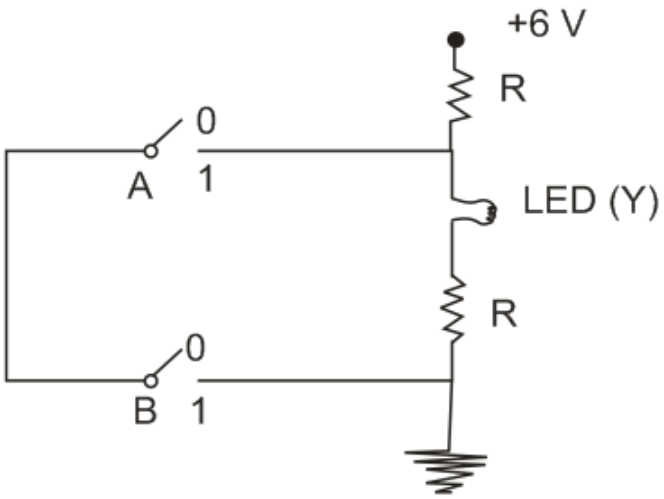
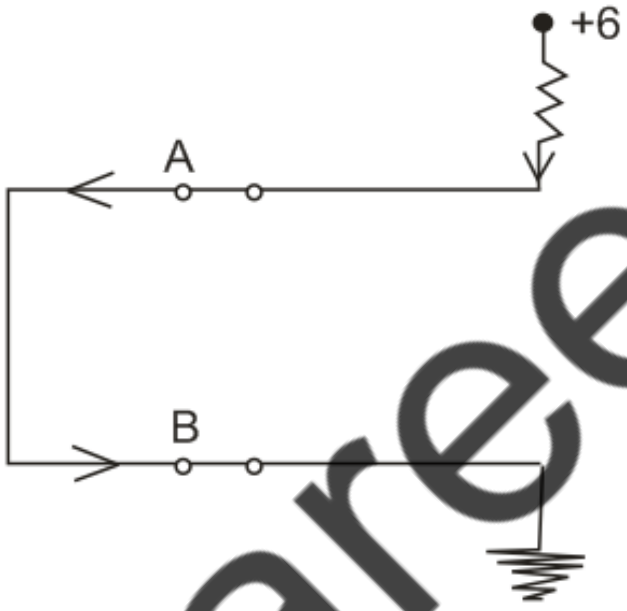


When both A and B is closed there is no current flow through LED

That is



In all other cases output is 1

A	B	Y
0	0	1
0	1	1
1	0	1

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1 1 0

NAND gate

Q. 2 A hollow metal sphere if radius R is uniformly charged . The electricity field due to the sphere at a distance r from the center :

Option 1:
Increases as r increases for r < R and for r > R

Option 2:
zero as r increases for r < R , decreases as r increases for r > R

Option 3:
zero as r increases for r < R , increases as r > R

Option 4:
decreases as r increases for r < R and for r > R

Correct Answer:
zero as r increases for r < R , decreases as r increases for r > R

Solution:

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Outside the sphere (P lies outside the sphere) -

$$E_{out} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} = \frac{\sigma R^2}{\epsilon_0 r^2}$$

$$V_{out} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r} = \frac{\sigma R^2}{\epsilon_0 r}$$

- wherein

σ - surface charge density.

At the surface of Sphere -

$$V = R$$

$$E_s = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^2} = \frac{\sigma}{\epsilon_0}$$

$$V_s = \frac{1}{4\pi\epsilon_0}\frac{Q}{R} = \frac{\sigma R}{\epsilon_0}$$

-

Inside the sphere (P lies inside the sphere) -

$$E_{in} = 0$$

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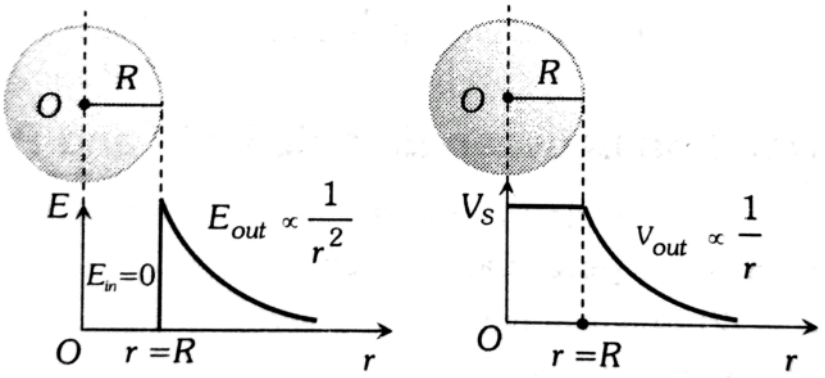
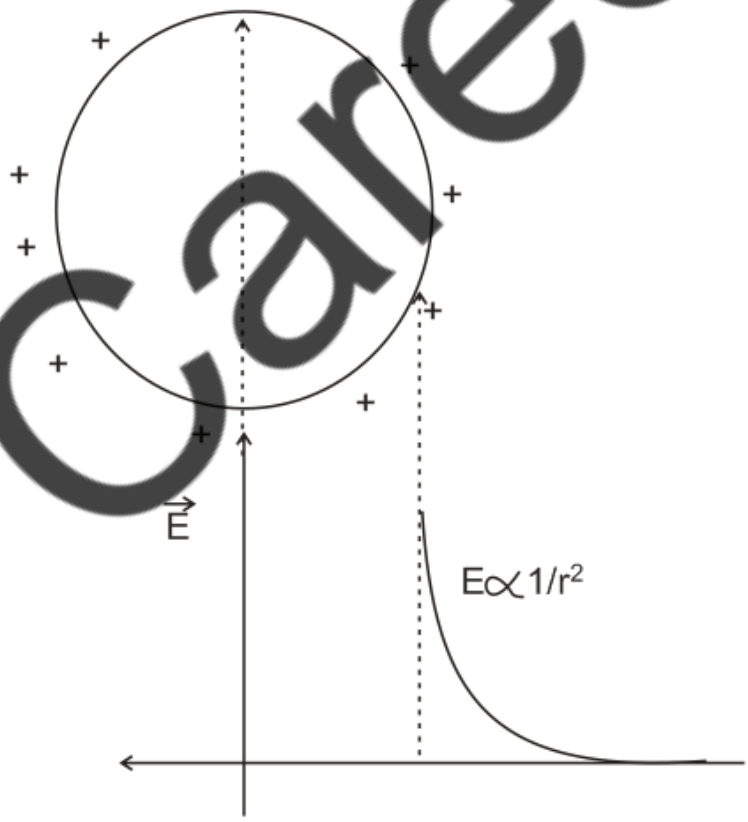
$$V_{in} = constant$$

- wherein



Inside sphere field is zero since there is no change inside , charge get distributed on the surface



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Q. 3 At a point A on the earth surface the angle of dip , $\delta = +25^\circ$, At a point B on the earth surface the angle of dip , $\delta = -25^\circ$, we can interpret that :

Option 1:
A and B are both located in northern hemisphere

Option 2:
A is located in southern hemisphere and B in northern hemisphere

Option 3:
A is located in northern hemisphere and B in southern hemipshere

Option 4:
A and B both are located in the southern hemisphere

Correct Answer:
A is located in northern hemisphere and B in southern hemipshere

Solution:

Angle of Inclination or Dip -

It is the angle between the direction of intensity of total magnetic field of earth and a horizontal line in the magnetic meridian.

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-

Angle of dip is zero at equator, positive at northen hemisphere and negative at southern hemisphere.
A is located in the northern hemisphere and B is located in southern hemisphere

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Q. 4 In a double slit experiment , when light of wavelength 400 nm was used , the angular width of the first minima formed on a screen palced 1 m away , was found to be 0.2° . what will be the angular width of the first minima , if the entire experimental apparatus is immersed in water ?

Option 1:
0.266°

Option 2:
0.15°

Option 3:
0.05°

Option 4:
0.1°

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Correct Answer:

0.15°

Solution:

Refractive index -

$$\mu = \frac{c}{v}$$

- wherein

$c =$ speed of light in vacuum.

$v =$ speed of light in medium.

Young Double Slit Experiment -

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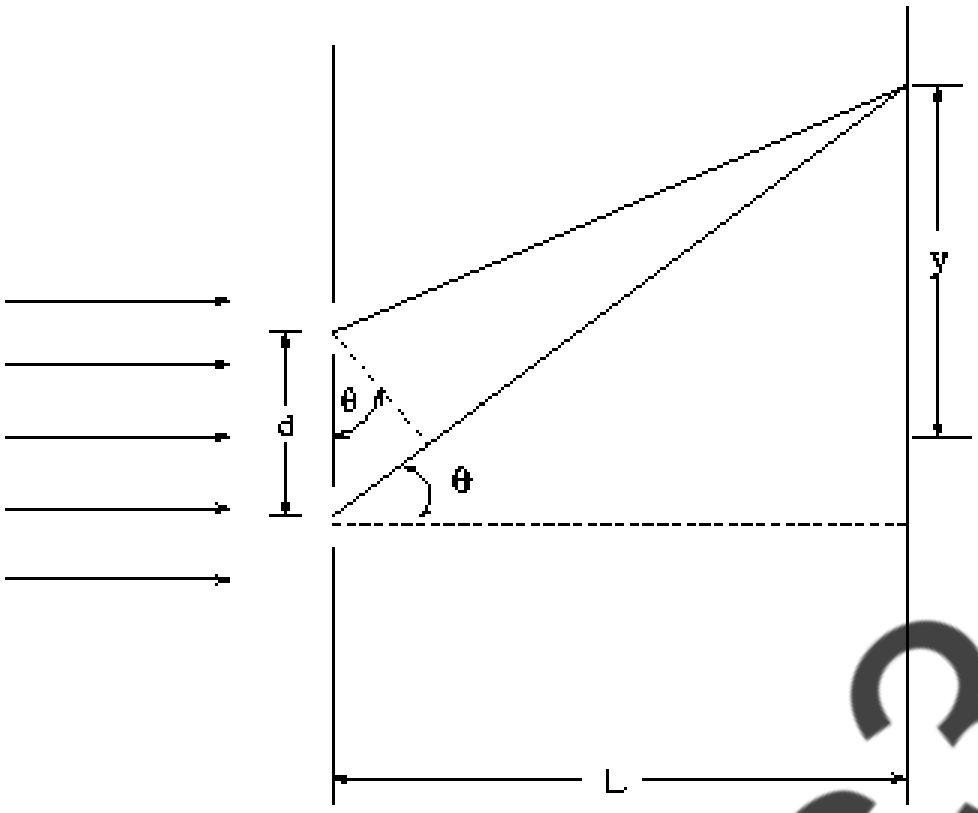

- wherein

$$y = \Delta x \cdot \left(\frac{D}{d}\right)$$

$y =$ Distance of a point on screen from central maxima

$\Delta x =$ Path difference at that point

Fringe Width in a medium -

$$\beta' = \frac{\beta}{\mu}$$

- wherein

Since wavelength in a medium become $\lambda = \frac{\lambda_o}{\mu}$

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Angular width =$\theta \ \alpha \ \ \lambda$

and $\mu = \frac{c}{v} = \frac{\lambda_1}{\lambda_2} = \frac{\theta_1}{\theta_2}$

$$\theta_2 = \frac{\theta_1}{\mu} \Rightarrow \frac{0.2 * 3}{4} = 0.15^0$$

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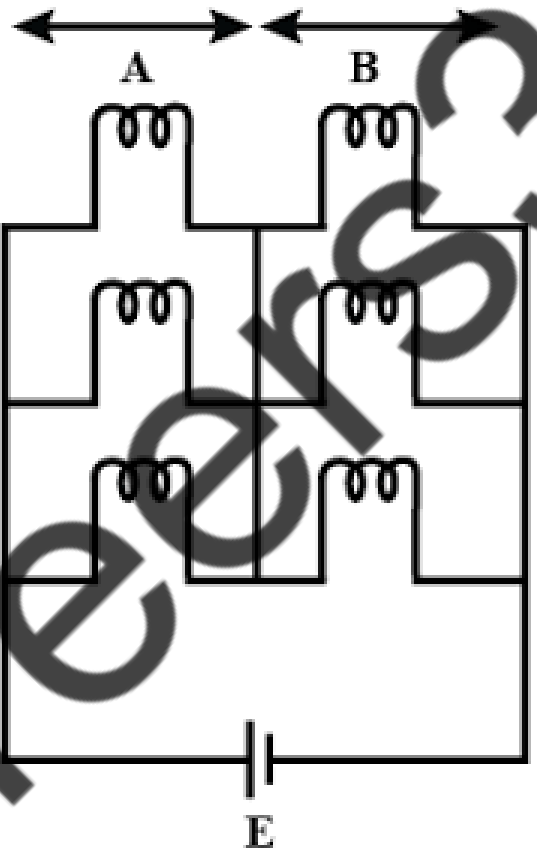
Q. 5 Six similar bulbs are connected as shown in the figure with a DC source of emf E, and zero internal resistance.

The ratio of power consumption by the bulbs when 1) all are glowing and 2) in the situation when two from section A and one from section B are glowing will be :



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Option 1:
4: 9

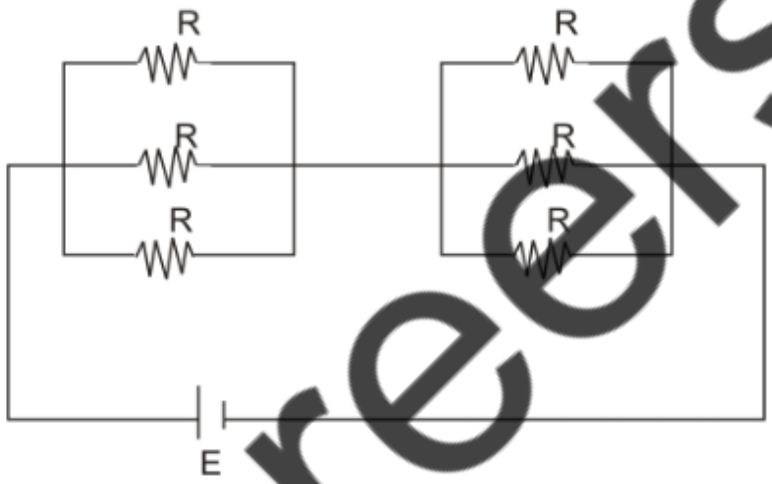
Option 2:
9 : 4

Option 3:
1 : 2

Option 4:
2 : 1

Correct Answer:
9 : 4

Solution:



$R_{eq} = 2R/3$

$P_1 = 3E^2/2R$

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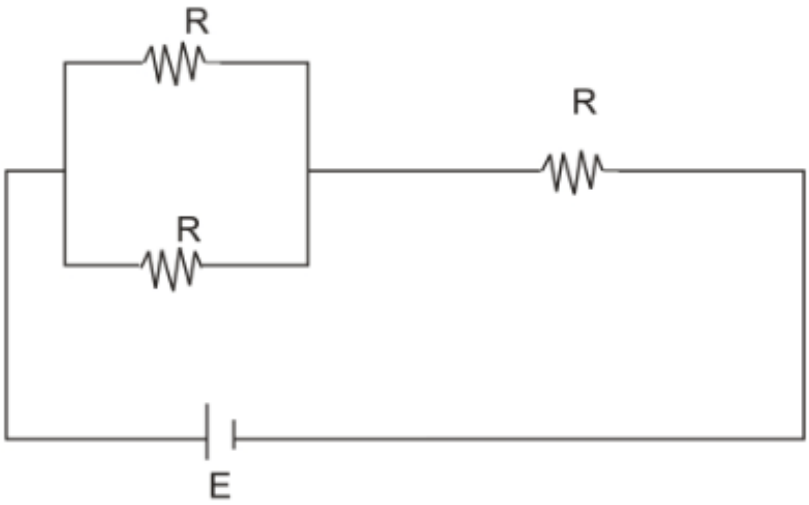

$$R_{eq} = 3R/2$$

$$P_2 = 2E^2/3R$$

$$P_1/P_2 = 9/4$$

Q. 6 In which of the following processes, the heat is neither absorbed nor released by a system?

Option 1:
isothermal

Option 2:
adiabatic

Option 3:
isobaric

Option 4:
isochoric

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Correct Answer:
adiabatic

Solution:
Adiabatic Process -

When a Thermodynamic System undergoes a change in such a way that no exchange of heat takes place.

- wherein

$$\Delta Q = 0$$

For the adiabatic process heat is neither absorbed, nor released by the system

Hence, the answer is the option (2).

Q. 7 A cylinder conductor of radius R is carrying a constant current , The plot of the magnitude of magnetic field , B with the distance d from , the centre of the conductor , is correctly represented by the figure :

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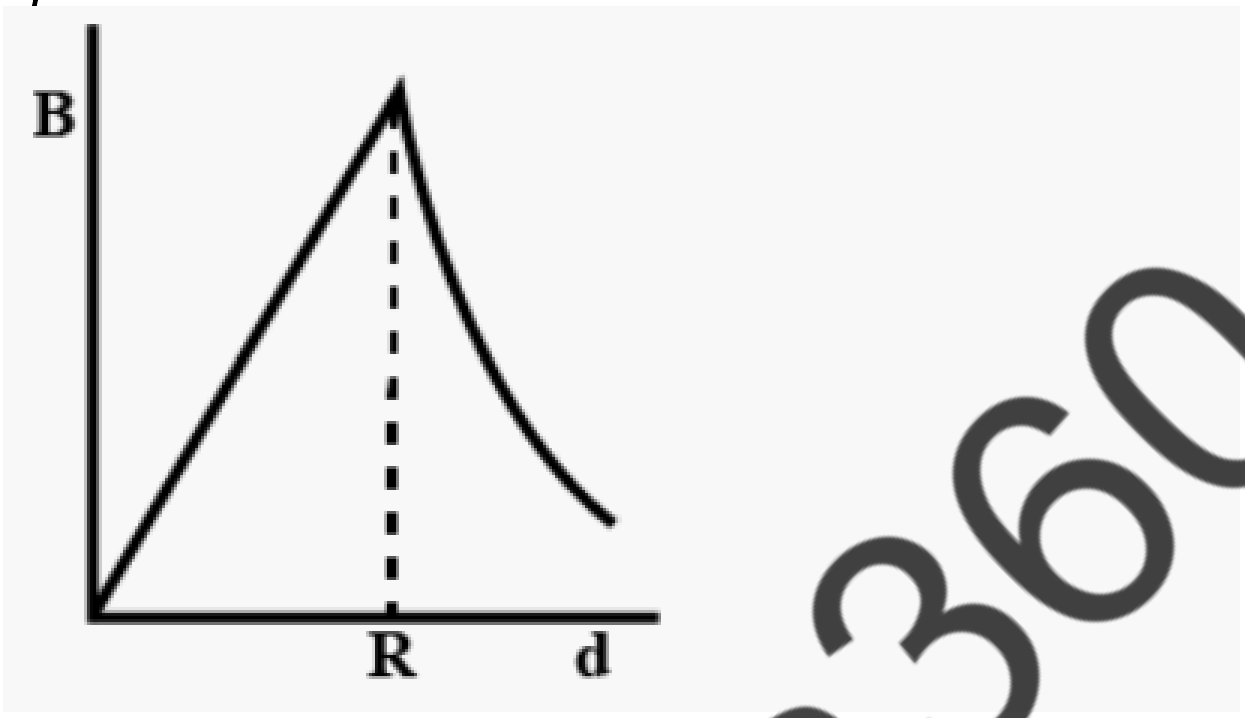
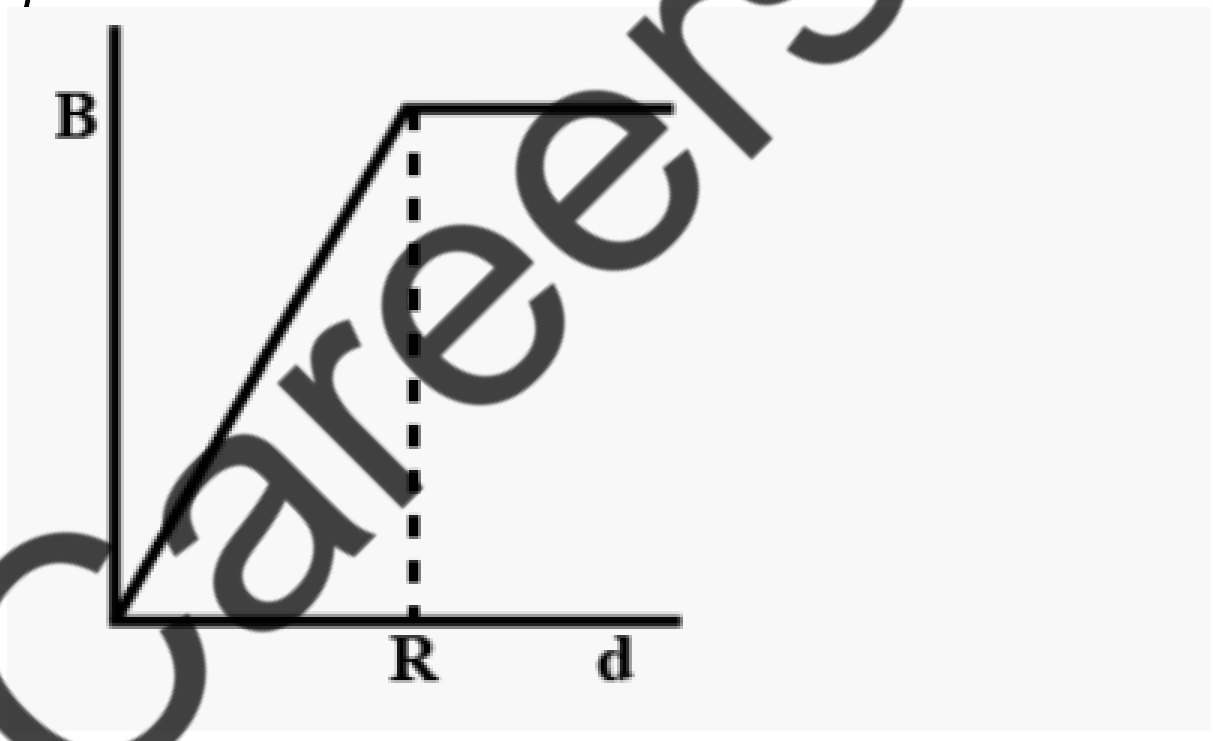
Option 1:



Option 2:



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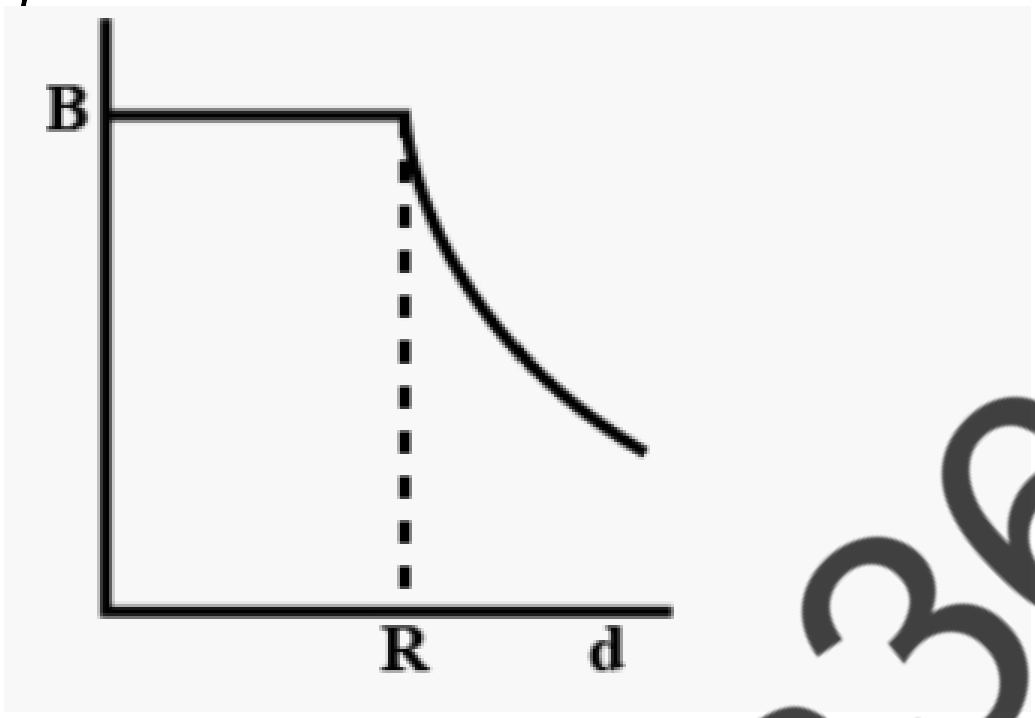
Option 3:



Option 4:

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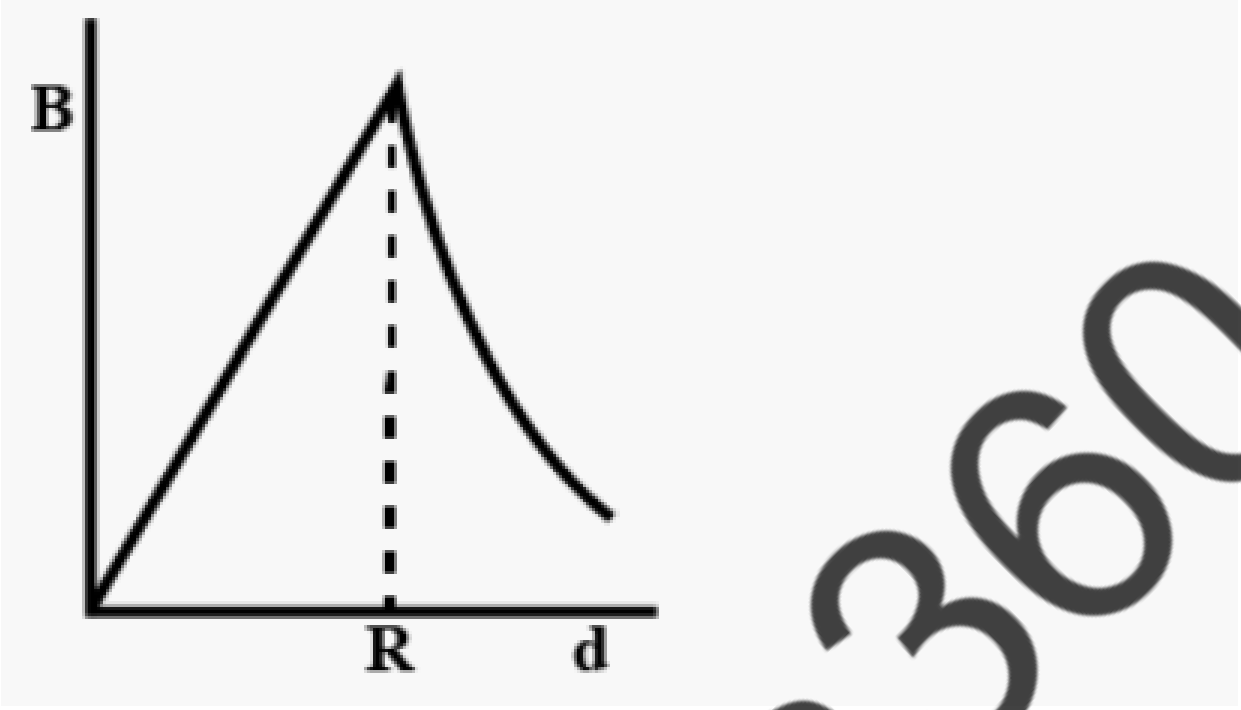
Correct Answer:

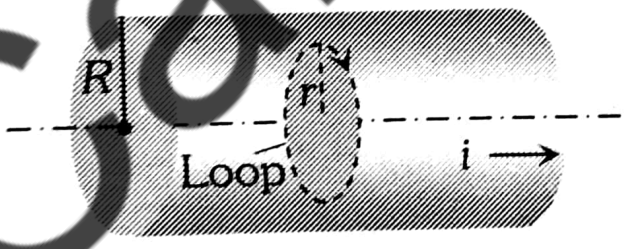


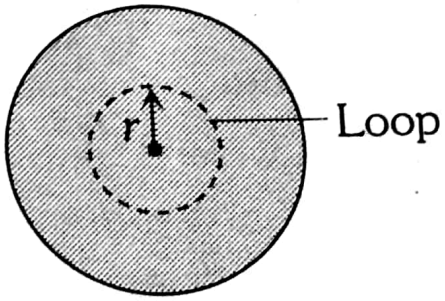
Solution:

Magnetic field Inside the solid cylinder -

B= $\frac{\mu_o}{2\pi}\frac{ir}{R^2}$

- wherein





Magnetic field outside the solid cylinder -

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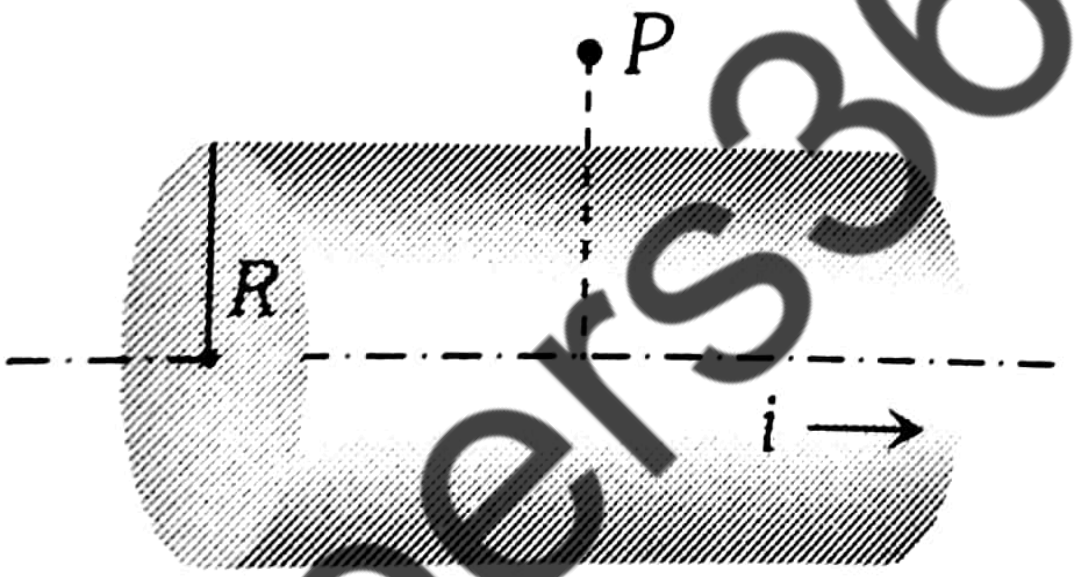
$$B_{surface} = \frac{\mu_o i}{2\pi R}$$

- wherein



In side cylinder B =

$$\frac{\mu_0 i r}{2\pi R^2}$$

$$B_{out} = \frac{\mu_0 i}{2\pi r}$$

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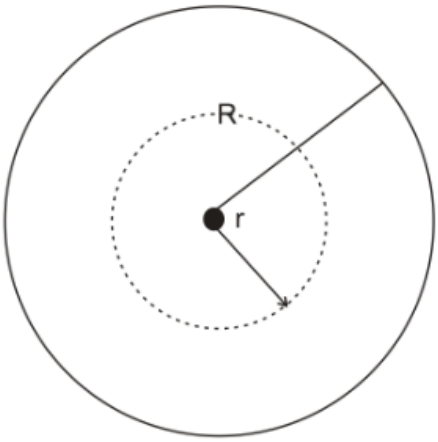
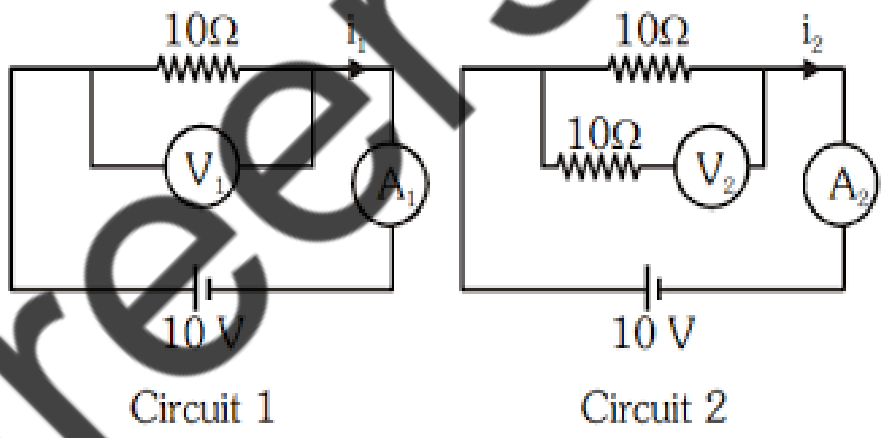


Q. 8 In the circuit shown below, the readings of the voltmeter and the ammeter will be :



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Option 1:

$V_2 > V_1 \ and \ i_1 = i_2$

Option 2:

$V_2 = V_1 \ and \ i_1 > i_2$

Option 3:

$V_2 = V_1 \ and \ i_1 = i_2$

Option 4:

$V_2 > V_1 \ and \ i_1 > i_2$

Correct Answer:

$V_2 = V_1 \ and \ i_1 = i_2$

Solution:

Ammeter -

It is a device used to measure current and is always connected in series.

Voltameter -

It is a device used to measure Potential differences and is always put in parallel with the circuit element.

Circuit 1

$$i_1 = 10/10 = 1A$$

$$V_1 = 10 * 1 = 10V$$

Circuit 2

The resistance of the voltmeter is very high. So a negligible amount of current flows through the voltmeter. So we can consider almost all currents flow through upper 10 Ω

$$\therefore V_1 \approx V_2$$

$$i_1 \approx i_2$$

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Q. 9 Which colour of light has the longest wavelength?

Option 1:
red

Option 2:
blue

Option 3:
green

Option 4:
violet

Correct Answer:
red

Solution:
Visible radiation -

Frequency Range 4×10^{11} Hz to 8×10^{14} Hz

- wherein

Wavelength Range 3750 A^o to 7500 A^o

Increasing order for wavelength= VIBGYOR

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Red colour has the highest wavelength. This can be easily detected by the human eye.

Hence, the answer is the option (1).

Q. 10 An increase in the temperature of a gas-filled in container would lead to :

Option 1:
increase in its mass

Option 2:
increase in its kinetic energy

Option 3:
decrease in its pressure

Option 4:
decrease in intermolecular distance

Correct Answer:
increase in its kinetic energy

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Solution:
At constant volume, the pressure P of a given mass of a gas is directly proportional to its absolute temperature T.

- wherein

$$P \alpha T$$

$$\Rightarrow \frac{P}{T} = constant$$

$$\Rightarrow \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

The kinetic energy of gas due to translation per mole -

$$E = \frac{3}{2}PV$$

$$= \frac{3}{2}RT$$

- wherein

R = Universal gas constant

T = temperature of gas

As the temperature of the gas increases, the particle gains kinetic energy and their speed increases.

Hence, the answer is the option (2).

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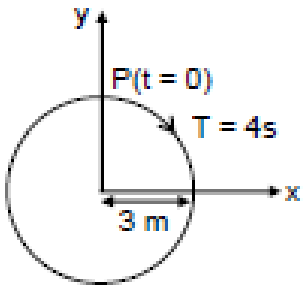
Q. 11 The radius of circle , the period of revolution , initial poisition and sense of revolution are indicated in fig.



y-projection of the radius vector of rotating particle P is

Option 1:

$y(t) = -3\cos 2\pi t, \ \ where \ \ y \ \ in \ \ m$

Option 2:

$y(t) = 4\sin\left(\frac{\pi t}{2}\right), \ \ where \ \ y \ \ in \ \ m$

Option 3:

$y(t) = 3\cos\left(\frac{3\pi t}{2}\right), \ \ where \ \ y \ \ in \ \ m$

Option 4:

$y(t) = 3\cos\left(\frac{\pi t}{2}\right), \ \ where \ \ y \ \ in \ \ m$

Correct Answer:

$y(t) = 3\cos\left(\frac{\pi t}{2}\right), \ \ where \ \ y \ \ in \ \ m$

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Solution:

At t = 0 , position is at 3 cm

and

$$\omega t = 2\pi f t = \frac{2\pi t}{T} = \frac{2\pi t}{4} = \frac{\pi t}{2}$$

$$y = 3\cos\left(\frac{\pi t}{2}\right)$$

Q. 12 Average velocity of a particle executing SHM in one complete vibration is :

Option 1:

$\frac{A\omega}{2}$

Option 2:

$A\omega$

Option 3:

$\frac{A\omega^2}{2}$

Option 4:

zero

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Correct Answer:
zero

Solution:
Equation of S.H.M. -

$$a = -\frac{d^2x}{dt^2} = -w^2x$$

$$w = \sqrt{\frac{k}{m}}$$

- wherein

$$x = A\sin(wt + \delta)$$

$$V = A\omega\cos(wt + \delta)$$

The average velocity of a particle executing SHM is zero

Average velocity = $a\omega$ (average of $\cos(\omega t + \delta)$ over period T)

and the average of the cos function over a complete cycle is zero

Hence, the answer is the option (4).

Q. 13 A solid cylinder of mass 2 Kg and radius 4 cm is rotating about its axis at the rate of 3 rpm . The torque required to stop after 2π revolution is :

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Option 1:

$2 \times 10^{-6} Nm$

Option 2:

$2 \times 10^{-3} Nm$

Option 3:

$12 \times 10^{-4} Nm$

Option 4:

$2 \times 10^{6} Nm$

Correct Answer:

$2 \times 10^{-6} Nm$

Solution:

Analogue of second law of motion for pure rotation -

$\vec{\tau} = I\,\alpha$

- wherein

Torque equation can be applied only about two point

(i) centre of motion.

(ii) point which has zero velocity/acceleration.

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$$\omega_2^2 = \omega_1^2 + 2\alpha\theta$$

$$0 = (2\pi * \frac{3}{60})^2 - 2 \times 2\pi \times 2\pi\alpha$$

$$\alpha = \frac{1}{800}$$

$$T = I\alpha = \frac{mr^2}{2}d$$

$$= 2 \times 10^{-6}$$

Q. 14 A block of mass 10 Kg is in contact against the inner wall of a hollow cylinder drum of radius 1 m . The coefficient of friction between the block and the iner wall of the cylinder is 0.1 . The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis will be

Option 1:
$\sqrt{10}rad/s$

Option 2:
$(10/2\pi)rad/s$

Option 3:
$10rad/s$

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Option 4:

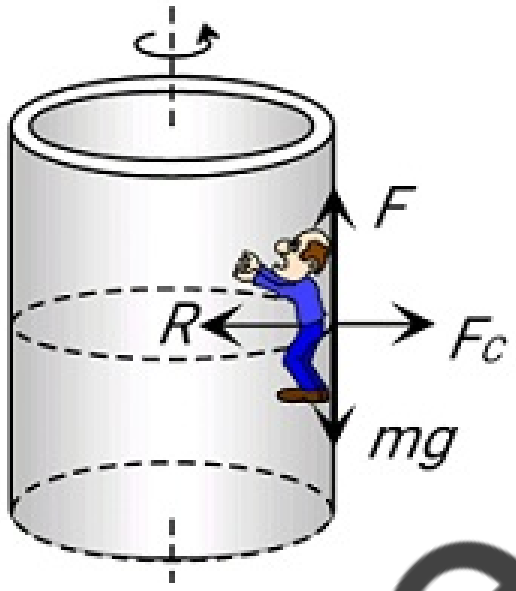
$10\pi rad/s$

Correct Answer:

$10rad/s$

Solution:

Sticking of Person with the wall of Rotor(Death well) -



F = weight of person (mg)

$$\mu R = mg$$

$$\mu F_c = mg$$

$$\mu m\omega_{min}^2 r = mg$$

$$\therefore \ \omega_{min} = \sqrt{\frac{g}{\mu r}}$$

- wherein

F = friction force

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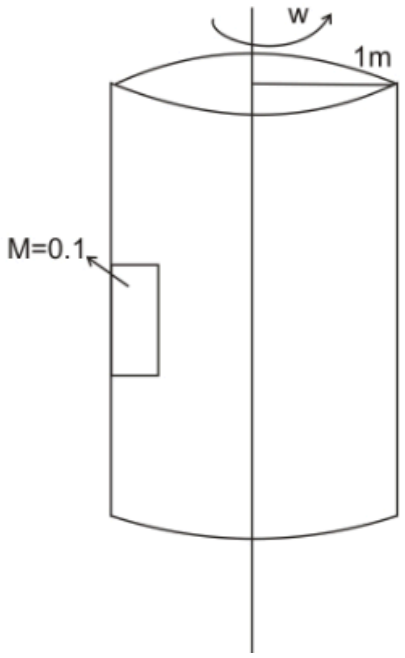
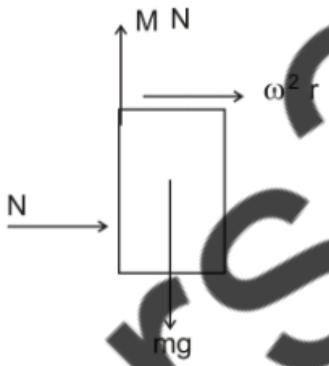
F_c = centrifugal force

W_{min} = minimum angular velocity

μ =coefficient of friction

r = radius of Rotor





$$mg = \mu N$$

$$N = m\omega^2 R$$

$$mg = \mu m\omega^2 R$$

$$\omega^2 = 10/0.1$$

$$\omega = 10$$

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Q. 15 The speed of a swimmer in still water is 20 m/s . The speed of river water is 10 m/s and is flowing due east . If he is standing on the south bank and wishes to cross he river along the sortest path , the angle at which he should make his strokes w.r.t north is given by :

Option 1:
30 west

Option 2:
0°

Option 3:
$60^{\circ} west$

Option 4:
$45^{\circ} west$

Correct Answer:
30 west

Solution:
Boat - River Problem -

To cross river in the shortest path

condition (velocity of boat along river flow must be zero)

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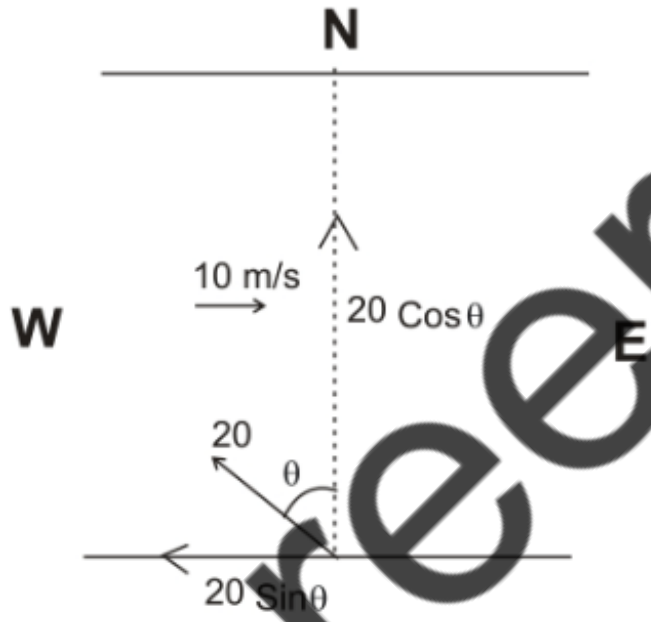
$$t = \frac{d}{\sqrt{v^2 - u^2}}$$

$d =$ width of river

v = Speed of Boat w.r.t. river

u = speed of river

-



$20\sin\theta = 10 \Rightarrow \theta = 30°west$

Q. 16 An 800-turn coil of the effective area $0.05m^2$ is kept perpendicular to a magnetic field $5 \times 10^{-5}T$. When the plane of the coil is rotated by 90 around any of its coplanar axes in 0.1 s, the emf induced in the coil will be :

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Option 1:
2 V

Option 2:
0.2 V

Option 3:
0.0002 V

Option 4:
0.02 V

Correct Answer:
0.02 V

Solution:
Rate of change of magnetic Flux -

$$\varepsilon = \frac{-d\phi}{dt}$$

- wherein

$d\phi \rightarrow \phi_2 - \phi_1$

$\phi_2 - \phi_1$ – change in flux

N = 800

$A = 0.05m^2$

$B = 5 \times 10^{-5}T$

case 1 B and A are parallel

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So $\phi_1 = BAN$

case 2 B and A are perpendicular

So $\phi_2 = 0$

$$emf = d\phi/dt = \frac{NBA\sin\omega t}{t} - 0$$

$$emf = \frac{800 \times 5 \times 10^{-5} \times \sin 90}{0.1}$$

$$= 0.02V$$

Hence the answer is the Option (4).

Q. 17 In an experiment, the percentage of error that occurred in the measurement of physical quantities A, B C, and D are 1 %, 2 %, 3 %, and 4 % respectively. Then the maximum percentage of error in the measurement X , there $X = \frac{A^2 B^{1/2}}{C^{1/3} D^3}$ will be :

Option 1:
(3/13) %

Option 2:
16 %

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Option 3:
- 10 %

Option 4:
10 %

Correct Answer:
16 %

Solution:
Error in quantity raised to some power -

$$x = \frac{a^n}{b^m} \rightarrow \frac{\Delta x}{x} = \pm\left(n\frac{\Delta a}{a} + m\frac{\Delta b}{b}\right)$$

- wherein

Δa= absolute error in measurement of a

Δb= absolute error in measurement of b

Δx= absolute error in measurement of x

$$\frac{\delta x}{x} = 2\frac{\Delta A}{A} + \frac{1}{2}\frac{\Delta B}{B} + \frac{1}{3}\frac{\Delta C}{C} + \frac{3\Delta D}{D}$$

$$= 16\%$$

Hence the answer is the option (2).

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Q. 18 The displacement of a particle executing simple harmonic motion is given by

$$y = A_0 + A\sin\omega t + B\ cos\omega t$$

Then the amplitude of its oscillation is given by :

Option 1:
$A_0 + \sqrt{A^2 + B^2}$

Option 2:
$\sqrt{A^2 + B^2}$

Option 3:
$\sqrt{A_0^2 + (A + B)^2}$

Option 4:
A+ B

Correct Answer:
$\sqrt{A^2 + B^2}$

Solution:
Equation of S.H.M. -

$$a = -\frac{d^2x}{dt^2} = -w^2x$$

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$$w = \sqrt{\frac{k}{m}}$$

- wherein

$$x = A\sin(wt + \delta)$$

$$y = A_0 + A\sin\omega t + B\ cos\omega t$$

max value of

$$(A\sin\omega t + B\ cos\omega t)\ \ is\ \ \sqrt{A^2 + B^2}$$

The particle executes SHM about its centre at

$$\alpha = A_0$$

$$with\ \ amplitude = \sqrt{A^2 + B^2}$$

Hence the answer is the Option (2)

Q. 19 An electron is accelerated through a potential difference of 10000 V . Its de Broglie wavelength is :

$(m_e) = 9 \times 10^{-31} Kg$

Option 1:

$12.2 \times 10^{-13} m$

Option 2:

$12.2 \times 10^{-12} m$

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Option 3:

$12.2 \times 10^{-14} m$

Option 4:

12.2 nm

Correct Answer:

$12.2 \times 10^{-12} m$

Solution:

De - Broglie wavelength of Electron -

$$\lambda_{electron} = \frac{12.27}{\sqrt{v}} A^{\circ}$$

where

$A^{\circ} - Angstrom$

$v - potential\ difference$

SO v=10000

SO $\lambda_{electron} = \frac{12.27}{\sqrt{10^4}} A^{\circ} = 12.27 * 10^{-12} m$

Hence the answer is the Option (2).

Q. 20 A mass m is attached to a thin wire and whirled in a vertical circle . The wire is most likely to break when :

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Option 1:
The mass is at the highest point

Option 2:
the wire is horizontal

Option 3:
The mass is at the lowest point

Option 4:
Inclined at an angle of 60 degree from vertical

Correct Answer:
The mass is at the lowest point

Solution:

Centripetal Force -

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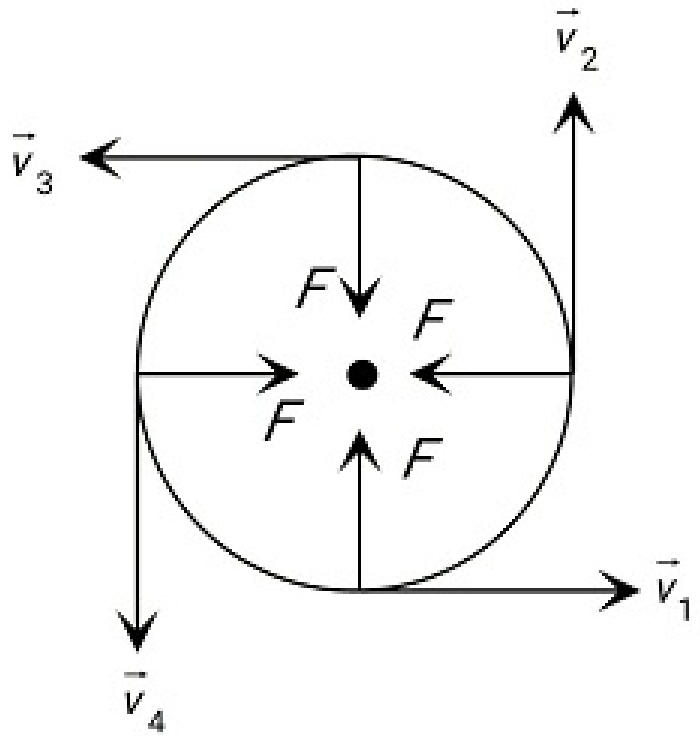

$$F = 4m\pi^2n^2r$$

$$F = \frac{4m\pi^2n^2r}{T^2}$$

F = Centripetal force

$\omega =$ Angular velocity

n = frequency

- wherein

Force acts on the body along the radius and towards centre.

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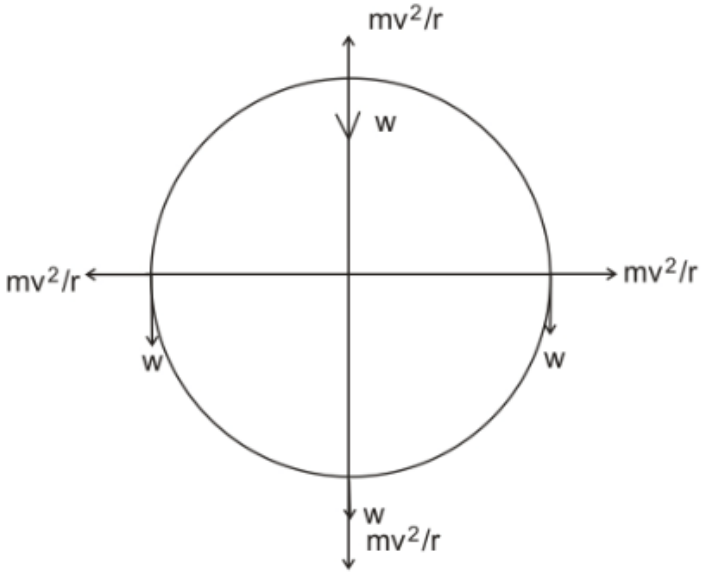

The resultant of weight and centrifugal force is maximum at bottom . so wire is mostly like to break at bottom

---

Q. 21 Two particles A and B are moving in uniform circular motion in concentric circles of radii $r_A \ and \ r_B$ with speed $v_A \ and \ v_B$ respectively. Their time period of rotation is the same. The ratio of the angular speed of A to that of B will be :

Option 1:

$r_A : r_B$

Option 2:

$v_A : v_B$

Option 3:

$r_B : r_A$

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Option 4:

1 : 1

Correct Answer:

1 : 1

Solution:

Relation of Angular Velocity frequency and Time period -

$$\omega = \frac{\theta}{t} = \frac{2\pi}{T} = 2\pi n$$

$$\omega = \frac{2\pi}{T} = 2\pi n$$

- wherein

$\omega =$ Angular Velocity

n = frequency

T = time period

For A and B

$Radius \quad r_A \quad r_B$

$velocity \quad V_A \quad V_B$

$angular \quad velocity \quad W_A \quad W_B$

$time \quad period \quad T_A \quad T_B$

$T = \frac{2\pi}{\omega}$ as

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$$as \;\; T_A = T_B$$

$$so \;\; W_A = W_B$$

Hence the answer is the Option (4).

Q. 22 A copper rod of 88 cm and an aluminum rod of unknown length have their increase in length independent of an increase in temperature. The length of the aluminum rod is :

$$(^{\alpha}C_u = 1.7 \times 10^{-5} K^{-1} \;\; and \;\; ^{\alpha}Al = 2.2 \times 10^{-5} K^{-1})$$

Option 1:
6.8 cm

Option 2:
113.9 cm

Option 3:
88 cm

Option 4:
68 cm

Correct Answer:
68 cm

Solution:
Change in Length -

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$$\Delta L = L_0(\alpha \Delta T)$$

- wherein

$$\Delta L = L - L_0$$

$L - L_0 =$Change in length

$$\Delta T = T - T_0$$

$T - T_0 =$ Change in temperature

For Copper $\rightarrow$ L = 88 cm

$$\Delta L = L_0 \alpha \Delta T$$

$$\Delta L \ to \ independent \ of \ \Delta T$$

$$\frac{\Delta L}{\Delta T} = L_0 \alpha = const$$

$$(L_0 \alpha)_{Cu} = (L\alpha)_{Al}$$

$$(1.7 \times 10^{-5}) \times 88 = (2.2 \times 10^{-5}) \times L_{Al}$$

$$L_{Al} = \frac{1.7 \times 4}{10^{-1}}$$

$$= 68cm$$

Hence the answer is the Option (4).

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Q. 23 Ionized hydrogen atoms and $\alpha - particle$ with the same momentum enter perpendicular to a constant magnetic field B. The ratio of the radii of their paths $r_H : r_\alpha$ will be :

Option 1:
2: 1

Option 2:
1 : 2

Option 3:
4 : 1

Option 4:
1 : 4

Correct Answer:
2: 1

Solution:
The radius of charged particle -

$$r = \frac{mv}{qB} = \frac{P}{qB} = \frac{\sqrt{2mk}}{qB} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$$

$$r = mv/qB = P/qB$$

p is the same for both

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$$H^+ \quad \Rightarrow q_{H^+} = 1e$$

$$He^{2+} \Rightarrow \quad q_{He^{2+}} = 2e$$

$$so \ \frac{r_H}{r_{He}} = \frac{q_{He^{2+}}}{q_{H^+}} = 2$$

Hence the answer is the option (1).

Q. 24 When a block of mass M is suspended by a long wire of length L , The length of the wire becomes (L+l) . The elastic potential energy stored in the extended wire is :

Option 1:
Mgl

Option 2:
MgL

Option 3:
1/2 Mgl

Option 4:
1/ MgL

Correct Answer:
1/2 Mgl

Solution:

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Work Done in Stretching Wire / Elastic P.E. -

$$= \frac{1}{2}\frac{YAl^2}{L} = \frac{1}{2}Fl$$

- wherein

L - Length of wire

l - increase in length

$$we \;\; have \; M_1, L_1, \Delta L = l$$

$$E = 1/2 \times F \times (\Delta L)$$

$$\Rightarrow E = 1/2 \times Mg \times l$$

Q. 25 For a p-type semiconductor, which of the following statements is true?

Option 1:
Electrons are majority carriers and trivalent atoms are the dopants .

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Option 2:
Holes are the majority carriers and trivalent atoms are the dopants

Option 3:
Holes are the majority carriers and pentavalent atoms are the dopants

Option 4:
Electron are the majority carriers and pentavalent atoms are the dopants

Correct Answer:
Holes are the majority carriers and trivalent atoms are the dopants

Solution:
p-type semiconductor -

Here the base semiconductor (i.e.pure Si or Ge) is mixed with tetravalent doping material (e.g. Al)

- wherein

1) $n_h >> n_e$

2) There is an acceptor level just above the valence band.

For P - type

Holes are majority carriers and trivalent atoms are dopants

Hence the answer is the option (2).

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Q. 26 The work done to raise a mass m from the surface of the earth to a height h, which is equal to the radius of the earth is :

Option 1:
mgR

Option 2:
2 mgR

Option 3:
1/2 mgR

Option 4:
3/2 mgR

Correct Answer:
1/2 mgR

Solution:

Gravitational Potential energy at a point -

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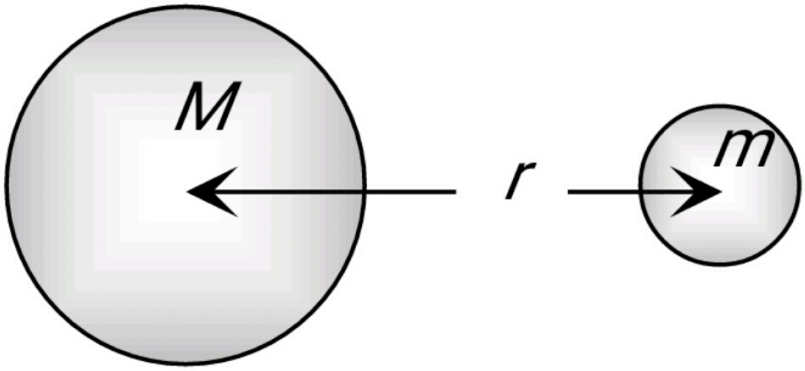

$$W = -\frac{GMm}{r}$$

$$U = -\frac{GMm}{r}$$

$U \rightarrow$ gravitational potential energy

$M \rightarrow$ Mass of source body

$m \rightarrow$ mass of test body

$r \rightarrow$ distance between two

- wherein

Always negative in the gravitational field because Force is attractive in nature.

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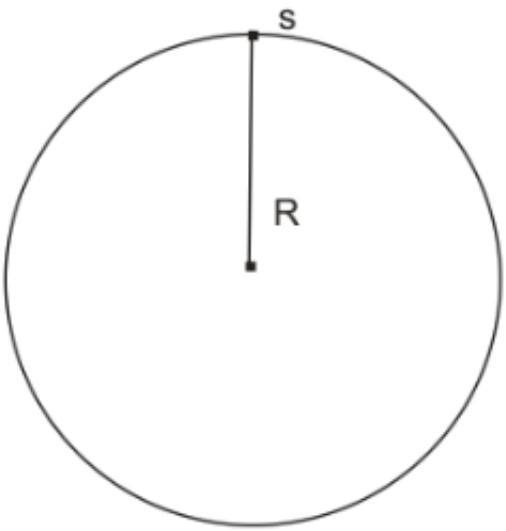

$$U_s = \frac{-Gmm}{R}$$

$$U_p = \frac{-Gmm}{2R}$$

$$\Delta U = \frac{Gmm}{R}\left(1 - \frac{1}{2}\right) =$$

$$\Delta U = \frac{1}{2}\frac{Gmm}{R}$$

$$W = \Delta U = \frac{GmmR}{2R * R} = \frac{1}{2}mgR$$

Q. 27 In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact what will be the angle of refraction?

Option 1:
180

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Option 2:
0

Option 3:
equal to angle of incidence

Option 4:
90

Correct Answer:
90

Solution:
Critical angle -

$\sin c$ = Refractive index of rarer medium / Refractive index of denser medium

- wherein

When the angle of incidence of travelling from a dense medium to a rarer medium is greater than the critical angle, no refraction occurs.

For internal reflection

$i = \theta_c, r = 90^\circ = angle\ of\ refraction$

Hence the answer is option (4).

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Q. 28 Two point charges A and B , having charges Q and - Q respectively , are placed at a certain distance apart ad force acting between them is F . If 25 % charge of A is transferres to B , then force between he charges becomes :

Option 1:
F

Option 2:
9F/16

Option 3:
16F/9

Option 4:
4F/3

Correct Answer:
9F/16

Solution:

Coulombic force -

$$F \propto Q_1 Q_2 = F \propto \frac{Q_1 Q_2}{r^2} = F = \frac{K Q_1 Q_2}{r^2}$$

- wherein

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K - proportionality Constant

Q_1 and Q_2 are two Point charge

$$F = \frac{-KQ^2}{d^2}$$

$$F' = \frac{k\left(\frac{3Q}{4}\frac{3Q}{4}\right)}{d^2}$$

$$\frac{F}{F'} = \frac{1}{\left(\frac{9}{16}\right)} = 16/9$$

$$F' = 9F/16$$

Q. 29 α particle constist of :

Option 1:
2 protons and 2 nuetrons only

Option 2:
2 electrons , 2 protons and 2 nuetrons

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Option 3:
3 electrons and 4 protons

Option 4:
2 protons only

Correct Answer:
2 protons and 2 nuetrons only

Solution:

α -decay -

$$_{Z}^{A}X \rightarrow _{Z-2}^{A-4}Y +_{2}^{4}He + Q$$

- wherein

$$Q\ value = (M_X - M_Y - M_{He})\, c^2$$

$$alpha\ particle \Rightarrow _4\ He^{2+} \Rightarrow Z = 4\ proton - 2\ neutron - 2$$

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Q. 30 Body A of mass 4m moving with speed u collides with another body B of mass 2m, at rest. The collision is head-on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is :

Option 1:
1/9

Option 2:
8/9

Option 3:
4/9

Option 4:
5/9

Correct Answer:
8/9

Solution:

Perfectly Elastic Collision -

Law of conservation of momentum and that of Kinetic Energy hold good.

- wherein

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$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$m_1, m_2 : masses$$

$$u_1, v_1 : initial\ and\ final\ velocity\ of\ the\ mass\ m_1$$

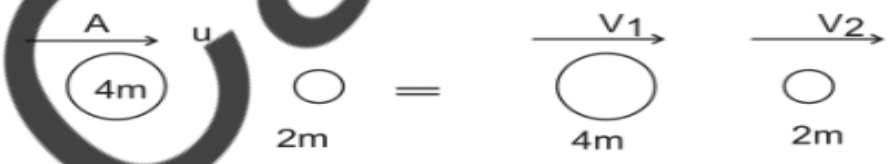
$$u_2, v_2 : initial\ and\ final\ velocity\ of\ the\ mass\ m_2$$

Perfectly Elastic Collision -

$$e = 1$$

- wherein

$$e\ :\ coefficient\ of\ restitution$$



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$$(4m)U = (4M)V_1 + 2mv_2 = 4U = 4V_1 + 2V_2$$

$$e = 1 = \frac{-(V_2 - V_1)}{(0 - u)}$$

$$V_2 - V_1 = U$$

$$4V_2 - 4V_1 = 4U$$

$$4V_1 + 2V_2 = 4U$$

$$- - - - - - - -$$

$$6V_2 = 8V$$

$$V_2 = 4/3U$$

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$$V_1 = V_2 - U$$

$$= (\frac{4}{3} - 1)U$$

$$(V_1 = \frac{1}{3}U)$$

$$for\ A$$

$$K_i = \frac{1}{2}mU^2$$

$$K_f = \frac{1}{2}mv_1^2 = \frac{1}{2}m(\frac{1}{9}U^2)$$

$$\frac{\Delta KE}{K_iO} = \frac{\frac{1}{2}mU^2(1 - 1/9)}{\frac{1}{2}mU^2}$$

$$fraction = 8/9$$

Q. 31 A body weighs 200 N on the surface of the earth how much will it weigh at half way down to the center of the earth ?

Option 1:
150 N

Option 2:
200 N

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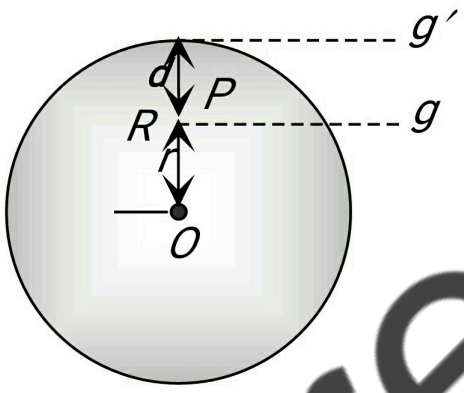
Option 3:
250 N

Option 4:
100 N

Correct Answer:
100 N

Solution:

Variation in 'g' with depth -



$$g' = g\left[1 - \frac{d}{R}\right]$$

$d \rightarrow$ depth from the surface of earth

$g' \rightarrow$ Value of g at depth 'd'

- wherein

'g' decreases on going below the surface of earth .

$g' \alpha (R - d)$

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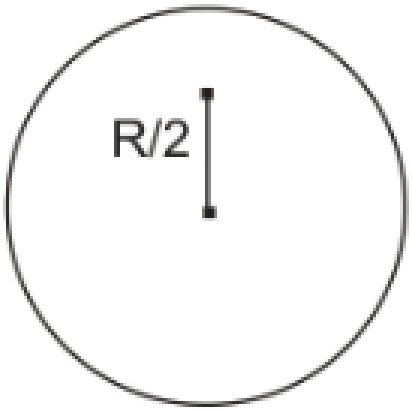

$$W_s = 200N = Mg_s$$

mg_p

$g_p = g_s(1 - h/R)\ \ h = R/2$

$g_p = g_s(1 - 1/2)$

$g_p = g_s/2$

$\frac{W_s}{W_p} = g_s/g_p = 2$

$W_p = W_s/2 = 200/2 = 100N$

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Q. 32 Pick the wrong answer in the context of a rainbow

Option 1:
When the light rays undergo two internal reflection in a water drop , a secondary rainbow is formed

Option 2:
The order of colors is reversed in the secondary rainbow

Option 3:
An observer can see a rainbow when his front is towards the sun

Option 4:
Rainbow is a combined effect of dispersion , refreaction and reflection of sunlight

Correct Answer:
An observer can see a rainbow when his front is towards the sun

Solution:
The Rainbow -

This is a phenomenon due to the combined effect of dispersion, refraction, and reflection of sunlight by spherical water droplets of rain. The conditions for observing a rainbow are that the sun should be shining in one part of the sky while it is raining in the opposite part of the sky. An observer can therefore see a rainbow only when his back is toward the sun.

Rainbows can seen when the back is towards the Sun.

Hence the answer is the option (3).

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Q. 33 A force F = 20 + 10 y acts on a particle in y - direction where F is in newton and y in meter Work done by this force to move the particle from y = 0 to y = 1 is :

Option 1:
30 J

Option 2:
5 J

Option 3:
25 J

Option 4:
20 J

Correct Answer:
25 J

Solution:
Definition of work done by variable force -

$$W = \int \vec{F} \cdot \vec{ds}$$

where

$\vec{F}$ is variable force and $\vec{ds}$ is a small displacement

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$$F = 20 + 10y$$

$$W = F \cdot ds$$

$$here\ W = \int_0^1 F \cdot dy = \int_0^1 (20 + 10y)dy$$

$$W = [20y]_0^1 + \left[\frac{10y^2}{2}\right]_0^1$$

$$W = 20 + 5 = 25J$$

Hence the answer is the Option (3).

Q. 34 A disc of radius 2 m and mass 100 Kg rolls on a horizontal floor. Its center of mass has a speed of 20 cm/s. How much work is needed to stop it?

Option 1:
3 J

Option 2:
30 Kj

Option 3:
2 J

Option 4:
1 J

Correct Answer:
3 J

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Solution:

The kinetic energy of a body in combined rotation and translation -

$$K.E = \frac{1}{2} I_{cm} w^2 + \frac{1}{2} m {v_0}^2$$

- wherein

$\frac{1}{2} m {v_0}^2$ = Translational kinetic energy of the 2center of mass

$\frac{1}{2} I_{cm} w^2$ = Rotational kinetic energy about the center of mass

$$W = \Delta KE$$

$$= \frac{1}{2} I W^2 + \frac{1}{2} m V^2$$

$$= \frac{1}{2} \times \frac{1}{2} m R^2 \omega^2 + \frac{1}{2} m V^2$$

$$(use\ V = \omega R)$$

$$= \frac{1}{4} m V^2 + \frac{1}{2} m V^2$$

$$W = \frac{3}{4} m V^2$$

$$W = \frac{3}{4} \times 100 \times \frac{20 \times 20}{100 \times 100}$$

$$W = 3J$$

Hence the answer is the option (1).

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Q. 35 A small hole of area of cross sectyion $2mm^2$ is present mear the botom of a fully filled open tank of heiht 2 m . Taking $g = 10m/s^2$, the rate of flow of water through the open hole would be nearly :

Option 1:
$12.6 \times 10^{-6}m^3/s$

Option 2:
$8.9 \times 10^{-6}m^3/s$

Option 3:
$2.23 \times 10^{-6}m^3/s$

Option 4:
$6.4 \times 10^{-6}m^3/s$

Correct Answer:
$12.6 \times 10^{-6}m^3/s$

Solution:

Bernoulli's Principle -

The total energy (Pressure energy, Potential energy and Kinetic energy) per unit volume or mass of an incompressiable and non viscous fluid in steady flow through a pipe remains constant.

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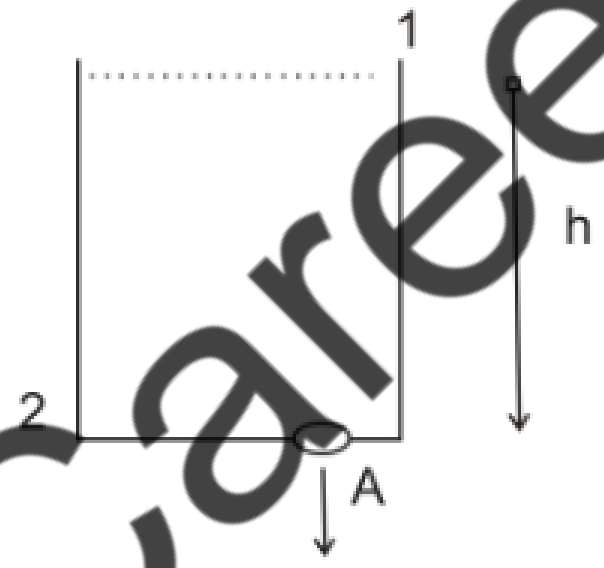
- wherein

$$P + \rho gh + \frac{1}{2}\rho v^2 = constant$$

$P \rightarrow Pressure\ energy$

$\rho gh \rightarrow Potential\ Energy$

$\frac{1}{2}\rho v^2 \rightarrow Kinetic\ Energy$



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$$A = 2mm^2$$

$$P_1 + \rho g h_1 + \frac{1}{2}\rho V_1^2 = P_2 + \rho g h_2 + \frac{1}{2}\rho V_2^2$$

$$V_2 = \sqrt{2gh_2} = \sqrt{2 \times 10 \times 2} = \sqrt{40}$$

$$rate\ of\ flow\ = AV$$

$$= (2 \times 10^{-6} \times \sqrt{40})$$

$$12.64 \times 10^{-6} m/s$$

Q. 36 When a object is shot from the bottom of a long smooth inclined plane kept at an angle $60°$ with horizontal , it can travel a distance x_1 along the plane . But when the inclination is decreased to $30°$ and the same object is shot with the same velocity , it can travel $x_2\ distance.\ Then\ x_1 : x_2\ will\ be:$

Option 1:
$1 : \sqrt{2}$

Option 2:
$\sqrt{2} : 1$

Option 3:
$1 : \sqrt{3}$

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Option 4:

$1 : 2\sqrt{3}$

Correct Answer:

$1 : \sqrt{3}$

Solution:

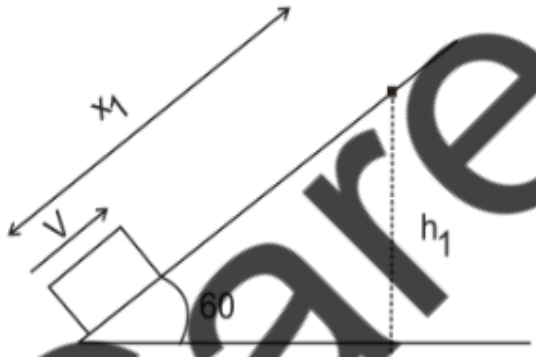
If only conservative forces act on a system, total mechnical energy remains constant -

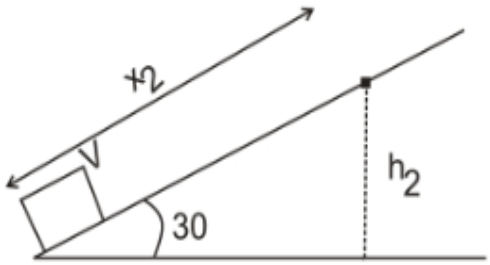
$$K + U = E\,(constant)$$

$$\Delta K + \Delta U = 0$$

$$\Delta K = -\Delta U$$

-





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$$\Delta KE = -\Delta PE$$

$$\frac{1}{2}mV^2 = mgh$$

$$h_1 = h_2$$

$$h_1 = x \sin 60$$

$$h_2 = x_2 \sin 30$$

$$\frac{x_1}{x_2} = \frac{\sin 30}{\sin 60}$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{3}/2} = 1/\sqrt{3}$$

Q. 37 A parallel palte capacitor of capacitance $20\mu F$ is being charged by a voltage source whose potential is changing at the rate of 3 V/s . The conduction current through the connecting wires , and the dispalcement current through the plates of the capacitor , would be resectively :

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Option 1:
zero, $60\mu A$

Option 2:
$60\mu A, 60\mu A$

Option 3:

$60\mu A, zero$

Option 4:

zero , zero

Correct Answer:

$60\mu A, 60\mu A$

Solution:

Maxwell's Displacement Current -

$$i_d = \varepsilon_o . \frac{d\Theta_E}{dt}$$

- wherein

i_d = Displacement Current

E_o = Electrical permittivity for vacuum

Θ_E = Total electric flux

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$$I_d = \varepsilon_0 \frac{d\theta_\varepsilon}{dt} = \frac{\varepsilon_0 d(E \cdot A)}{dt} \quad .....(\because E \cdot d = V$$

$$C = \frac{\varepsilon_0 A}{d})$$

$$= \frac{\varepsilon_0 d(VA)}{ddt}$$

$$I_D = C\frac{dv}{dt} = 20 \times 20 \times 10^{-6} \times 3$$

$$I_D = 60\mu A$$

$$For\ charged\ capacitor\ \ I_D = I_C$$

$$I_C = 60\mu A$$

Q. 38 The unit of thermal conductivity is :

Option 1:
JmK^{-1}

Option 2:
$Jm^{-1}K^{-1}$

Option 3:
WmK^{-1}

Option 4:
$Wm^{-1}K^{-1}$

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Correct Answer:

$Wm^{-1}K^{-1}$

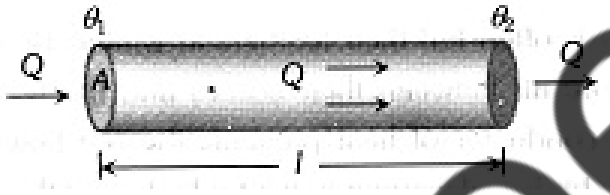
Solution:

Thermal Conductivity -

$$Q = \frac{KA(\theta_1 - \theta_2)t}{l}$$

K = thermal conductivity

- wherein



$$Q = \frac{KA(\theta_1 - \theta_2)t}{l}$$

$$K = \frac{Ql}{A\theta * t}$$

$unit \rightarrow W/mk$

$Wm^{-1}k^{-1}$

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Q. 39 Which of the following acts as a circuit protection device ?

Option 1:
conductor

Option 2:
inductor

Option 3:
switch

Option 4:
fuse

Correct Answer:
fuse

Solution:

Current through the circuit -

$i = 0$

- wherein

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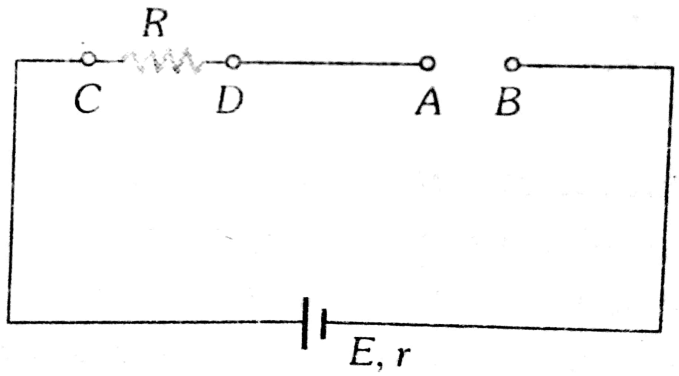

Answer is fuse as it will work as open circuit.So current passing will be zero.

Q. 40 A soap bubble , having radius of 1 mm is blown from, a detergent solution having a surface tension , of $2.5 \times 10^{-2} N/m$. The pressure inside the bubble equals at a point Z_0 below the free surface of water in a container of water = $10^3 K\, g/m^3$, the value of Z_0 is

Option 1:
100 cm

Option 2:
10 cm

Option 3:
1 cm

Option 4:
0.5 cm

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Correct Answer:
1 cm

Solution:

Change in Pressure of bubble in air -

$$\Delta P = \left(\frac{2T}{R}\right) \times 2 = \frac{4T}{R}$$

- wherein

T- Temperature

R- Radius

$$r = 10^{-3}$$

$$T = 2.5 \times 10^{-2} N/m$$

$$\Delta P = \frac{4T}{r}$$

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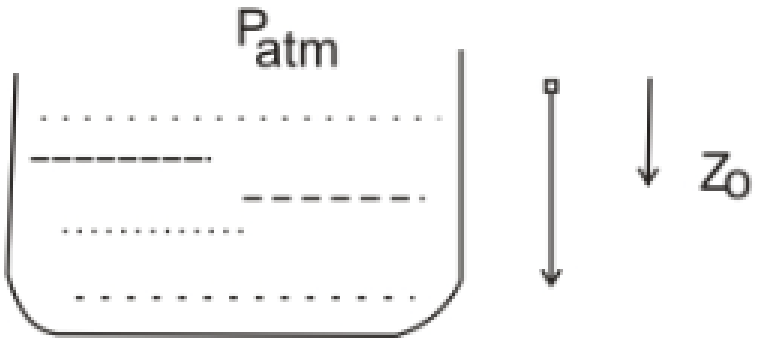

$$P_{z_0} = P_A + \rho g z_0$$

$$P_{z_0} - P_A = \delta g z_0 = 4T/r$$

$$Z_0 = \frac{4T}{r} \times (1/\rho y)$$

$$= \frac{4 \times 2.5 \times 10^{-2} \times 1}{10^{-3} \times 10^3 \times 10}$$

$$z_0 = 10^{-2}$$

$$z_0 = 1cm$$

Q. 41 The total energy of an electron in an atom in an orbit is -3.4 eV. Its kinetic and potential energies are respectively

Option 1:
-3.4 e V, -3.4 eV

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Option 2:
-3.4 e V , - 6.8 e V

Option 3:
3.4 e V , - 6.8 e V

Option 4:
3.4 e V , 3.4 e V

Correct Answer:
3.4 e V , - 6.8 e V

Solution:
Relation between K.E., P.E. & total energy -

$$K.E. = (Total\ energy)$$

$$= \frac{1}{2}(Potential\ energy)$$

- wherein

Potential Energy and total energy are negative. While kinetic energy is positive

$$E = -KE$$

$$2E = PE$$

$$KE = 3.4eV$$

$$PE = -6.8eV$$

Hence the answer is option (3).

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Q. 42 Two similar thin equiconvex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1 . When the space between the two lenses is filled with glycerin then the equivalent focal length is F_2 . The ratio $F_1 : F_2$

Option 1:
2 : 1

Option 2:
1 : 2

Option 3:
2 : 3

Option 4:
3 : 4

Correct Answer:
1 : 2

Solution:
Lens placed close to each other -

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} - - - - - - - + \frac{1}{f_n}$$

$\frac{1}{f_{eq}}$ =Equivalent focal length.

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- wherein

$f_1, f_2, - - - - - - - f_n$ are focal lenght of lenc 1, 2, 3, -----n

Lensmaker's Formula -

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

- wherein

$\mu_1 =$ the refractive index of the medium of the object

$\mu_2 =$ the refractive index of the lens

$R_1 \, and \, R_2$ are the radius of curvature of two surface

$$1/f = (u-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$R_1 = R_2$

$1/f = (u-1)(2/R)$

$F_1 = f/2$

$1/f' = (u-1)\left(\frac{1}{-R} - \frac{1}{R}\right)$

$1/f' = -(u-1)(2/R) = -1/f$

$1/F_2 = 1/f + 1/f + 1/f' = 1/f$

$$\Rightarrow \frac{F_1}{F_2} = 1/2$$

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Hence the answer is the Option (2).

Q. 43 Two parallel infinite line charges with linear charge densities $+\lambda C/m \ and \ -\lambda C/m$ are placed at a distance of 2R in free space. what is the electric field mid-way between the two line charges?

Option 1:
zero

Option 2:
$\frac{2\lambda}{\pi\epsilon_0 R}N/C$

Option 3:
$\frac{\lambda}{\pi\epsilon_0 R}N/C$

Option 4:
$\frac{\lambda}{2\pi\epsilon_0 R}N/C$

Correct Answer:
$\frac{\lambda}{\pi\epsilon_0 R}N/C$

Solution:
If the wire is infinitely long -

$l \rightarrow \alpha, \quad \alpha = \beta = 90^\circ$

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$$E_x = \frac{2k\lambda}{r},\ E_y = 0,\ \ E_{net} = \frac{\lambda}{2\pi\varepsilon_0 r}$$

$$V = \frac{-\lambda}{2\pi\varepsilon_0}\log_e r + c$$

$$|E_1| = |E_2|$$

$$E_1 = \frac{\lambda}{2\pi R\epsilon_0}$$

$$E_{net} = E_1 + E_2$$

$$E_{net} = \frac{\lambda}{\pi R\epsilon_0}$$

Hence the answer is the option (3).

Q. 44 In which of the following devices, the eddy current effect is not used?

Option 1:
induction furnace

Option 2:
magnetic braking in train

Option 3:
electromagnet

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Option 4:
electric heater

Correct Answer:
electric heater

Solution:
Eddy Current -

When a changing magnetic flux is applied to a bulk piece of conducting metal then circulating current is produced.

- wherein

i.e. These are circulating currents like eddies in matter.

The answer is the electric heater because an Electric heater is based on the heating effects of current.

Hence the answer is option (4).

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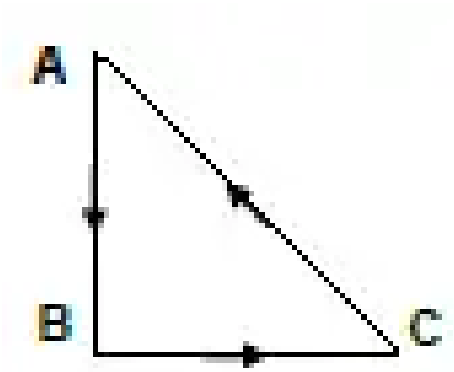
Q. 45 A particle moving with velocity $\vec{v}$ is acted by three forces shown by the vector triangle PQR . The velocity of the particle will :



Option 1:
increase

Option 2:
derease

Option 3:
remain constant

Option 4:
change acording to the smallest force QR

Correct Answer:
remain constant

Solution:

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Equilibrium of Concurrent Forces -

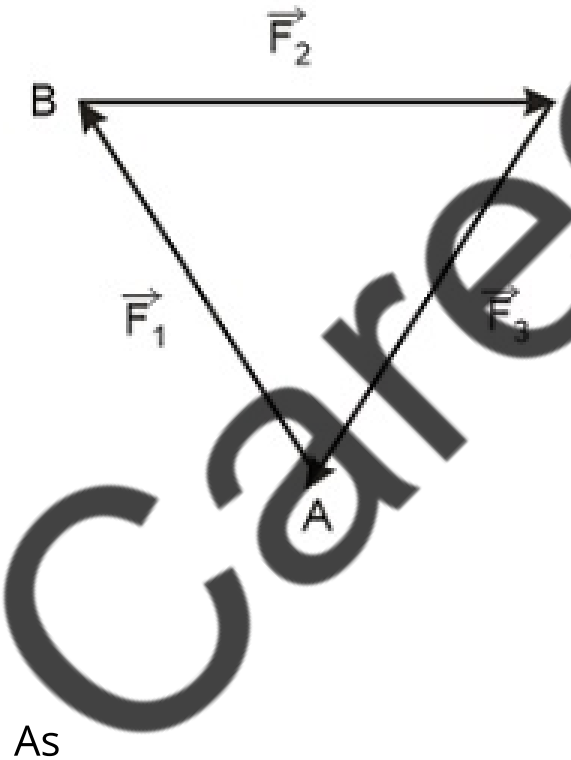
If all the forces working on a body are acting on the same point then they are said to be concurrent.

- wherein

$$\sum \vec{F_{net}} = 0$$

$$or \sum \vec{F}_x = 0, \sum \vec{F}_y = 0, \sum \vec{F}_z = 0$$

$*$ Three forces will be in equilibrium if they are represented by three sides of triangle taken in order



As

$$F_{net} = 0$$

$$so \ V \ remains \ constant$$

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Chemistry

Q. 1 For the second period elements, the **correct increasing** order of first ionization enthalpy is

Option 1:
Li < Be < B < C < N < O < F < Ne

Option 2:
Li < B < Be < C < O < N < F < Ne

Option 3:
Li < B < Be < C < N < O < F < Ne

Option 4:
Li < Be < B < C < O < N < F < Ne

Correct Answer:
Li < B < Be < C < O < N < F < Ne

Solution:
Ionization enthalpy - It is defined as the minimum energy required to remove the electron from the outermost shell of an isolated gaseous atom (X) to form the gaseous ion.

$$X_{(g)} \rightarrow X^{+}_{(g)} + e^{-}$$

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On moving left to right in the period, the atomic radii of elements decrease as the effective nuclear charge increases. Hence, the Ionization energy increases over the period. For N and O, Nitrogen has a filled ($2p^3$) electronic configuration while O ($2p^4$) has a partial. Hence, the ionization energy of N is more than that of O. For B and Be, In B electron is to be removed from the p- subshell ($2s^2 2p^1$), and in Be, it is from ($2s^2$). Since the penetration power of s>p>d>f. More energy is required to take an electron out from Be than B.

$$Li < B < Be < C < O < N < F < Ne$$

$$Li - 1s^2, 2s^1, \quad Be - 1s^2, 2s^2$$

$$C - 1s^2, 2s^2, 2p^2$$

$$N - 1s^2, 2s^2, 2p^3, \quad O - 1s^2, 2s^2, 2p^4$$

$$F - 1s^2, 2s^2, 2p^5, \quad Ne - 1s^2, 2s^2, 2p^6$$

Stability order of the orbitals with electrons

$$2p^6 > 2p^5(\text{due to F}) > 2p^3 > 2p^4 > 2p^2 > 2s^2 > 2p^1 > 2s^1$$

Fully Filled > Half Filled > Partially Filled

Hence, the answer is the option (2).

Q. 2 The method used to remove the temporary hardness of water is:

Option 1:
Calgon's Method

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Option 2:
Clark's Method

Option 3:
Ion-exchange method

Option 4:
Synthetic Resins method

Correct Answer:
Clark's Method

Solution:
Clark's method is used for removing the temporary hardness of the water by adding Slaked Lime $Ca(OH)_2$.

$$\mathrm{Ca\,(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + 2H_2O}$$
$$\mathrm{Mg(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + Mg(OH)_2 \downarrow + 2H_2O}$$

These precipitates can be removed by Filtration.

Hence, option number (2) is correct.

Q. 3 Which of the following is an amphoteric hydroxide?

Option 1:
$Sr(OH)_2$

Option 2:
$Ca(OH)_2$

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Option 3:
$Mg(OH)_2$

Option 4:
$Be(OH)_2$

Correct Answer:
$Be(OH)_2$

Solution:
Basic nature of hydroxides of alkaline metals -

Except $Be(OH)_2$ all hydroxides are basic in nature. this character increases down the group. The basic character increases down the group because the polarizing power of cation decreases down the group and hence, the covalent character.
$Be(OH)_2$ is amphoteric.

Due to the weak M-OH bond which breaks easily heterolytically to yield OH^-.

An Amphoteric hydroxide can neutralize acids and bases.

$$\text{Acid: } Be(OH)_2 + 6H^+ \longrightarrow 2Be^{2+} + 2H_2O$$
$$\text{Base: } Be(OH)_2 + 2OH^- \longrightarrow Be(OH)_4^{2-}$$

Hence, option number (4) is correct.

Q. 4 Among the following, the narrow-spectrum antibiotic is:

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Option 1:
penicillin G

Option 2:
ampicillin

Option 3:
amoxycillin

Option 4:
chloramphenicol

Correct Answer:
penicillin G

Solution:
As we have learnt,

Narrow spectrum antibiotics are effective against either gram-positive or gram-negative bacteria but not both.

Penicillin G is a narrow-spectrum antibiotic. It is used for the specific infection when the causative organism is known and will not kill as many of the normal microorganisms as a broad-spectrum antibiotic in the body.

Therefore, **Option(1) is correct**

Q. 5 Which mixture of the solution will lead to the formation of negatively charged colloidal $[AgI]I^-$ sol. ?

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Option 1:
50 mL of 1 M $AgNO_3$ + 50 mL of 1.5 M KI

Option 2:
50 mL of 1 M $AgNO_3$ + 50 mL of 2 M KI

Option 3:
50 mL of 2 M $AgNO_3$ + 50 mL of 1.5 M KI

Option 4:
50 mL of 0.1 M $AgNO_3$ + 50 mL of 0.1 M KI

Correct Answer:
50 mL of 1 M $AgNO_3$ + 50 mL of 2 M KI

Solution:
The reaction will be

$$\mathrm{AgNO_3 + KI \longrightarrow AgI + KNO_3}$$

$$\mathrm{AgI + KI \longrightarrow [AgI]I^- + K^+}$$

Negatively charged colloid

So, the overall reaction will be

$$\mathrm{AgNO_3 + 2KI \longrightarrow [AgI]I^- + KNO_3 + K^+}$$

Now, From the above reaction

For the formation of 1 mol of Negatively charged colloidal $\mathrm{[AgI]I^-}$, we Need 2 mol of KI.

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Or the mole ratio required will be $= \frac{\text{mol of } AgNO_3}{\text{mol of KI}} = \frac{1}{2}$

From the above options,

only $50 \text{ mL of } 1M \text{ } AgNO_3 + 50 \text{ mL of } 2M \text{ KI}$ has the same ratio.

$$\frac{\text{mol of } AgNO_3}{\text{mol of KI}} = \frac{50mL \times 1M}{50mL \times 2M}$$

$$\frac{\text{mol of } AgNO_3}{\text{mol of KI}} = \frac{1}{2}$$

So, Option 2 is correct.
Note - In the official answer key by NTA both options 1 and 2 were answers. 1

Q. 6 Conjugate base for Bronsted acids H_2O and HF are :

Option 1:
OH^- and H_2F^+, respectively

Option 2:
H_3O^+ and F^-, respectively

Option 3:
OH^- and F^-, respectively

Option 4:
H_3O^+ and H_2F^+, respectively

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Correct Answer:
OH^- and F^-, respectively

Solution:
Bronsted - Lowry Acids and Bases - According to Bronsted Lowry's theory, acid is a substance that is capable of donating a hydrogen ion H^+, and bases are substances that are capable of accepting H^+ ions.

The conjugate base of $H_2O \longrightarrow H^+ + OH^-$

and $HF \longrightarrow H^+ + F^- (\text{base})$

So, OH^- and F^- are the answer.

Hence, the answer is the option (3).

Q. 7 The number of sigma (σ) and pi(π) bonds in pent-2-en-4-yne is:

Option 1:
10 σ bonds and 3 π bonds

Option 2:
8 σ bonds and 5 π bonds

Option 3:
11 σ bonds and 2 π bonds

Option 4:
13 σ bonds and no π bonds

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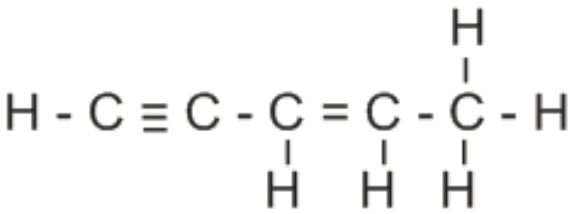
Correct Answer:

10 σ bonds and 3 π bonds

Solution:

Option 1 (correct answer)

Structure of pent-2-en-4-yne

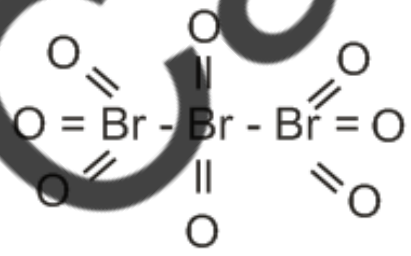


Here there are 10 σ bonds $= 6\ -(C-H)$ and $4(C-C)$ bonds

And 3 π bonds $= 2[C \equiv C]$ and $1[C=C]$

Q. 8 The **correct** structure of tribromooctaoxide is:

Option 1:



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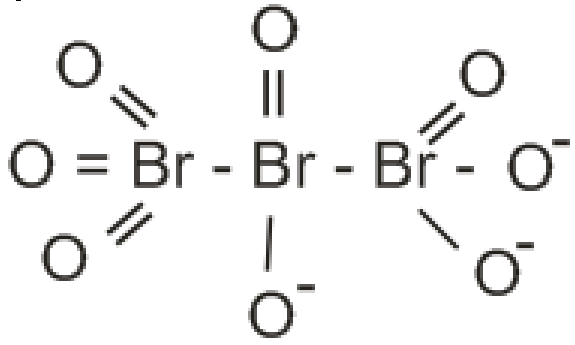
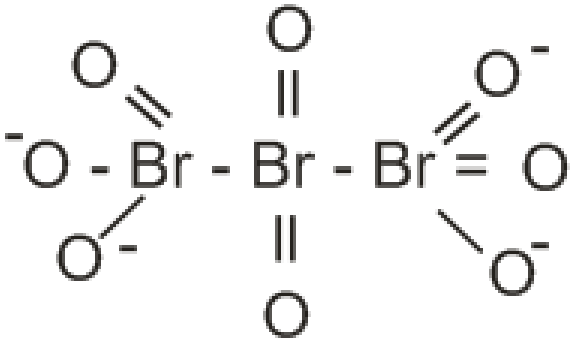
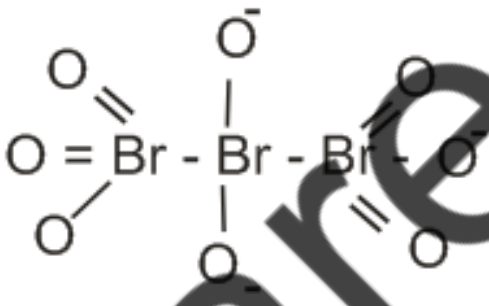
Option 2:



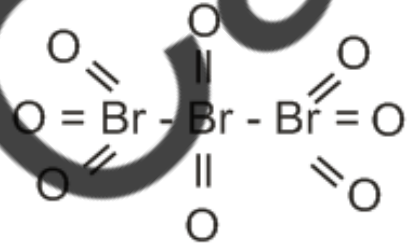
Option 3:



Option 4:



Correct Answer:



Solution:

Structure of tribromooctaoxide is: (not ion)

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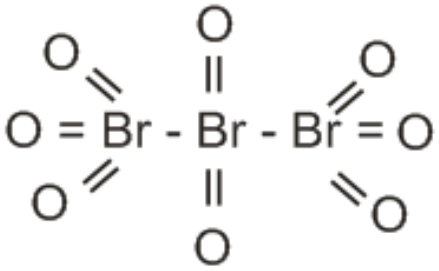

The correct answer is option 1.

Q. 9 pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$ is:

Option 1:
0.5×10^{-15}

Option 2:
0.25×10^{-10}

Option 3:
0.125×10^{-15}

Option 4:
0.5×10^{-10}

Correct Answer:
0.5×10^{-15}

Solution:
$Ca(OH)_2 \rightleftharpoons Ca^{2+} + 2OH^-$

Given pH = 9

So,

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$$pH + pOH = 14$$

$$\Rightarrow pOH = 14 - 9 = 5$$

$$\Rightarrow -\log[OH^-] = 5$$

$$\Rightarrow [OH^-] = 10^{-5}$$

So,

$$[Ca^{2+}] = \frac{[OH^-]}{2} = \frac{10^{-5}}{2}$$

And K_{sp} will be

$$K_{sp} = [Ca^{2+}][OH^-]^2$$

$$K_{sp} = \frac{10^{-5}}{2} \times (10^{-5})^2$$

$$K_{sp} = 0.5 \times 10^{-15} mol^3/L^3$$

Hence, the answer is the option (1).

Q. 10 The **correct** order of basic strength of methyl substituted amines in an aqueous solution is:

Option 1:

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

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Option 2:

$(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$

Option 3:

$(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$

Option 4:

$CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

Correct Answer:

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

Solution:

Comparision of Basic nature in aliphatic amines in gaseous phase -

The +I effect of the alkyl group, which increase the electron density on N atom, facilitates the nitrogen atom to use its unshared pair of electrons for bond formation. Thus R - NH_2 is expected to be more basic than NH_3. Going by the same arguments the basic strength of amines is expected to increase with the increase in the number of alkyl groups attached to the nitrogen atom.

- wherein

$R_3N > R_2NH > RNH_2 > NH_3$

Comparision of Basic nature in aliphatic amines in aqueous phase. -

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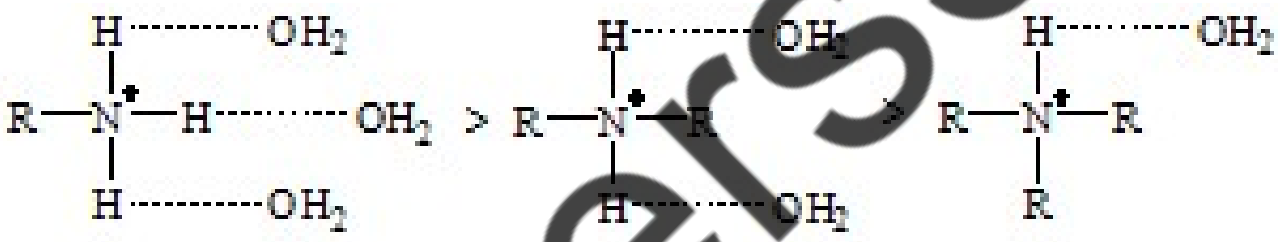
The basic strength of an amine in aqueous medium is determined not only on the basic of +I effect of the alkyl groups attached to the nitrogen atom but also the solvation of alkyl ammonium ion formed by uptake of proton.

The solvation of alkyl ammonium ion follows the above order. Both the +I effect and solvation of amine make basic order of aniline in aqueous medium as

$$2^\circ > 1^\circ > 3^\circ > NH_3$$

- wherein



Option 1

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In aqueous solution the order of basicity changes

$$(CH_3)_2NH > CH_3NH > (CH_3)_3NH$$

Amines in water solution exist as ammonium ions. Due to H-bonding 1° amine and 2° amine are more basic than 3° amine.

Q. 11 For a cell involving one electron $E^{\circ}_{cell} = 0.59V$ at 298K, the equilibrium constant for the cell reaction is:

[Given that $\frac{2.303RT}{F} = 0.059V$ at $T = 298K$]

Option 1:
1.0×10^{2}

Option 2:
1.0×10^{5}

Option 3:
1.0×10^{10}

Option 4:
1.0×10^{30}

Correct Answer:
1.0×10^{10}

Solution:
We know,

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$$E^{o}_{cell} = \frac{2.303RT}{nF} \log K_{eq}$$

So,

$$\log K_{eq} = \frac{E^{o}_{cell}}{\frac{2.303RT}{nF}}$$

$$\log K_{eq} = \frac{0.59}{\frac{0.059}{1}} \qquad [\text{and (One electron) } n = 1]$$

$$\log K_{eq} = \frac{0.59}{0.059}$$

$$\log K_{eq} = 10$$

$$\log K_{eq} = 10 \log_{10} 10$$

$$\log K_{eq} = \log_{10} 10^{10}$$

$$K_{eq} = 1 \times 10^{10}$$

Hence, the answer is the option (3).

Q. 12 Among the following, the one that is not a greenhouse gas is:

Option 1:
nitrous oxide

Option 2:
methane

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Option 3:
ozone

Option 4:
sulphur dioxide

Correct Answer:
sulphur dioxide

Solution:

Greenhouse gases -

Emissions that rise into the atmosphere and trap the sun's energy, keeping heat from escaping.

- wherein

CO_2, CH_4, O_3, CFC_5 watervapour , Nitrous oxide.

N_2O (Nitrous oxide)

CH_4 (Methane)

O_3 (Ozone)

All the above are greenhouse gases but SO_2 is not greenhouse gas.

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Q. 13 The mixture that forms maximum boiling azeotrope is:

Option 1:
Water + Nitric Acid

Option 2:
Ethanol + Water

Option 3:
Acetone + Carbon disulphide

Option 4:
Heptane + Octane

Correct Answer:
Water + Nitric Acid

Solution:
The Solutions that show a large negative deviation from Raoult's law form maximum boiling azeotrope at a specific combination.

Nitric acid and water, 68% Nitric acid + 32% Water by mass with a boiling point of 393.5 K, is an example of this class of azeotrope.

Ethanol and water show a positive deviation.

Acetone and Carbon disulfide show a positive deviation.

Heptane and Octane show no deviation because they behave as ideal solutions.

Hence, the answer is the option (1).

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Q. 14 Which one is malachite from the following?

Option 1:
$CuFeS_2$

Option 2:
$Cu(OH)_2$

Option 3:
Fe_3O_4

Option 4:
$CuCO_3 \cdot Cu(OH)_2$

Correct Answer:
$CuCO_3 \cdot Cu(OH)_2$

Solution:
Chalcopyrite - $CuFeS_2$

Cupic hydroxide $Cu(OH)_2$

Magnetite Fe_3O_4

Malachite $CuCO_3.Cu(OH)_2$

Hence, Option 4 is the correct answer.

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Q. 15 Match the following:

(a) Pure Nitrogen	(i) Chlorine
(b) Haber process	(ii) Sulphuric Acid
(c) Contact process	(iii) Ammonia
(d) Deacon's process	(iv) Sodium azide or Barium Azide

Which of the following is the **correct** option?

(a) (b) (c) (d)

Option 1:
(i) (ii) (iii) (iv)

Option 2:
(ii) (iv) (i) (iii)

Option 3:
(iii) (iv) (ii) (i)

Option 4:
(iv) (iii) (ii) (i)

Correct Answer:
(iv) (iii) (ii) (i)

Solution:

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a) Pure Nitrogen $\Rightarrow$ Sodium azide (NaN_3) or Barium azide $(Ba(N_3)_2)$

b) Haber process $\Rightarrow$ Making ammonia

$$N_2 + 3H_2 \leftrightharpoons 2NH_3$$

c) Contact process $\Rightarrow$ producing sulphuric acid (H_2SO_4)

d) Deacon's process $\Rightarrow$ Produces Cl_2 gas

$$4HCl + O_2 \rightarrow 2Cl_2 + 2H_2O$$

The correct answer is option 4.

Q. 16 Which is the **correct** thermal stability order for $\mathrm{H_2E\ (E = O, S, Se, Te\ and\ Po)}$?

Option 1:

$\mathrm{H_2S < H_2O < H_2Se < H_2Te < H_2Po}$

Option 2:

$\mathrm{H_2O < H_2S < H_2Se < H_2Te < H_2Po}$

Option 3:

$\mathrm{H_2Po < H_2Te < H_2Se < H_2S < H_2O}$

Option 4:

$\mathrm{H_2Se < H_2Te < H_2Po < H_2O < H_2S}$

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Correct Answer:

$\mathrm{H_2Po < H_2Te < H_2Se < H_2S < H_2O}$

Solution:

The thermal stability of hydrides of group 16 elements decreases down the group. It is because the E-H bond dissociation energy decreases down the group with an increase in the size of a central atom.

Hence, the answer is the option (3).

Q. 17 Identify the **incorrect** statement related to PCl_5 from the following:

Option 1:

Three equatorial $P-Cl$ bonds make an angle of 120^o with each other

Option 2:

Two axial $P-Cl$ bonds make an angle of 180^o with each other

Option 3:

Axial $P-Cl$ bonds are longer than equatorial $P-Cl$ bonds

Option 4:

PCl_5 molecule is non-reactive

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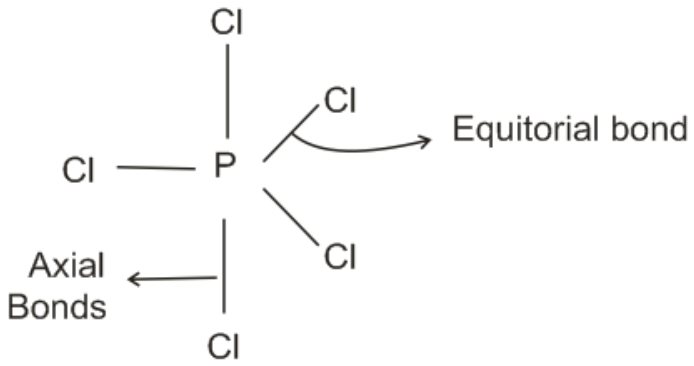
Correct Answer:

PCl_5 molecule is non-reactive

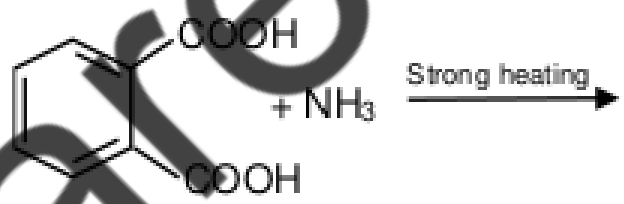
Solution:



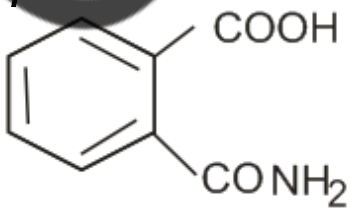
In PCl_5, there are two types of bonds that are axial bonds and equatorial bonds. It is found that axial bonds are longer than equatorial bonds. Due to unequal bond PCl_5 is unstable and highly reactive.

The correct answer is option 4.

Q. 18 The major product of the following reaction is:



Option 1:



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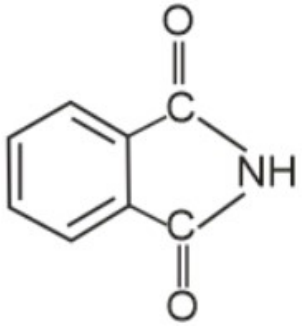
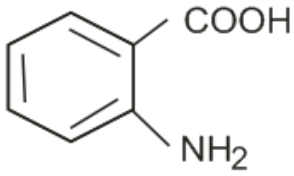
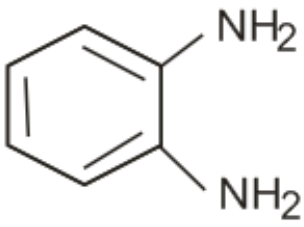
Option 2:



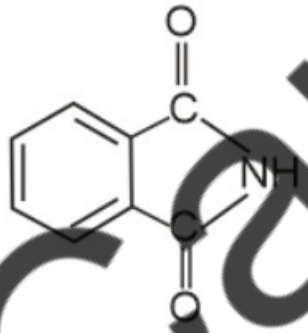
Option 3:



Option 4:



Correct Answer:



Solution:

The product will be-

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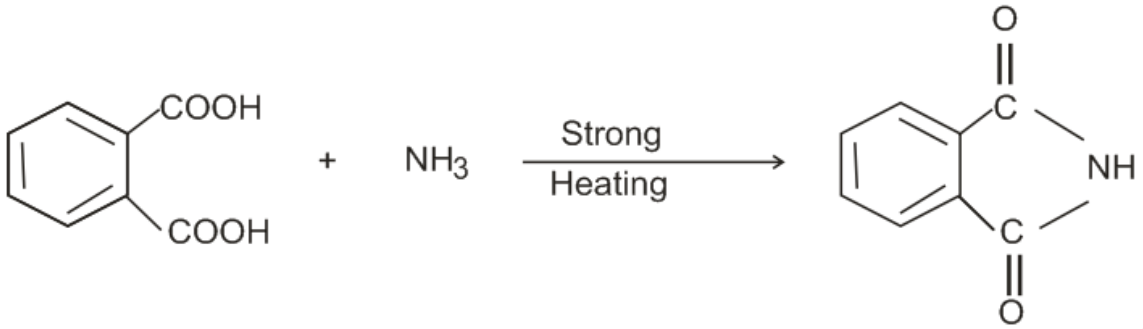
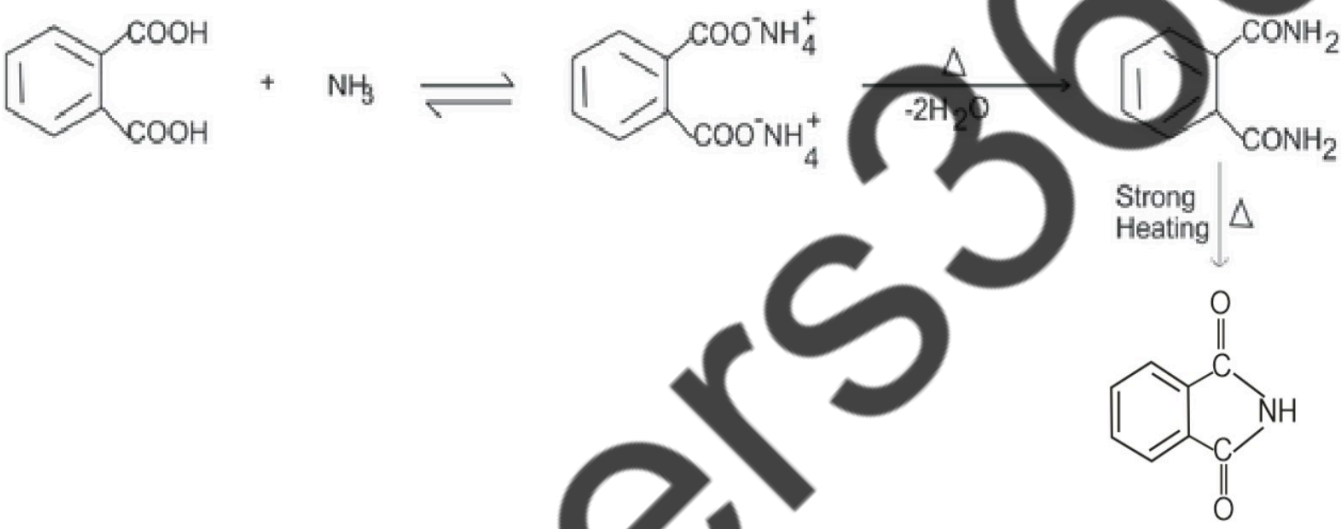


Phthalimide

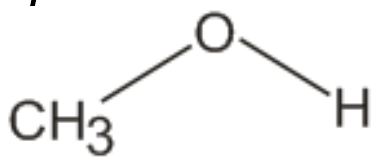
The reaction will be -



Q. 19 The compound that is most difficult to protonate is:

Option 1:

Option 2:



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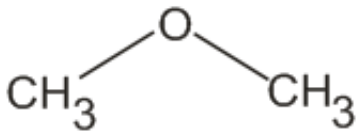
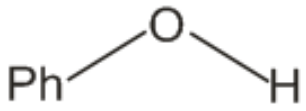
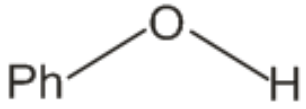
Option 3:



Option 4:



Correct Answer:



Solution:

Protonation of phenol is difficult.

Reason:- In alcohols the electron density on oxygen is high. $(R-O-H)$

due to the localisation of its lone pair and e^- donating nature of the alkyl group. So, their protonation is easy.

However, in phenols $(Ph-O-H)$ the lone pair of oxygen is in resonance with the benzene ring and thus is delocalised.

So, the protonation of phenol is difficult.

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Q. 20 The manganate and permanganate ions are tetrahedral, due to:

Option 1:
The π-bond involves overlap of p-orbitals of oxygen with d-orbitals of manganese.

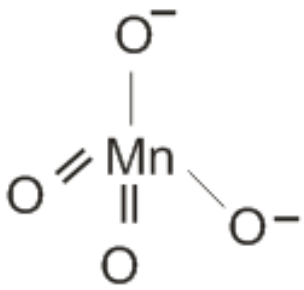
Option 2:
There is no π-bonding.

Option 3:
The π-bonding involves the overlap of p-orbitals of oxygen with p-orbitals of manganese.

Option 4:
The π—bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese.

Correct Answer:
The π-bond involves overlap of p-orbitals of oxygen with d-orbitals of manganese.

Solution:

Manganate ion:- MnO_4^{2-}  [Green, paramagnetic]

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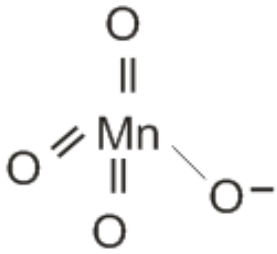
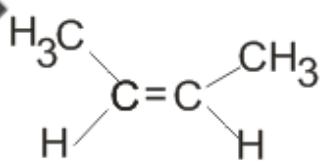
Permanganate ion:- MnO_4^-  [purple, diamagnetic]

$\Rightarrow$These ions are tetrahedral in shape.

$\Rightarrow$The π-bonding takes place by the overlap of p-orbitals of oxygen with d-orbitals of manganese.

Hence, the option number (1) is correct.

Q. 21 The most suitable reagent for the following conversion is:

$H_3C - C \equiv C - CH_3 \rightarrow$ 

Cis-2-butene

Option 1:
Na/liquid NH_3

Option 2:
H_2, Pd/C, Quinoline

Option 3:
Zn/ HCl

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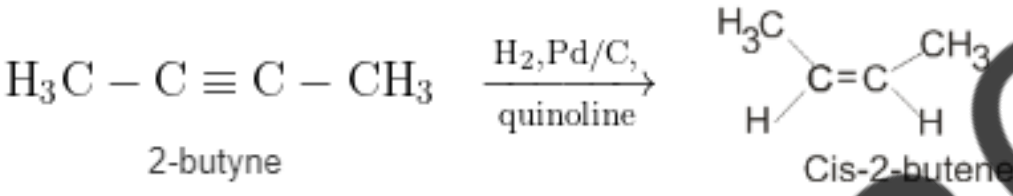
Option 4:

Hg^{2+}/H^{+}, H_2O

Correct Answer:

H_2, Pd/C, Quinoline

Solution:

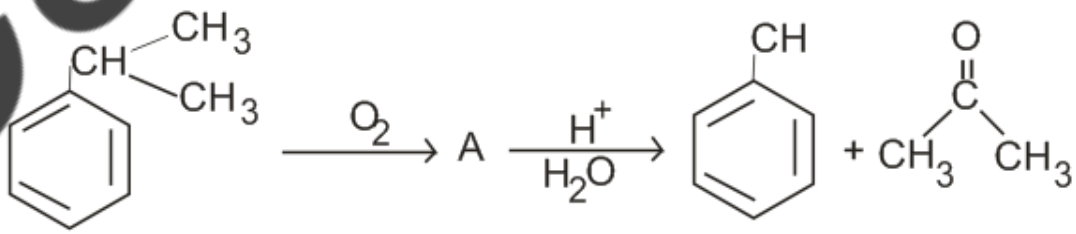


$H_2, Pd/C, \text{Quinoline}$:- Lindlar's Catalyst.

It is a palladium catalyst poisoned with traces lead and quinoline, that reduces its activity such that it can only reduce alkynes, not alkanes. It always gives the cis-alkene in contrast to Na/NH_3, which gives the trans-alkenes.

Option 2 is correct.

Q. 22 The structure of intermediate A, in the following reaction, is:



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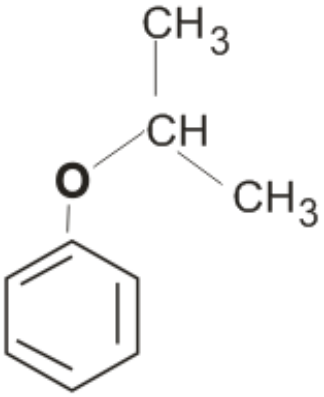
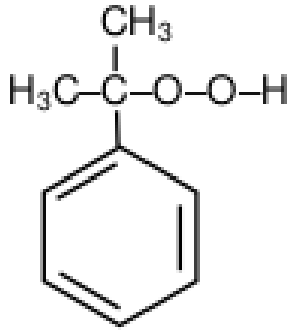
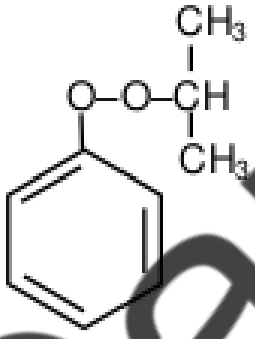
Option 1:



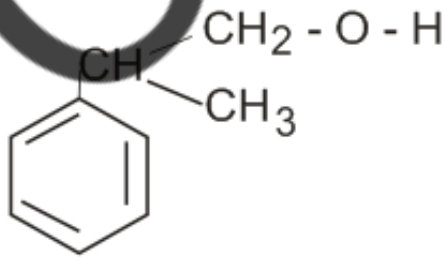
Option 2:



Option 3:



Option 4:



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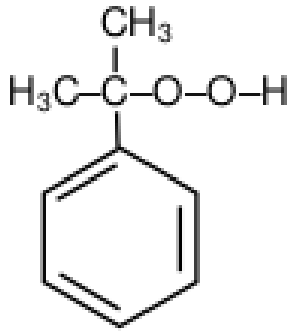
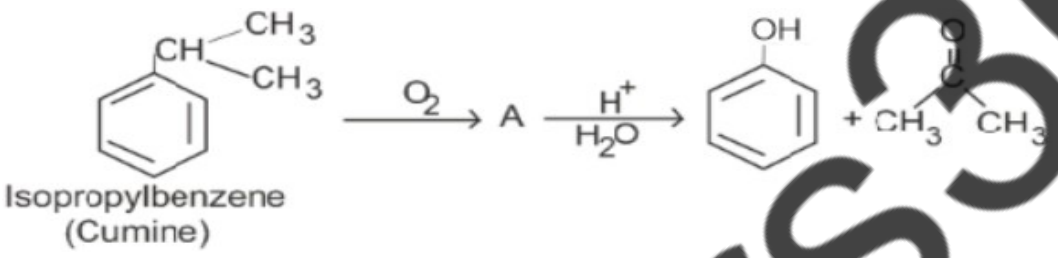
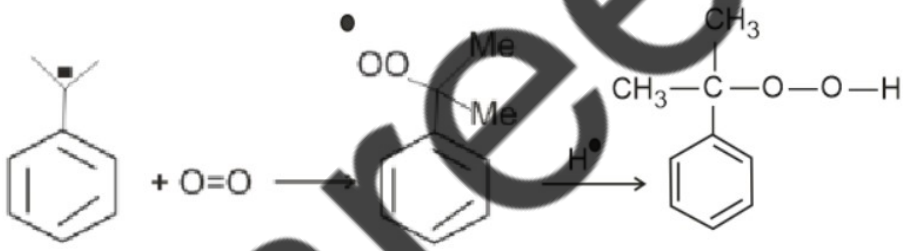
Correct Answer:



Solution:



Cumene is oxidized in air, which removes the tertiary benzylic hydrogen from cumene and hence forms cumene radical.



The cumene radical then bonds with an oxygen molecule to give cumene peroxide radical, which in turn forms cumene hydroperoxide by abstracting benzylic hydrogen from the cumene molecule.

Option 2 is correct.

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Q. 23 If the rate constant for a first order reaction is k, the time (t) required for the completion of 99% of the reaction is given by:

Option 1:

$$t_{\frac{1}{2}} = \frac{0.693}{k}$$

Option 2:

$$t_{\frac{1}{2}} = \frac{6.909}{k}$$

Option 3:

$$t_{\frac{1}{2}} = \frac{4.606}{k}$$

Option 4:

$$t_{\frac{1}{2}} = \frac{2.303}{k}$$

Correct Answer:

$$t_{\frac{1}{2}} = \frac{4.606}{k}$$

Solution:

For first-order reaction,

$$t_{\frac{1}{2}} = \frac{2.303}{k} \log_{10} \frac{a}{a-x}$$

$$a = 100\% \quad x = 99\%$$

$$a - x = 1\%$$

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$$t_{\frac{1}{2}} = \frac{2.303}{k} \log_{10} \frac{100}{1} = \frac{2.303}{k} \log_{10} 10^2$$

$$t_{\frac{1}{2}} = \frac{2.303}{k} \times 2 = \frac{4.606}{k}$$

Hence, the answer is the option (3).

Q. 24 Which of the following reactions are disproportionation reactions?

(a) $2Cu^{+} \rightarrow Cu^{2+} + Cu^{0}$

(b)
$3MnO_4^{2-} + 4H^{+} \rightarrow MnO_2 + 2H_2O + 2MnO_4^{-}$

(c) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

(d)
$2MnO_4^{-} + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^{\oplus}$

Select the **correct** option from the following:

Option 1:
(a) and (b) only

Option 2:
(a), (b) and (c)

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Option 3:
(a), (c) and (d)

Option 4:
(a) and (d) only

Correct Answer:
(a) and (b) only

Solution:
Disproportionation Reactions - A special type of redox reaction in which one oxidation state is simultaneously oxidized and reduced. One of the reacting substances in a disproportionation reaction always contains an element that can exist in at least three oxidation states.

$$2H_2\,O_2(aq) \rightarrow 2H_2\,O(l) + O_2(g)$$

+1 -1 +1 -2 0

Option 1

Disproportionation is a type of redox reaction in which a species is simultaneously reduced and oxidized forming two different products.

∴ (a) $2Cu^{+} \rightarrow Cu^{2+} + Cu^{0}$

(b) $3MnO_4^{2-} + 4H^{+} \rightarrow 2MnO_2 + 2H_2O$

$$3\overset{+6}{Mn}O_4^{2-} + 4H^{+} \rightarrow 2\overset{+7}{Mn}O_4^{-} + \overset{+4}{Mn}O_2 + 2H_2O$$

Hence, the answer is the option (1).

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Q. 25 In which case change in entropy is negative?

Option 1:
Evapouration of water

Option 2:
Expansion of a gas at constant temperature

Option 3:
Sublimation of solid to gas

Option 4:
$2H_{(g)} \rightarrow H_{2(g)}$

Correct Answer:
$2H_{(g)} \rightarrow H_{2(g)}$

Solution:
$2H_{(g)} \rightarrow H_{2(g)}$

degree of randomness decreases

$\therefore$ entropy decreases and is negative.

Hence, the answer is the option (4).

Q. 26 The biodegradable polymer is:

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Option 1:
nylon-6,6

Option 2:
nylon 2 - nylon 6

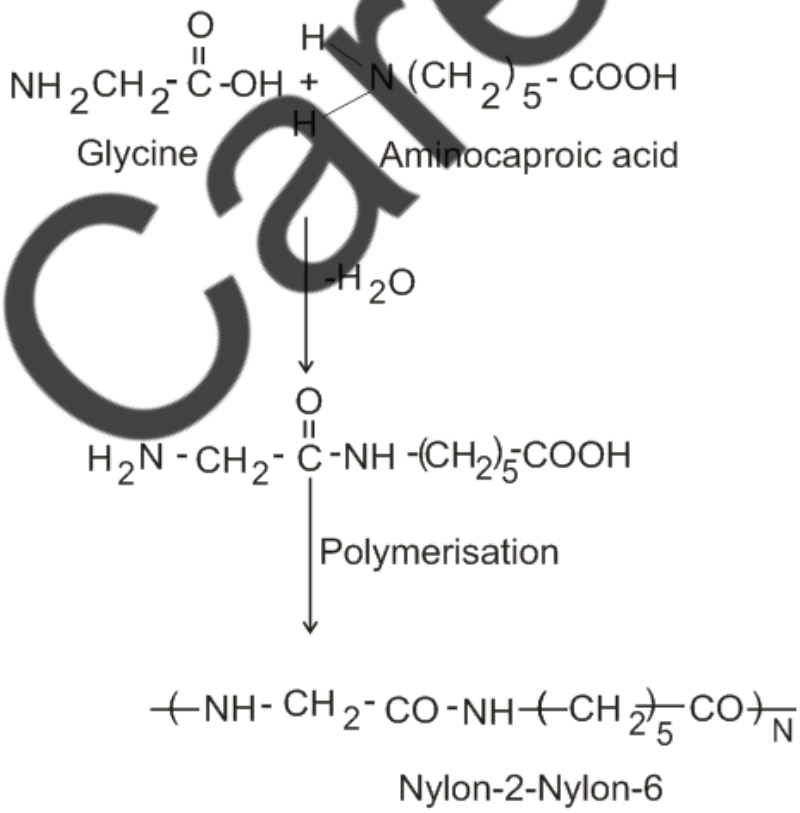
Option 3:
nylon-6

Option 4:
Buna-S

Correct Answer:
nylon 2 - nylon 6

Solution:
The biodegradable polymer is Nylon-2-Nylon-6

monomers: Glycine, aminocaproic acid



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Q. 27 A gas at 350K and 15 bar have a molar volume 20 percent smaller than that of an ideal gas under the same conditions. The **correct** option for the gas and its compressibility factor (Z) is:

Option 1:
Z>1 and attractive forces are dominant

Option 2:
Z>1 and repulsive forces are dominant

Option 3:
Z<1 and attractive forces are dominant

Option 4:
Z<1 and repulsive forces are dominant

Correct Answer:
Z<1 and attractive forces are dominant

Solution:
Compressibility Factor -

$$Z = \frac{V_{real}}{V_{ideal}}$$

$V_{real} = Z \times V_{ideal}$

The molar volume of gas is 20% smaller than the ideal gas.

for ideal gas $V_{real} = V_m$, since Z=1

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$\therefore\ V_m < V_{\text{ideal}}$

put the value of V_{real}

$Z \times V_{ideal} < V_{ideal}$

$Z<1$

When Z < 1: Gas shows negative deviation and is more compressible than expected from ideal behavior. Z < 1 corresponds to attractive forces between molecules.

Hence, option 3 is correct

Q. 28 A compound is formed by cation C and anion A. The Anions form hexagonal close packed lattice and the cations occupy 75% of the octahedral void. The formula of the compound is

Option 1:
C_2A_3

Option 2:
C_3A_2

Option 3:
C_3A_4

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Option 4:
C_4A_3

Correct Answer:
C_3A_4

Solution:
We know,

Anions A: No of ions per unit in HCP = 6

Cations C: No of octahedral voids in HCP = 6

Cations C occupy 75% of the octahedral void.

So, $75\ \%\ \text{of}\ 6 = 6 \times 75/100 = 4.5$

Then, the formula of compound

$\mathrm{C_{4.5}A_6} = \mathrm{C_3A_4}$

Therefore, the correct option is (3).

Q. 29 Enzymes that utilize ATP in phosphate transfer require an alkaline earth metal (M) as the cofactor M is:

Option 1:
Be

Option 2:
Mg

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Option 3:
Ca

Option 4:
Sr

Correct Answer:
Mg

Solution:
Enzymes -

These are biological catalysts that enhances the rate of reaction by lowering the activation energy.

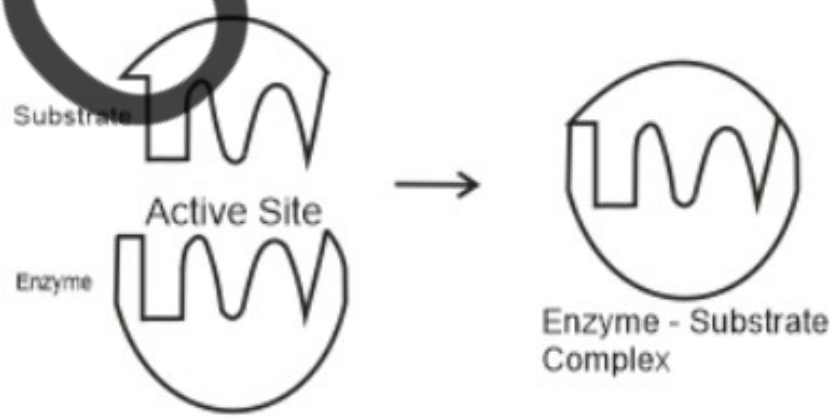
- wherein

Eg. Maltose amylase, carbonic anhydrase,etc.

Lock and key hypothesis of enzyme action -

Enzymes molecules act like a lock whereas substrate acts like a key . Substrate fits in the active site of the enzyme and forms a complex which decompose to yield products

- wherein



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Option 2

Mg is the alkaline earth metal as the co-factor. (FACT)

Q. 30 4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The **correct** option is:

Option 1:
5f > 6p > 5p > 4d

Option 2:
6p > 5f > 5p > 4d

Option 3:
6p > 5f > 4d > 5p

Option 4:
5f > 6p > 4d > 5p

Correct Answer:
5f > 6p > 5p > 4d

Solution:
Aufbau Principle -

In the ground state of the atoms, the orbitals are filled in order of their increasing energies.

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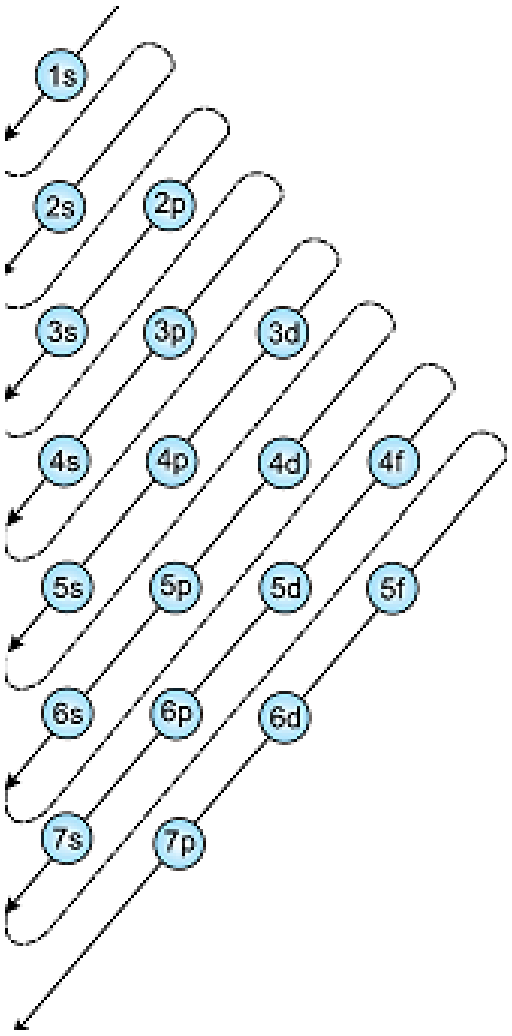

Option 1

	5f	>	6p	>	5p	>	4d
$(n+l)=$	5 + 3		6 + 1		5 + 1		4 + 2
value	8		7		6		6

The energy of various orbitals are compared with $(n+l)$ rule.

The orbitals having a lower value of (n + l) have lower energy. If the value of $(n+l)$ is the same for two orbitals, then the orbital with a lower value of 'n' would have lower energy and filled first.

Therefore, the correct option is (1).

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Q. 31 For the cell reaction

$$2Fe^{3+}(aq) + 2I^{-}(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$$

$E^{o}_{cell} = 0.24V$ at 298K. The standard Gibbs energy $(\Delta_r G^{o})$ of the cell reaction is:

[Given that Faraday constant F = 96500 C mol^{-1}]

Option 1:
-46.32 kJ mol^{-1}

Option 2:
-23.16 kJ mol^{-1}

Option 3:
46.32 kJ mol^{-1}

Option 4:
23.16 kJ mol^{-1}

Correct Answer:
-46.32 kJ mol^{-1}

Solution:
We know that

$$\Delta G^{o} = -nFE^{o}_{cell}$$

$$\Rightarrow \Delta G^{o} = -2 \times 96500 \times 0.24$$

$$\Rightarrow \Delta G^{o} = -46.32\ kJmol^{-1}$$

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Hence, the answer is the option (1).

Q. 32 For an ideal solution, the **correct** option is:

Option 1:
$\Delta_{mix} S = 0$ at constant T and P

Option 2:
$\Delta_{mix} V \neq 0$ at constant T and P

Option 3:
$\Delta_{mix} H = 0$ at constant T and P

Option 4:
$\Delta_{mix} G = 0$ at constant T and P

Correct Answer:
$\Delta_{mix} H = 0$ at constant T and P

Solution:
For an ideal solution, the Gibbs free energy is always negative, thus their mixing is always spontaneous. The volume change of the mixing of ideal solutions is zero.

An ideal solution is one in which the enthalpy of the solution is zero

$\therefore \Delta_{mix} H = 0$ at constant T and P

Hence, the answer is the option (3).

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Q. 33 The non-essential amino acid among the following is:

Option 1:
valine

Option 2:
leucine

Option 3:
alanine

Option 4:
lysine

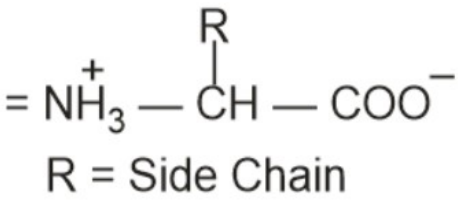
Correct Answer:
alanine

Solution:

Amino acids -

Compounds containing carboxyl and amino groups

- wherein



20 amino acids present in protein -

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Name	One letter code
Glycine -	G
Alanine -	A
Valine -	V
Leucine -	L
Isoleucine -	I
Arginine -	R
Lysine -	K
Glutamic acid -	E
Aspartic acid -	D
Glutamine -	Q
Asparagive -	N
Threonene -	T
Soine -	S
Cysteine -	C
Methionine -	M
Phenylalanine -	F
Tyrosine -	Y
Tryptophan -	W

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Histidine - H

Proline - P

-

Non essential amino acids are those which produce in the human body itself.

Essential amino acids are those which do not produce in the human body & we have to take them from outside sources.

∴ Alanine is the non-essential amino acid.

Option 3 is correct.

Q. 34 What is the **correct** electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory?

Option 1:

$t_{2g}^4 e_g^2$

Option 2:

$t_{2g}^6 e_g^0$

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Option 3:

$e^3 t_2^3$

Option 4:

$e^4 t_2^2$

Correct Answer:

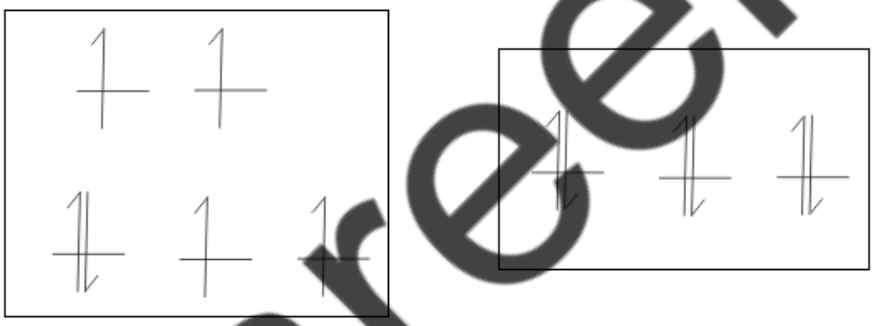
$t_{2g}^6 e_g^0$

Solution:

Splitting occurs as follows for octahedral

For Fe^{2+} (d^6), electron distribution occurs as follows:

For High spin complex and low spin complexes, the respective arrangement of electrons will be



CN^- being a strong field ligand casusedcausede splitting and forms a low spin complex with electronic configuration. $t_{2g}^6 e_g^0$ is the correct distribution.

Option 2 is correct.

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Q. 35 Which of the following is **incorrect** statement?

Option 1:
PbF_4 is covalent in nature

Option 2:
$SiCl_4$ is easily hydrolysed

Option 3:
GeX_4 (X = F, Cl, Br, I) is more stable than GeX_2

Option 4:
SnF_4 is ionic in nature

Correct Answer:
PbF_4 is covalent in nature

Solution:
Fajans Rules -

It accounts for the covalent character in ionic compounds covalency is favored by

1. Small cation
2. Large anion and
3. Large charge on either ion

Condition for the formation of ionic bonds -

1. Low ionization potential of cation
2. High electron affinity of anion
3. high lattice energy

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(i) PbF_4 is not covalent. Because the Pb atom is very large & F atom is very small, according to Fajan's rule, PbF_4 is not covalent.

(ii) $SiCl_4$ is not a very stable compound because of steric hindrance thus can be hydrolyzed very easily

(iii) GeX_4 is more stable than GeX_2 as in GeX_4 all the orbitals are fully filled.

(iv) SnF_4 is ionic. Because the F atom is very small & Sn atom is very large, according to Fajan's rule, it is ionic.

Hence, the answer is the option (1).

Q. 36 Match the Xenon compound in **Column-I** with its structure in C**olumn-II** and assign the correct code:

Column-I	**Column-II**
(a) XeF_4	(i) pyramidal
(b) XeF_6	(ii) Square planar
(c) $XeOF_4$	(iii) distorted octahedral
(d) XeO_3	(iv) square pyramidal

Code:

(a) (b) (c) (d)

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Option 1:
(i) (ii) (iii) (iv)

Option 2:
(ii) (iii) (iv) (i)

Option 3:
(ii) (iii) (i) (iv)

Option 4:
(iii) (iv) (i) (ii)

Correct Answer:
(ii) (iii) (iv) (i)

Solution:
Hybridization -

$$SN = \frac{H + M - Q}{2}$$

$$SN_{XeF_4} = \frac{8+4}{2} = 6 \quad sp^3d^2$$

$XeF_4 \longrightarrow$ is square planar with 2 lone pairs and 4 bond pairs.

$$SN_{XeF_6} = \frac{8+6}{2} = 7 \quad sp^3d^3$$

$XeF_6 \longrightarrow$ is a distorted octahedral because of the lone pair present.

$$SN_{XeOF_4} = \frac{8+4}{2} = 6 \quad sp^3d^2$$

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$XeOF_4 \longrightarrow$ It is square pyramidal with one lone pair & its hybridization is sp^3d^2

$XeO_3 \longrightarrow$ Hybridization is sp^3 & its shape is pyramidal with one lone pair.

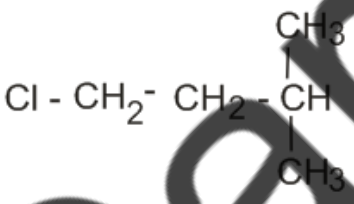
$$SN_{XeO_3} = \frac{8+0}{4} = 4 \quad sp^3$$

$\therefore$ (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

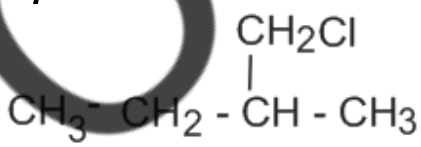
Hence, the answer is the option (2).

Q. 37 An alkane "A" on reaction with O_3 and Zn-H_2O gives propane and ethanal in equimolar ratio. Addition of HCl to Alkane "A" gives "B" as the major product. The structure of the product "B" is

Option 1:



Option 2:



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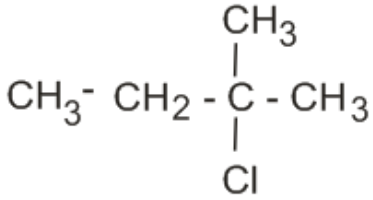
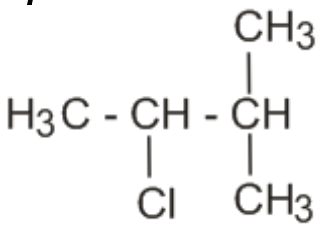
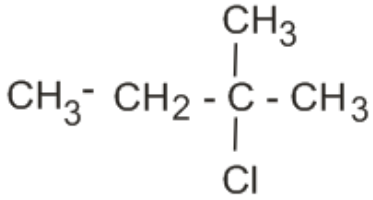
Option 3:



Option 4:



Correct Answer:



Solution:

This oxidation of alkene occurs at the double bond. Thus, propane & ethanol must combine to give the alkene

(Propanone)

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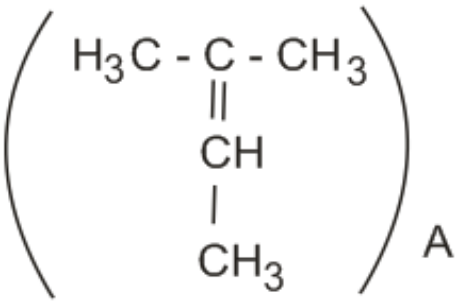
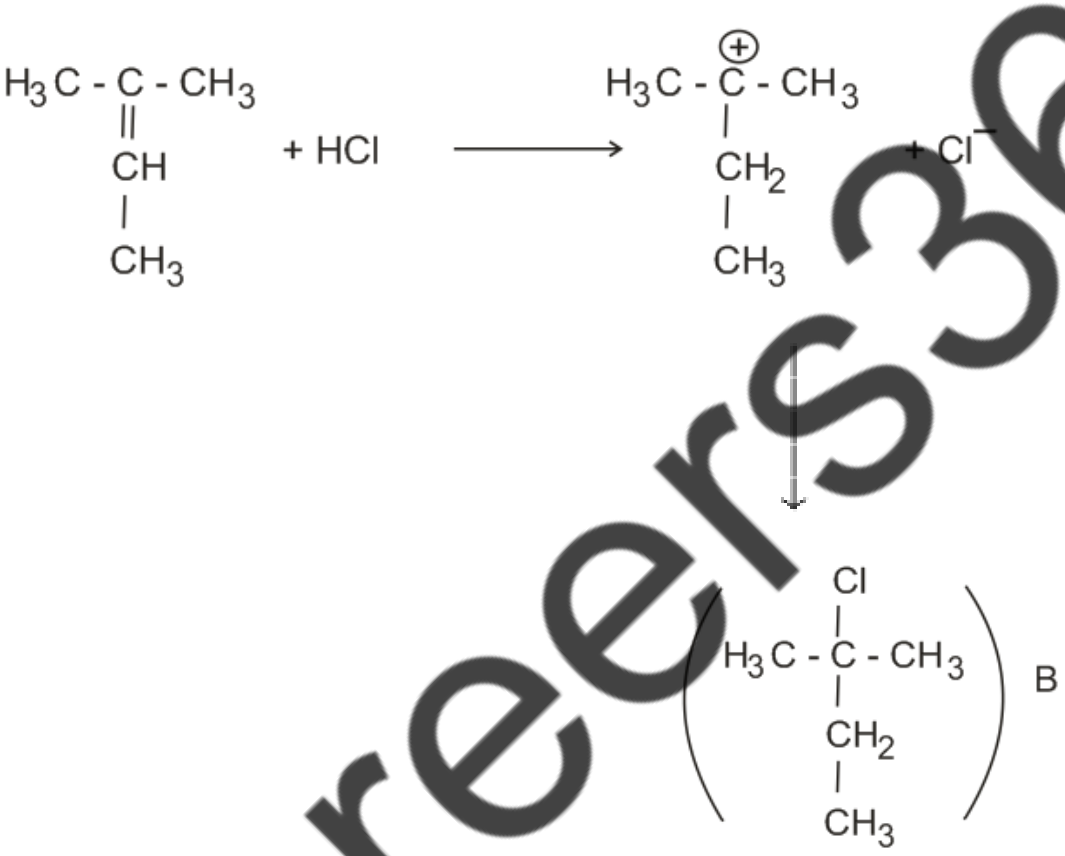


Now this reacts with HCl to form B as follows:



Option 3 is correct.

Q. 38 For the chemical reaction,

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

the **correct** option is:

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Option 1:

$$-\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$$

Option 2:

$$-\frac{d[N_2]}{dt} = 2\frac{d[NH_3]}{dt}$$

Option 3:

$$-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

Option 4:

$$3\frac{d[H_2]}{dt} = 2\frac{d[NH_3]}{dt}$$

Correct Answer:

$$-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

Solution:

For a chemical reaction:

$$aA + bB \longrightarrow cC + dD$$

The rate is always given as:

$$-\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$$

$\therefore$ for the reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g),$$

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Thus,

$$-\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

Hence, the answer is the option (3).

Q. 39 Under the isothermal condition, a gas at 300K expands from 0.1L to 0.25L against a constant external pressure of 2 bar. The work done by the gas is:

[Given that 1L bar = 100 J]

Option 1:
-30 J

Option 2:
5 kJ

Option 3:
25 J

Option 4:
30 J

Correct Answer:
30 J

Solution:
Work (W) - Any type of energy transfer between the system & surroundings that is not due to temperature difference is known as work.

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It is considered an ordered form of energy.

Work Done for Irreversible Isothermal Expansion of an ideal gas -

$$W = -P_{ext}\,(V_f - V_i)$$

P_{ext} may be equal to P_f or may not be equal to P_f, but work done is always calculated by P_{ext}

We know that the work done is given as:

$$W = -P\Delta V$$

Now,

$$P_{at} = 2bar$$

$$V_i = 0.1L$$

$$V_f = 0.25L$$

$$\therefore W = -2bar(0.25 - 0.1)$$
$$\Rightarrow W = -2 \times 0.15$$
$$\Rightarrow W = -0.30Lbar$$
$$\Rightarrow W = -30J$$

Work done by the gas = 30J

Hence, the answer is the option (4).

Q. 40 Among the following, the reaction that proceeds through an electrophilic substitution is:

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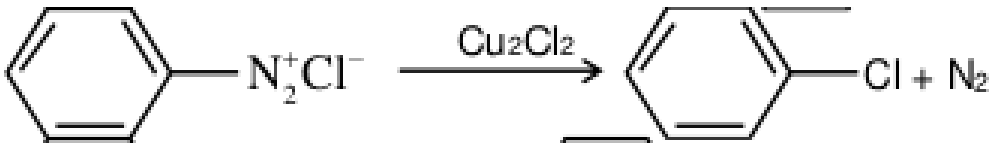
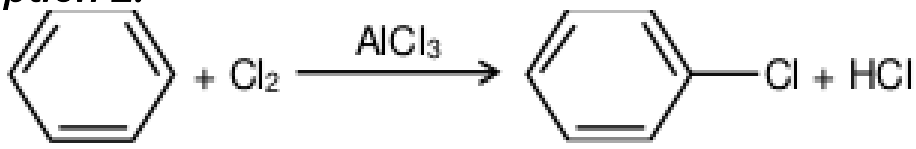
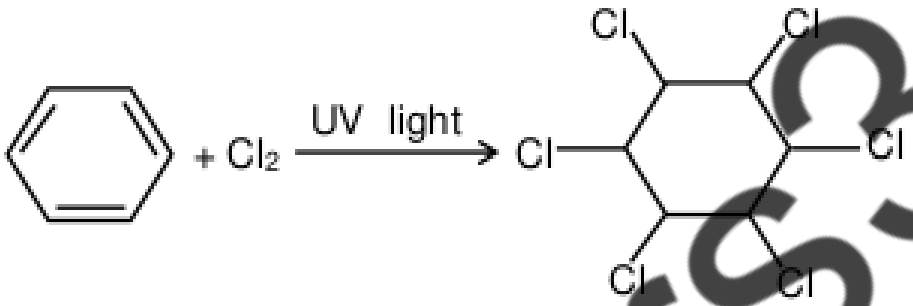
Option 1:



Option 2:

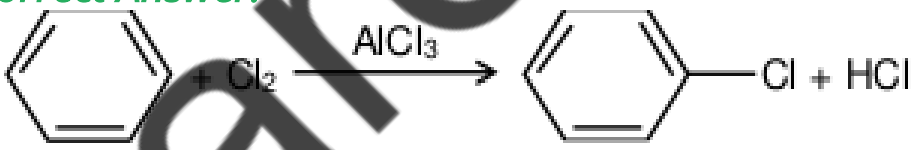


Option 3:



Option 4:

Correct Answer:



Solution:

The electrophilic substitution reaction is:

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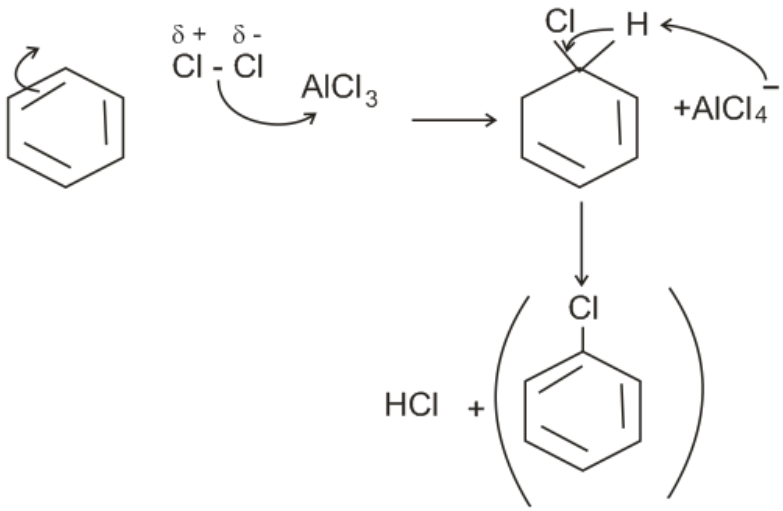

Option 2 is correct

Q. 41 Which of the following diatomic molecular species has only π-bonds according to Molecular Orbital Theory?

Option 1:
O_2

Option 2:
N_2

Option 3:
C_2

Option 4:
Be_2

Correct Answer:
C_2

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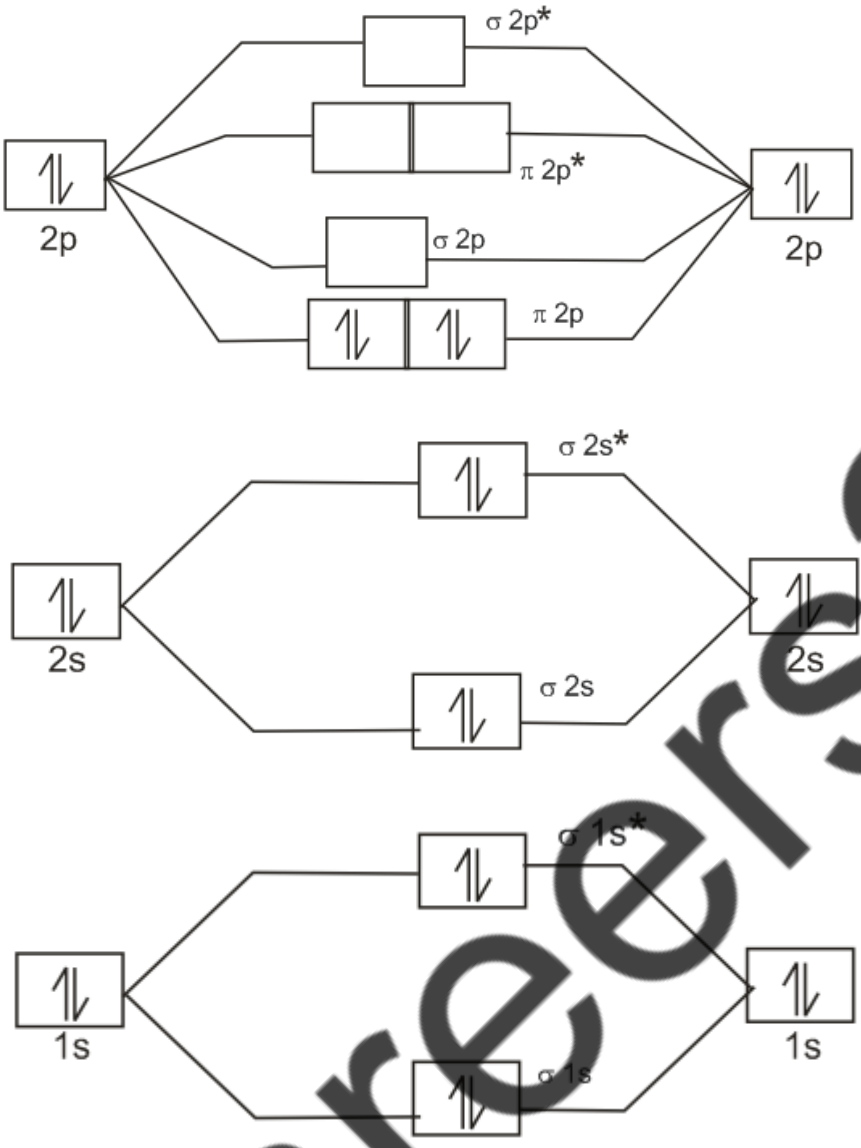
Solution:

Molecular orbital diagram for C_2:



$$C_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$O_2 - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1$$

$$N_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^2$$

$$Be_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2$$

$\therefore$ both are $\pi 2p_x^2$ & $\pi 2p_y^2$ bonds.

Hence **Option 3 is correct.**

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Q. 42 Which of the following species is **not** stable?

Option 1:
$[SiF_6]^{2-}$

Option 2:
$[GeCl_6]^{2-}$

Option 3:
$[Sn(OH)_6]^{2-}$

Option 4:
$[SiCl_6]^{2-}$

Correct Answer:
$[SiCl_6]^{2-}$

Solution:
In given molecules, $[SiCl_6]^{2-}$ is not stable. This is because six large chlorine atoms surround a relatively smaller silicon atom, due to a larger steric hindrance, the molecule gets less stable. In $[SiF_6]^{2-}$, the size of F^- is less hence, lesser repulsions and the molecule is stable. in case of $[GeCl_6]^{2-}\ and\ [Sn(OH)_6]^{2-}$ size of Germanium and Tin atoms is large enough to accommodate six anions.

Hence, the answer is the option (4).

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Q. 43 Which will make basic buffer?

Option 1:

$50mL\ of\ 0.1M NaOH + 25mL\ of\ 0.1M CH_3COOH$

Option 2:

$100mL\ of\ 0.1M CH_3COOH + 100mL\ of\ 0.1M NaOH$

Option 3:

$100mL\ of\ 0.1M HCl + 200mL\ of\ 0.1M NH_4OH$

Option 4:

$100mL\ of\ 0.1M HCl + 100mL\ of\ 0.1M NaOH$

Correct Answer:

$100mL\ of\ 0.1M HCl + 200mL\ of\ 0.1M NH_4OH$

Solution:

A basic Buffer is a solution that has salt & weak base.

$$\therefore \underset{50ml}{NaOH} + \underset{25ml}{CH_3COOH} \longrightarrow \underset{salt}{CH_3COONa} + H_2O + \underset{(strong\ base)}{NaOH}$$

$$\therefore \underset{100ml}{CH_3COOH} + \underset{100ml}{NaOH} \longrightarrow \underset{salt}{CH_3COONa} + H_2O$$

$$\therefore \underset{200ml}{NH_4OH} + \underset{100ml}{HCl} \longrightarrow \underset{salt}{NH_4Cl} + H_2O + \underset{(weak\ base)}{NH_4OH}$$

$$\therefore \underset{100ml}{HCl} + \underset{100ml}{NaOH} \longrightarrow \underset{salt}{NaCl} + H_2O$$

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Solution 1 is basic because of the presence of a strong base.
Solution 2 is slightly basic due to the presence of a salt a weak acid and a strong base.

Solution 3 is basic because of the presence of a weak base and its conjugate acid.

Solution 4 will be neutral as the only salt of strong acid and base is present.

Hence, the answer is the option (3).

Q. 44 Which of the following transitions in the spectrum of hydrogen atoms falls in the visible region?

Option 1:
Lyman Series

Option 2:
Balmer Series

Option 3:
Paschen Series

Option 4:
Brackett Series

Correct Answer:
Balmer Series

Solution:

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Balmer Series Spectrum lies in the visible region.

$$\frac{1}{\lambda} = \mathrm{RZ}^2 \left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2} \right)$$

Where $\mathrm{n}_1 = 2 \text{ and } \mathrm{n}_2 = 3, 4, 5, 6.....$

The visible spectrum of light from hydrogen displays four wavelengths, 410 nm, 434 nm, 486 nm & 656 nm, that correspond to emissions of photons by electrons in the excited state transitioning to quantum number 2.

Hence, the answer is the option (2).

Q. 45 The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is:

Option 1:
10

Option 2:
20

Option 3:
30

Option 4:
40

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Correct Answer:
30

Solution:
The reaction is :

$$N_2 + H_2 \longrightarrow NH_3$$

For 20 moles of ammonia, the balanced equation is:

$$10\ N_2 + 30\ H_2 \longrightarrow 20\ NH_3$$

So, 30 moles of H_2 are required.

Hence, the answer is the option (3).

Biology

Q. 1 Convesion of glucose to glucose - 6 - phosphate, the first irrevesible reaction of glycolysis, is catalysed by:

Option 1:
Aldolase

Option 2:
Hexokinase

Option 3:
Enolase

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Option 4:
Phosphofructokinase

Correct Answer:
Hexokinase

Solution:
Glycolysis or EMP pathuray -

Occurs in cytoplasm, common to both aerobic and anaerobic respiration. One molecule of glusose is reduced to two molecules of pyruvate.

$Glucose + 2NAD^+ + 2ADP + 2H_3PO_4 \rightarrow 2Pyruvate + 2NADH + 2H^+ + 2ATP$

Net gain of ATP = 2ATP + 6ATP (from 2NADH) = 8ATP

- wherein

Given by Embden, Mayerhof and Parnas

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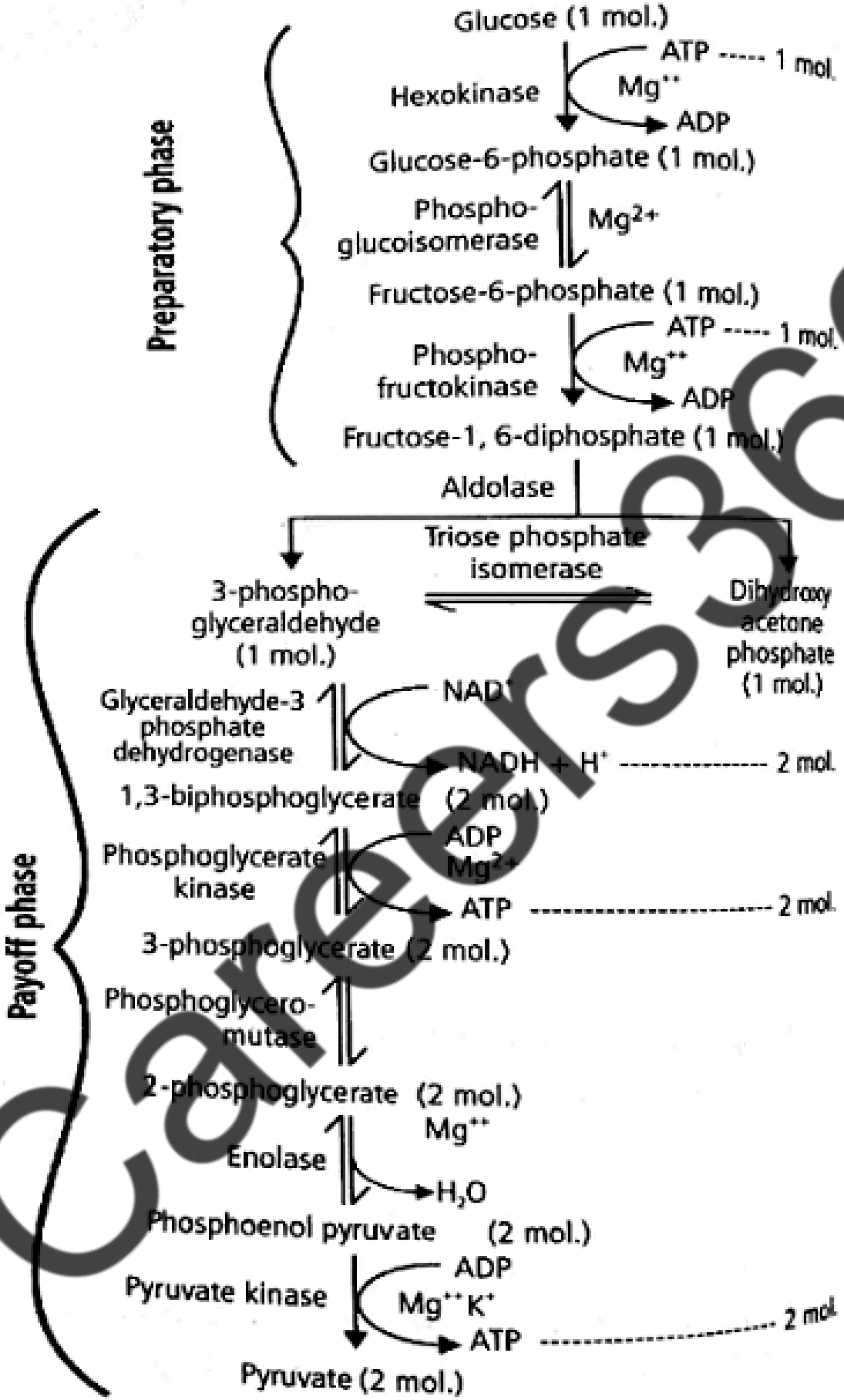
Preparatory phase
Payoff phase
Glucose (1 mol.)
ATP ----- 1 mol.
Hexokinase
Mg++
ADP
Glucose-6-phosphate (1 mol.)
Phospho-glucoisomerase
Mg2+
Fructose-6-phosphate (1 mol.)
ATP ----- 1 mol.
Phospho-fructokinase
Mg++
ADP
Fructose-1, 6-diphosphate (1 mol.)
Aldolase
Triose phosphate isomerase
3-phospho-glyceraldehyde (1 mol.)
Dihydroxy acetone phosphate (1 mol.)
Glyceraldehyde-3 phosphate dehydrogenase
NAD+
NADH + H+ ---------- 2 mol.
1,3-biphosphoglycerate (2 mol.)
Phosphoglycerate kinase
ADP
Mg2+
ATP ---------- 2 mol.
3-phosphoglycerate (2 mol.)
Phosphoglycero-mutase
2-phosphoglycerate (2 mol.)
Mg++
Enolase
H2O
Phosphoenol pyruvate (2 mol.)
Pyruvate kinase
ADP
Mg++ K+
ATP ---------- 2 mol.
Pyruvate (2 mol.)

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During glycolysis, glucose is phosphorlayted by a family of enzymes called hexokinase to form glucose - 6 - phosphate. This reaction required ATP and it is an irrevesible reaction. Hexokinase is a kind of transferase enzymes.

Q. 2 What is the site of perception of photoperiod necessary for induction of flowering in plants?

Option 1:
Pulvinus

Option 2:
Shoot apex

Option 3:
Leaves

Option 4:
Lateral buds

Correct Answer:
Leaves

Solution:

Photoperiodism -

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The response of plants to changes in relative lengths of day and night

-

The photoperiodic stimulus is preceived by the leaves. As a result of this a floral hormone is produced in leaves which is then transported to apical tip, subsequently causing the initation of floral primodia. The photoperiodic stimulus is preceived by the leaves was demonstrated by the experiments on *Xanthium pennsylvanicum*. Hence, the correct answer is leaves.

Q. 3 Which of the following is true for Golden rice?

Option 1:
It is a Vitamin A enriched, with a gene from daffodil.

Option 2:
It is pest resistant, with gene from *Bacillus thuringiensis*

Option 3:
It is drought tolerant, developed using *Agrobacterium* vector

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Option 4:

It has a yellow grains, because of a gene introduced from a primitive variety of rice.

Correct Answer:

It is a Vitamin A enriched, with a gene from daffodil.

Solution:

Golden Rice -

Vitamin A enriched rice was produced to increase nutrient quantity.

- wherein

Golden rice are not commercially available.

Golden rice was developed by IRRI to address vitamin A deficiency which is a serious public health problem affecting millions of children.

Golden rice was a genetically engineered to produce high level of a beta - carotene biosynthesis genes:

1. psy (phytoene synthase) from daffodil

2. crel (phytoene destauarse) from soil bacterium *Erwinia uredovera*

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The psy and crEl genus were transferred into rice genome under endosperm specific promoter, so that they are only expressed in endosperm.

Thus, golden rice is a vitamin A enriched, with a gene from daffodil.

Q. 4 Identify the correct pair representing the causative agent of typhoid fever and the confirmatory test for typhoid.

Option 1:
Plasmodium vivax / UTI test

Option 2:
Streptococcus pneumoniae / Widal test

Option 3:
Salmonella typhi / Anthrone test

Option 4:
Salmonella typhi / Widal test

Correct Answer:
Salmonella typhi / Widal test

Solution:

Widal test -

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Widal test is taken to confirm whether an individual is affected with typhoid.

- wherein

A female named Mary Mallon nicknamed Typhoid Mary, She was a cook by profession and was a typhoid carrier who continued to spread typhoid for several years through the food she prepared.

Typhoid is a systematic bacterial infection caused by *Salmonella typhi* usually through ingestion of contaminated food or water. Diagnosis of thyphoid can be done by widal test.

Plasmodium vivax causes malaria, while *Streptococcus pneumoniae* is responsible for causing Pneumonia.

UTI test is used for diagnose of urinary tract infectious while anthrone test is a method for quantitative analysis of carbohydrates.

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Q. 5 Colostrum, the yellowish fluid, secreted by mother during the initial days of lactation is a very essential to mpart immunity to the new born baby infants because it contains:

Option 1:
Natural killer cells

Option 2:
Monocytes

Option 3:
Macrophages

Option 4:
Immunoglobulin A

Correct Answer:
Immunoglobulin A

Solution:

Colostrum -

The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant.

- wherein

The foetus also receives some antibodies from their mother, through the placenta during pregnancy. These are some examples of passive immunity.

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Colostrum is the first milk secreted by at the time of parturition. It contains more lactalbumin and lactoprotein and is also rich in antibodies such as immunoglobin A that confers passive immunity to the new bron.

Colostrum mainly contains

1. Nutritional factors - energy, protein, vitamin, mineral. Crabohydrates fats

2. Immune factors - IgA, IgG, IgM, IgD, IgE, lactoferrin, cytokines, etc.

3. Growth factors.

Q. 6 It takes very long time for pineapple plants to flower. Which combination of hormones can be applied to artificially induce flowering in pineapple plants through out the year to increase the yields?

Option 1:
Auxin and Ethyelene

Option 2:
Gibberellin and Cytokinin

Option 3:
Gibberellin and Abscisic acid

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Option 4:
Cytokinnin and Abscisic acid

Correct Answer:
Auxin and Ethyelene

Solution:

Auxin and Ethylene can induce flowering in pineapple.

Other hormones like gibberellins are involved in seed germination, cytokinins in cell division, senescence etc. and abscisic acid in stress tolerance and stomatal movements.

Q. 7 DNA precipitation out of mixture of biomolecules can be achieved by treatment with:

Option 1:
Isopropanol

Option 2:
Chilled ethanol

Option 3:
Methanol at room temperature

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Option 4:
Chilled chloroform

Correct Answer:
Chilled ethanol

Solution:
DNA -

It is a polynucleotide, in which the sugar is found as deoxyribose. Hence, the name Deoxyribonucleic acid. (DNA)

-In a mixture of biomolecules, DNA can be precipitated by adding chilled ethanol, RNA by adding isopropanal and protein is not precipitated but is recovered from supernatent.

Q. 8 What triggers activation of protoxins to active Bt toxin of *Bacillus thuringinensis* in boll worm?

Option 1:
Body temperature

Option 2:
Moist surface of midnight

Option 3:
Alkaline pH of gut

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Option 4:
Acidic pH of stomach

Correct Answer:
Alkaline pH of gut

Solution:

Bacillus Thuringiensis -

Bacillus Thuringiensis is a Gram-positive bacteria; During sporulation this bacteria forms crystals of proteins or (cry protein), which are coded by cry genes (Cry 1 AC) Cry genes are used to make insecticide transgenic plants.

- wherein

When insects ingest crystal proteins , the alkaline pH of gut denature the insoluble protein into soluble. Soluble proteins makes gut porous and cause death of insect (Lepidopteran)

The toxins present in *Bacillus thuringinesis* remain as inactive protoxins which when enters gut of insect gets activated. The alkaline pH of insect gut activates the protoxin into active toxin.

The toxin then binds to specific receptors in the gut and causes the gut wall to breakdown. The insectidies as the bacteria proliferates.

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Q. 9 The frequency of recombination between gene pairs on the same chromosome as a measure of the distance between the genes was explained by:

Option 1:
T.H. Morgan

Option 2:
Gergor . J. Mendel

Option 3:
Alfered Sturtevant

Option 4:
Sutton Boveri

Correct Answer:
Alfered Sturtevant

Solution:

Recombination -

Morgan experimented and found that when the two genes in a hybrid cross situated on same chromosone, the proportion of parental gene combination were much higher than the non-parental type..

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Based on this observation, he coined the term recombination.

- wherein

To describe the generation of non-parental gene combinations

Morgan was the first one to hypothesize the frequency of recombination was related to the disatnce between the genes on a chromosome. However, Sturtevant was the first one to explain that the proportion of the crossover during crossing over could be used as an index of the distance between genes. Based on this, he also constructed the genetic map of fruit flies.

Q. 10 Which of the following is the most important cause for animals and plants being driven to the extinction?

Option 1:
Habitat loss and frangementation

Option 2:
Drought and floods

Option 3:
Economic exploitation

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Option 4:
Alien species invasion

Correct Answer:
Habitat loss and frangementation

Solution:

As we have already discussed in Cause of biodiversity losses -

The major causes of the biodiversity loss include habitat loss and fragmentation, alien species invasion, climate change, economic exploitation natural disasters like drought, floods, etc. Among these , the most important cause of extinction is habitat loss and fragmentation. It is caused by deforestation, overpopulation, pollution and global warming. Habitat loss distrups the complex interactions among the species and also kill or force out many plants. With the fragmentation of forest, the species occupying deeper parts are first to disappear.

Hence, the correct option is (a).

Q. 11 Identify the cells whose secretion protects the lining of gastro - intestinal tract from various enzymes.

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Option 1:
Cheif cells

Option 2:
Goblet cells

Option 3:
Oxyntic cells

Option 4:
Duodenal cells

Correct Answer:
Goblet cells

Solution:

Goblet Cells -

The intestinal mucosal epithelium has goblet cells which secrete mucus.

- wherein

The secretions of the brush border cells of the mucosa alongwith the secretions of the goblet cells constitute the intestinal juice or succus entericus

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Secretion of globet cells i.e. mucus protects the GI tract from various enzyme.

Q. 12 Match the following Column I and Column II

Column I	Column II
a. P - wave	i. Depolarization of ventricles
b. QRS complex	ii. Repolarization of ventricles
c. T - wave	iii. Coronary ischemia
d. Reduction in the size of T - wave	iv. Depolarization of atria

Option 1:
a-iii, b-i, c-ii, d-iv

Option 2:
a-ii, b-iv, c-i, d-iii

Option 3:
a-iii, b-iv, c-i, d-iii

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Option 4:
a-iv, b-i, c-ii, d- iii

Correct Answer:
a-iv, b-i, c-ii, d- iii

Solution:

ECG -

Electrocardiogram

- wherein

ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle.

P - wave - Depolarization of atria

QRS complex - Depolarization of ventricles

T - wave - Repolarization of ventricles

Reduction in the size of T - wave - Coronary Ischemia

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Q. 13 Match the following structures with their respective loaction in organs:

a. Crypts of Lieberkuhn	i. Pancreas
b. Glisson's Capsule	ii. Duodenum
c. Islets of Langerhans	iii. small intestine
d. Brunner's gland	iv. Liver

Option 1:
a-iii, b-i, c-ii, d-iv

Option 2:
a-ii, b-iv, c-i, d-iii

Option 3:
a-iii, b-iv, c-i, d-ii

Option 4:
a-iii, b-ii, c-i, d-iv

Correct Answer:
a-iii, b-iv, c-i, d-ii

Solution:
Regions in Small Intestine -

Structures	**Location**

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Crypts of Lieberkuhn	Small intestine
Glisson's capsule	Liver
Islets of Langerhans	Pancreas
Brunner's gland	Duodenum

Q. 14 Cells in G_0 phase:

Option 1:
exit the cell cycle

Option 2:
enter the cell cycle

Option 3:
suspend the cell cycle

Option 4:
terminate the cell cycle

Correct Answer:
exit the cell cycle

Solution:

Quiescent Stage (Go) -

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Few cells that do not divide exit G phase to enter an inactive stage called G_o

-

The G_0 phase is a period in the cell cycle in which cells exit the cell cycle and enters a quiscent state. In this stage cells neither divide nor prepare to divide.

Q. 15 The Earth Summit held in Rio de Janerio in 1992 was called:

Option 1:
to reduce CO_2 emissions and global warming

Option 2:
for conservation of biodiversity and sustainable utilization of its benefits.

Option 3:
to assess threat posed to native species by invasive weed species.

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Option 4:
for immediate steps to discontinue use of CFCs that were damaging the ozone layer.

Correct Answer:
for conservation of biodiversity and sustainable utilization of its benefits.

Solution:

As we have already discussed in The Earth Summit -

- The historic Convention on Biological Diversity ('The Earth Summit') held in Rio de Janeiro in 1992, called upon all nations to take appropriate measures for conservation of biodiversity and sustainable utilisation of its benefits.

Hence, the correct option is (b).

Q. 16 Which of the following glucose transporter is insulin - dependent?

Option 1:
GLUT - I

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Option 2:
GLUT II

Option 3:
GLUT III

Option 4:
GLUT IV

Correct Answer:
GLUT IV

Solution:
Insulin -

Insulin is a peptide hormone, which plays a major role in the regulation of glucose homeostasis

- wherein

Insulin acts mainly on hepatocytes and adipocytes (cells of adipose tissue), and enhances cellular glucose uptake and utilization

Among the options given, GLUT -I is insulin independent while GLUT -IV is insulin dependent. The fuctions and roles of GLUT - II and GLUT - III are not completely known GLUT - IV is responsible for the majority of transport of glucose into muscle and adipose cells.

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Q. 17 Which of the statement given below is not true about formation of Annual Rings in tree?

Option 1:
Annual rings is a combination of spring wood and autumn wood produced in a year.

Option 2:
Differential activity of cambium causes light and dark bands of tissue - early and late wood respectively

Option 3:
Activity of cambium depends upon variation in climate

Option 4:
Annual rings are not prominent in trees of temperate region.

Correct Answer:
Annual rings are not prominent in trees of temperate region.

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Solution:

Annual rings are most prominent in trees of temperate region.

Annual rings refer to an annual formation of wood in plants, consisting of two concentric layers, one of spring wood and others of autumn wood. These layers are formed due to differential activity of cambium caused by variation in climate. These rings are most prominent in trees of temperate regions as these have distinct growing periods and periods when there is no to little growth.

Q. 18 Match the following hormones with the respective disease:

a. Insulin	i. Addisons's
b. Thyroxin	ii.Diabetes insipidus
c. Corticoids	iii. Acromegaly
d. Growth hormone	iv. Goitre
	v. Diabetes mellitus

Select the correct option

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Option 1:
a - v, b-i, c - ii, d-iii

Option 2:
a-ii, b-iv, c-iii, d-i

Option 3:
a-v, b-iv, c-i, d-iii

Option 4:
a-ii, b-iv, c-i, d-iii

Correct Answer:
a-v, b-iv, c-i, d-iii

Solution:

Role of Thyroid hormones -

Thyroid hormones play an important role in the regulation of the basal metabolic rate.

- wherein

These hormones also support the process of red blood cell formation.

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- Insulin deficiency caused diabetes mellius
- Over production of thyroxin causes goitre
- Deficiency of corticoids such as cortisol causes Addisions disease
- Hyper secretion of growth hormone causes acromegaly.

Q. 19 In some plants, the female gamete develops into embryo without fertilization. This phenomenon is known as:

Option 1:
Autogamy

Option 2:
Parthenocarpy

Option 3:
Syngamy

Option 4:
Parthenogenesis

Correct Answer:
Parthenogenesis

Solution:
Parthenogenesis -

In a few organisms, the female gamete undergoes development to form new organisms without fertilisation.

- wherein

Eg., rotifers, honeybees and even some lizards and birds (turkey).

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When female gamete does not undergo fertilization and gets developed into embryo the process is called parhenogenesis.It occurs in some plants and many animals.

Parthenocrapy is formation of fruits without fertilization.

Autogamy refers to self - pollination.

Syngamy refers to physical union of male and female gamete.

Q. 20 Which of the following ecological pyramid is generally inverted?

Option 1:
Pyramid of numbers in grassland

Option 2:
Pyramid of energy

Option 3:
Pyramid of biomass in forest

Option 4:
Pyramid of biomass in a sea

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Correct Answer:
Pyramid of biomass in a sea

Solution:

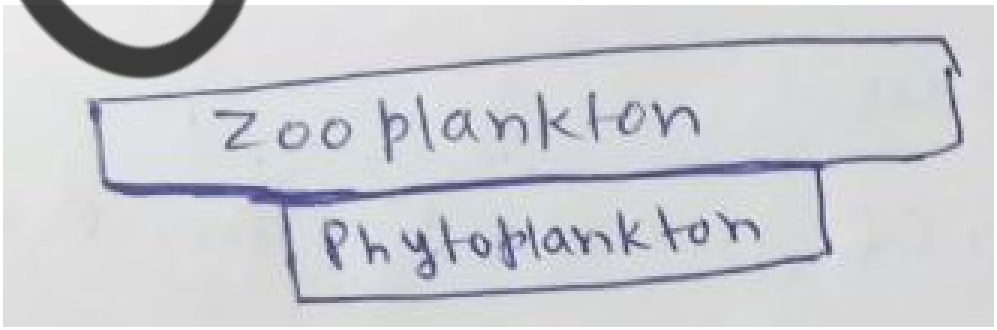
Ecological pyramids -

The number, biomass and energy of organisms gradually decrease from the producer level to the consumer level. This can be represented in the form of a pyramid called ecological pyramid.

- wherein

Ecological pyramid is the graphic representation of the number, biomass, and energy of the successive trophic levels of an ecosystem.

Pyramid of biomass in sea is generally inverted. This is because the producers are tiny phytoplanktons that have negligible biomass as compared to the organisms at the higher trophic levels.



Pyramid of biomass in sea

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Q. 21 Extrtusion of second polar body from egg nucleus occurs:

Option 1:
after entry of sperm but before fertilization

Option 2:
after fertilization

Option 3:
before entry of sperm into ovum

Option 4:
simultaneously with first cleavge

Correct Answer:
after entry of sperm but before fertilization

Solution:

Second Polar body -

This induces the completion of the meiotic division of the secondary oocyte. The second meiotic division is also unequal and results in the formation of a second polar body and a haploid ovum (ootid) .

-

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Extrusion of second polar body from egg nucleus occurs during the entry of sperm but before fertilization. It occurs as a consequence of oocycte activation which is triggered by fertilizing spermatozoa.

Q. 22 Pinus seed cannot germinate and establish without fungal associaton. This is because:

Option 1:
its embryo is immature

Option 2:
it has obligate association with mycorrhizae

Option 3:
it has a very hard seed coat

Option 4:
its seed contain inhibitors that prevents germinatiom.

Correct Answer:
it has obligate association with mycorrhizae

Solution:

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Mycorrhiza -

Mycorrhiza is a symbiotic association of a fungus with a root system. The fungi is extremely helpful for water and mineral absorption for the plants. Pinus seed have obligate association with mycorrhizae. They germinate only in the presence of fungus.

-

Mycorrhiza is a symbiotic association of a fungus with a root system. The fungi is extremely helpful for water and mineral absorption for the plants. Pinus seed have obligate association with mycorrhizae. They germinate only in the presence of fungus.

Q. 23 Which of the following factors is responsible for the formation of concentrated urine?

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Option 1:
Low levels of antidiuretic hormone

Option 2:
Maintaining hypersmolarity towards inner medullary interstitium in the kidneys.

Option 3:
Secretion of erythroprotein by Juxtaglomerular complex.

Option 4:
Hydroststic pressure during glomerular filtration.

Correct Answer:
Maintaining hypersmolarity towards inner medullary interstitium in the kidneys.

Solution:

Processes involved in Urine formation -

Reabsorption

- wherein

A comparison of the volume of the filtrate formed per day (180 litres per day) with that of the urine released (1.5 litres), suggest that nearly 99 per cent of the filtrate has to be reabsorbed by the renal tubules. This process is called reabsorption.

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The factors responsible for forming a concentrated urine are:

1. High level of ADH which increases the permeability of distal tubules and collecting ducts to the water, thereby allowing these tubular segements to reabsorbs water.

2. Hypersomolarity toward inner medullary interstitium that provides osmotic gradient necessary for the water reabsorption.

Thus, option 2 is correct.

Q. 24 In *Artirrhinum* (Snapdragon), a red flower was crossed with a white flower and in F_1 generation, pink flowers were obtained. When pink flower were selfed, the F_2 generation showed white, red and pink flowers. Choose the incorrect statement from the following:

Option 1:
This experiment does not follow the Principle of Dominace

Option 2:
Pink colour in F_1generation is due to incomplete dominace.

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Option 3:

Ratio of F_2 is $\frac{1}{4}$(red):$\frac{2}{4}$(Pink) : $\frac{1}{4}$(white)

Option 4:

Law of Segregation does not apply in this experiment.

Correct Answer:

Law of Segregation does not apply in this experiment.

Solution:

Incomplete dominance -

This can be understood by the inheritance of flower color in (dog flower) - Snapdragon.

- wherein

The F_1 had a phenotype that aid not resemble either of the two parents and was in between them.

In heritance of pink flower colour in Antirrhium is a example of incomplete dominance. This is a variation of dominance , wherien the dominant - recessive relationship is not a completely followed in a hetorozygote neither red nor white colour is developed. Instead

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F_1 represent pink coloured flowers. In F_2 generation , the red, pink and white colour flowers are formed in the ratio of 1 : 2 : 1 respectively.

Law of segregation does apply in this experiment as each gamete contains an allele and allele segregation normally takes place. Law of Dominance is not applied.

Q. 25 Which part of the brain is responsible for thermoregulation?

Option 1:
Cerebrum

Option 2:
Hypothalamus

Option 3:
Corpus callosum

Option 4:
Medulla oblongata

Correct Answer:
Hypothalamus

Solution:

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Hypothalamus -

Another very important part of the brain called hypothalamus lies at the base of the thalamus.

- wherein

The hypothalamus contains a number of centres which control body temperature, urge for eating and drinking. It also contains several groups of neurosecretory cells, which secrete hormones called hypothalamic hormones.

Hypothalamus is the part of the brain involved in the thermoregulation. Cerebrum is involved in interpreting touch, vision, hearing, speech , reasoning movement, etc.

Corpus callosum joins two hemispheres of cerebrum.

Medulla oblongata helps regulate breathing, heart funvtion, digestion, etc.

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Q. 26 Which of the following sexually transmitted diseases is not completely curable?

Option 1:
Gonorrhoea

Option 2:
Genital warts

Option 3:
Genital herpes

Option 4:
Chlamydisasis

Correct Answer:
Genital herpes

Solution:

Herpes -

Herpes is an infection caused by HSV (herpes simplex virus). This virus affects the external genitalia, anal region, mucosal surfaces, and skin in other parts of the body. There is no cure available for it.

-

Herpes is a type of sexually transmitted disease and if is extremely contagious and the bad thing is it stay with you for life. Genital herpes is usually caused by HSV-2.

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Q. 27 Respiratory Quotient(RQ) value of Tripalmitin is:

Option 1:
0.9

Option 2:
0.7

Option 3:
0.07

Option 4:
0.09

Correct Answer:
0.7

Solution:

Respiratory quotient (RQ) -

Ratio of volume of CO_2 evolved to the volume of oxygen consumed per unit time per unit weight at standard temperature and pressure.

- wherein

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$$RQ = \frac{CO_2\ evolved}{O_2\ consumed}$$

To identify the RQ value, firstly the ratio of the volume of carbon dioxide given off in respiration to that of the O_2 consumed that has a value near 1 when the organism is burning chiefly carbohydrates, near 0.7 when chiefly fats, and near 0.8 when chiefly proteins but sometimes exceeding 1 when crabohydrates are being changed to fats.

Q. 28 Select the correct group of biocontrol agents:

Option 1:
Bacillus thuringiensis , Tobacco mosiac virus , Aphids

Option 2:
Trichoderma, Baculovirus, Bacillus thuringiensis

Option 3:
Oscillatoria , Rhizobium , Trichoderma

Option 4:
Nostoc , Azospririllum, Nucleopolyhedrovirus

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Correct Answer:
Trichoderma, Baculovirus, Bacillus thuringiensis

Solution:

As discussed in Microbes as Biocontrol Agents -

Correct answer is *Tricoderma, Bacillus thuringiensis, Baculatarius*

- *Trichoderma* - It is a potent biocontrol agent and it is used extensively for soil born diseases.
- *Baculavirus* - It has narrow based insecticididal properties as they kills the insects and arthropoda
- *Bacillus thurengenisis* - It is a natural spore forming bacterium that occurs in soil, and it produce the protein crystal that is toxic for insects.

Hence, the correct option is (b).

Q. 29 Which one of the following statements regulating post - fertilization development in flowering plants is incorrect?

Option 1:
Ovary develpos into the fruit

Option 2:
Zygote develops into embryo

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Option 3:
Central cell develops into endosperm

Option 4:
Ovules develop innto embryo sac

Correct Answer:
Ovules develop innto embryo sac

Solution:

Post-fertilisation : structures and events -

Once the double fertilisation has taken place, events of endosperm and embryo development, maturation of ovule(s) into seed(s) and ovary into fruit, are collectively termed post-fertilisation events.

-

After fertilization, the ovary of the flowers usually develps into the fruit while ovules develop into seeds.

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Q. 30 Concanavalin A is:

Option 1:
an alkaloid

Option 2:
an essential oil

Option 3:
a lectin

Option 4:
a pigment

Correct Answer:
a lectin

Solution:
Extracellular coats - mucoprotein or glycoprotein -

A glycocalyx or mucous coat of protein and mucopolysaccharide, cover epithelial lining of gastrointestinal tract.

Concanavalin A is a lectin, i.e, carbohydrates biding protein which is extracted from Jack - bean. It binds specifically to sugars , glycoproteins and glycolipids.

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Q. 31 Match the following organisms with the products they produce:

a. *Lactobacillus*	*i.* Cheese
b. Saccharomyces cerevisiae	ii. Curd
c. Apergillus niger	iii Citric acid
d. Acetobacter aceti	iv. Bread
	v. Acetic acid

Option 1:
a-ii, b-iv, c- v, d-iii

Option 2:
a-ii, b-iv, c-iii, d-v

Option 3:
a-iii, b-iv, c-v, d-i

Option 4:
a-ii, b-i, c-iii, d-v

Correct Answer:
a-ii, b-iv, c-iii, d-v

Solution:

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Microbes in household products – lactic acid bacteria (LAB) -

Micro-organisms such as Lactobacillus and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd.

- wherein

During growth, the LAB produce acids that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B12.

Microorganisms	*Industrial product obtained*
Lactobacillus	Curd
saccharomyces cerevisiae	Bread
Aspergillus niger	Citric acid
Acetobacter aceti	Acetic acid

Hence, the correct option is (b).

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Q. 32 _____ statements:

A. Coenzymes or metal ion that is tightly bound to enzymes protein is called prosthetic group.

B. A complete catalytic active enzymes with its bound prosthetic group is called apoenzyme.

Option 1:
Both A and B are true

Option 2:
A is true but B is false

Option 3:
Both A and B are false

Option 4:
A is false but B is true

Correct Answer:
A is true but B is false

Solution:

-

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Prosthetic group which can be a coenzymes or metal ion and is attached to enzymes protein. On other hand apoenzymes is an inactive enzymes is an inactive enzymes whose actival is dependent on binding of an inorganic cofactor.

Thus statement A is correct and B is incorrect.

Q. 33 Correct sequence of phases in cell cycle is

Option 1:
$M \rightarrow G_1 \rightarrow G_2 \rightarrow S$

Option 2:
$G_1 \rightarrow G_2 \rightarrow S \rightarrow M$

Option 3:
$S \rightarrow G_1 \rightarrow G_2 \rightarrow M$

Option 4:
$G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

Correct Answer:
$G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

Solution:

Interphase -

Interphase represents the phase between two successive M-phases.

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- wherein

The interphase lasts more than 95% of the duration of cell cycle, also called as resting phase.

The cell cycle is divided into two phases interphase and M- phase. During interphase cell grows and make a copy of its DNA. While during M phase , cell seprates its DNA into two sets and divides its cytoplasm forming two new cells. Interphase is divided into three subphases i.e. G_1 phase, S phase and G_2 phase.

Thus, the sequence of the phase of cell cycle is $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

Q. 34 *Thiobacillus* is a group of bacteria helpful in carrying out:

Option 1:
Nitrogen fixation

Option 2:
Chemoautotrophic fixation

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Option 3:
Nitrification

Option 4:
Denitrification

Correct Answer:
Denitrification

Solution:

Denitrification -

Nitrates are reduced to gaseous compounds of nitrogen by Pseudomonas denitrificans, Thiobacillus denitrificans, Micrococcis denitrificans

- wherein

$$NO_3 \rightarrow NO_2 \rightarrow NO \rightarrow N_2O \rightarrow N_2$$

Thiobacillus denitrificans is a bacteria invovled in denitrification. It's actions result in convesion of nitrates in soil to free atmospheric nitrogen.

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Q. 35 Select the incorrect statement:

Option 1:
Inbreeding increases homozygosity

Option 2:
Inbreeding is essential to evolve purelines in any animal

Option 3:
Inbreeding selects harmful recessive genes tat reduce fertility and productivity.

Option 4:
Inbreeding helps in accumulation of superior genes and elimination of undesirable genes.

Correct Answer:
Inbreeding helps in accumulation of superior genes and elimination of undesirable genes.

Solution:

Disadvantages of Inbreeding -

Inbreeding depression

- wherein

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Continued inbreeding, especially close inbreeding, usually reduces fertility and even productivity. This is called inbreeding depression.

Disadvantages of Inbreeding -

Inbreeding depression

- wherein

Continued inbreeding, especially close inbreeding, usually reduces fertility and even productivity. This is called inbreeding depression.

Inbreeding is the process of mating of more closely related individuals within the same breed for 4-6 generations. Inbreeding increases homozygosity and is essential to evolve a pureline in any animal. It exposes harmful recessive genes that can be eliminated by the selection. Inbreeding also hep in accumilation of superior genes and elimination only to a certain level. After a few generation it starts reducing the fertilty.

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Q. 36 What map unit (Centi morgan) is adopted in the construction of genetic maps?

Option 1:
A unit of distance between two expressed genes, representing 10% cross over.

Option 2:
A unit of distance between two expressed genes, representing 100% cross over.

Option 3:
A unit of distance between gene on chromosomes, representing 1% cross over

Option 4:
A unit of distance between genes on chromosomes, representing 50% cross over.

Correct Answer:
A unit of distance between gene on chromosomes, representing 1% cross over

Solution:

Genetic Maps -

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Genetic maps are widely used as a starting point in the sequence of whole genomes.

- wherein

In a genetic map, the frequency of recombination between genes pairs on the same chromosomes is a measure of the distance between genes, and 'mapped' their positions on the chromosomes.

A map unit refers to a unit of distance between genes on chromsomes that represent 1% cross over

---

Q. 37 Which one of the following is not a method of in - situ conservation of biodivesity?

Option 1:
Biosphere reserve

Option 2:
Wildlife sanctuary

Option 3:
Botanical garden

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Option 4:
Sacred grove

Correct Answer:
Botanical garden

Solution:

As In-situ Conservation of Biodiversity - Sanctuaries and Biosphere Reserves -

The methods of biodivesity conservation are of two types in-situ conservation and ex-situ conservation. In-situ conservation incudes protected areas such as national parks, Sanctury, biosphere reserve , etc. Ex- situ conservation methods involve seed bank, botanical gardens, zoological parks, aquariums, etc.

Hence, the correct option is (c).

Q. 38

Placentation, in which ovules develop on the inner wall of the ovary or in the peripheral part is:

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Option 1:
Basal

Option 2:
Axile

Option 3:
Parietal

Option 4:
Free central

Correct Answer:
Parietal

Solution:

Placentation -

The arrangements of ovules of placenta within the ovary is called placentation.

-

The arranfgemment of ovules within, the ovary is known as placentaion. The placentations are of different types - marginal, axile, parietal, basal, central and free - central. The ovules develpo in the inner wall of the ovary or on peripheral part in parietal

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placentation. In this type, ovary is one chambered but becomes two chambered due to the presence of false septum called replum, e.g, mustard.

Q. 39 Due to increasing air - brone allergens and pollutnats, many people in urban areas are suffering from respiratory disorder causing wheezing due to:

Option 1:
benign growth on mucous lining of nasal cavity

Option 2:
inflammation of bronchi and bronchioles

Option 3:
proliferation of fibrous tissues and damage of the alveolar walls

Option 4:
reduction in the secretion of surfacants by pneumocyctes.

Correct Answer:
inflammation of bronchi and bronchioles

Solution:

Causes of air pollution -

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Industries: Combustion of fossil fuels like coal, petroleum, etc. Industrial smoke is the main source of pollution.

- wherein

Automobiles: The combustion of petrol and diesel in automobiles releases harmful gases into the air. They also produce dust.

Wheezing is a high - pitched, coarse whistling sound that can occur when a person is breathing. The main cause of wheezing is inflammation of bronchi and bronchioles which led by increasing air - brone allergens and pollutants.

Q. 40 Which of the following statements is correct?

Option 1:
Cornea is an external, transparent and protective proteinacious covering of the eye - ball

Option 2:
Cornea consists of dense connective tissue of elastin and can repair itself

Option 3:
Corena is convex, transparent layer which is higly vascularised.

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Option 4:

Corena consists of dense matrix of collegen and is the most senstive portion of eye.

Correct Answer:

Cornea is an external, transparent and protective proteinacious covering of the eye - ball

Solution:

Cornea -

The anteriorportion of this layer is called the cornea.

- wherein

The middle layer, choroid, contains many blood vessels and looks bluish in colour. The choroid layer is thin over the posterior two-thirds of the eye ball, but it becomes thick in the anterior part to form the ciliary body.

Cornea is the eye outermost layer. It is external and transparent cornea is protective in function and proteinaceous in nature. It contains no blood vessels.

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Q. 41 Purines found both in DNA and RNA are:

Option 1:
Adenine and thymine

Option 2:
Adenine and guanine

Option 3:
Guanine and cytosine

Option 4:
Cytosine and thymine

Correct Answer:
Adenine and guanine

Solution:

Nucleotide -

When a phosphaste group to $5'$- oH of a nucleoside through phosphoester linkage.

- wherein

A corresponding nucleotide (or deoxynucleotide depending upon of sugar present) is formed.

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Nucleic acids are composed of 3 components:

Nitrogenous bases,

5 carbon sugars, and

one or more phospahte groups.

Nitrogenous bases are of two kinds i.e, purines and pyrimidines.

Adenine and guanine are purines while cytosine, thymine and uracil are pyrimidrines. In both the DNA and RNA purine are same while among pyrimidines, thymine is present in DNA and uracil in RNA.

Q. 42 Expressed Sequence Tags (ESTs) refers to:

Option 1:
Genes expressed as RNA

Option 2:
Polypeptide expression

Option 3:
DNA polymorphism

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Option 4:
Novel DNA sequence

Correct Answer:
Genes expressed as RNA

Solution:

DNA fingerprinting -

DNA fingerprinting is a very quick way to compare the DNA sequence of any two individuals.

-

Expressed sequence tags are genes that are expressed as RNA. These are used to identifying all the genes of genome.

Q. 43 Phloem in gymnosperms lacks:

Option 1:
Albuminous cells and sieve cells

Option 2:
Sieve tube only

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Option 3:
Companion cells only

Option 4:
Both sieves tubes and companion cells

Correct Answer:
Both sieves tubes and companion cells

Solution:
The pholem of gymnosperm possess sieve cells. Seive tubes and companion cells are absent in the phloem of gymnosperm.

Q. 44 What is the genetic disorder in which an individual has an overall masculine development, gynaecomastia, and sterile?

Option 1:
Turner's Syndrome

Option 2:
Klinefelter's syndrome

Option 3:
Edward syndrome

Option 4:
Down's syndrome

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Correct Answer:
Klinefelter's syndrome

Solution:

Klinefelter's syndrome -

It is caused due to the presence of an additional copy of x-chromosome resulting into a karyotype of 47,XXY.

-

Klinfelter syndrome is caused by the presence of an extra X chromosomes in males, wherein the genetic composition is 44+XXY. The total no. of chromosmes is 47. The symptoms of this syndrome are sterility, development of breasts and masculine development.

Q. 45 Grass leaves curl inwards during very dry weather. Select the most appropriate reason for the following:

Option 1:
Closure of stomata

Option 2:
Flaccidity of bulliform cells

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Option 3:
Shrinkage of aire spaces in spongy mesophyll

Option 4:
Tyloses in vessels

Correct Answer:
Flaccidity of bulliform cells

Solution:

Opening and closing of stomata -

This is governed by the change in osmotic pressure or turgidity of guard cells.

- wherein

When guard cells are turgid, stomata open and when guard cells are flaccid, stomata close.

The rolling or curling of grass leaves is mainly done due to bulliform cells. These are large bubble - shaped cells distributed between epidermal cells of the leaf. During stress conditions (such as dry weather) they become flaccid and causes the leaves to curl or roll. They unroll the leaves by gaining turgidity once stress conditions are over.

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Q. 46 Consider the following features:

a. Organs system level of organization

b. Bilateral symmerty

c. True colemates with segmentation of body

Select the correct option of animal group which possess all the above characteristics.

Option 1:
Annelida, Arthropoda and Chordata

Option 2:
Annelida, Arthropoda and Mollusca

Option 3:
Arthropoda, Mollusca and Chordata

Option 4:
Annelida, Mollusca and Chordata

Correct Answer:
Annelida, Arthropoda and Chordata

Solution:

Metameric Segmentation -

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Body externally and internally divided into segments with a serial repetition of at least some organs.

- wherein

Eg. Annelia, Arthropoda and chordate.

Annelida, Arthropoda and Chordata are the phyla that show segmentation. They have organ sysyem level of organization, bilateral symmerty and true coelom. Molluscans also share these feratures but do not show segmentation.

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Q. 47 Under which of the following conditions will there be no change in the reading frame of the following mRNA?

5' AACAGCGGUGCUAUU 3'

Option 1:
Insertion of G at 5^{th} position

Option 2:
Deletion of G from 5^{th} position

Option 3:
Inseration of A and G at 4^{th} and 5^{th} position respectively

Option 4:
Deletion of GGU from 7^{th} ,8^{th} and 9^{th} positions

Correct Answer:
Deletion of GGU from 7^{th} ,8^{th} and 9^{th} positions

Solution:

Frameshift Mutation -

a mutation caused by the addition or deletion of a base pair or base pairs in the DNA of a gene resulting in the translation of the genetic code in an unnatural reading frame from the position of the mutation to the end of the gene.

-

Insertion or deletion of one or two bases changes the reading frame from the position of insertion or deletion. Hence, option 1,2 and 3 would cause change in reading frme. Deletion of three bases delets a codon and reading frame remains unaltered from that point onwards.

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Q. 48 Select the correct option:

Option 1:
8^{th} ,9^{th} and 10^{th} pairs of the ribs articulate directly with the sternum.

Option 2:
11^{th} and 12^{th} pairs of ribs are connected to the sternum with the help of hylaine cartilage.

Option 3:
Each rib is a flat thin bone and all the ribes areconnected dorsally to the thorace vertebrae and ventrally to the sternum

Option 4:
There are seven pairs of vetebramnal three pairs of vertebrochondral and two pairs of vertebral ribs.

Correct Answer:
There are seven pairs of vetebramnal three pairs of vertebrochondral and two pairs of vertebral ribs.

Solution:

Structure of Ribs -

Dorsally, the ribs are attached to the thoracic vertebrae and ventrally connected to the sternum with the help of hyaline cartilage.

-

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Q. 49 The shorter and longer arm of a sub-metacentric chromosomes are referred as:

Option 1:
s- arm and l-arm respectively

Option 2:
p - arm and q- arm respectively

Option 3:
q- arm and p - arm respectively

Option 4:
m - arm and n - arm respectively

Correct Answer:
p - arm and q- arm respectively

Solution:

Types of chromosomes -

Based on the position of centromere, the chromosomes can be classified into four types:

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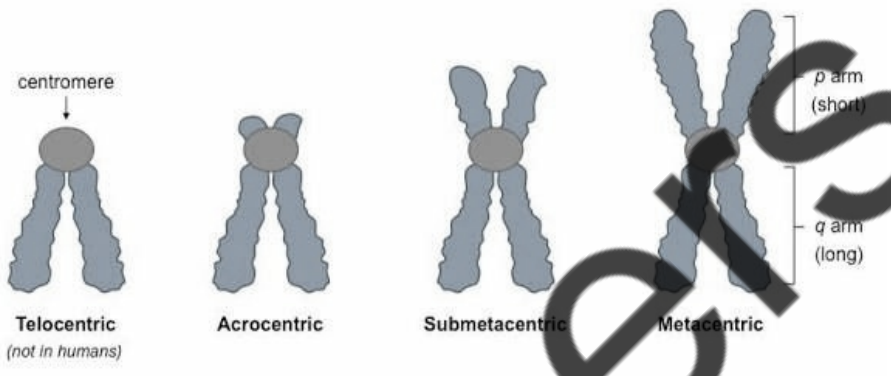
1) Metacentric chromosome- middle centromere forming two equal arms.

2) Sub-metacentric- centromere slightly away from the center.

3)Acrocentric- centromere is situated close to its end

4)Telocentric- terminal centromere

- wherein



In submetacentric chromosomes the centromere is palced at submedian. The arms become unequal in length. The short arm is called the p- arm and the long one is called q-arm.

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Q. 50 Xylem translocates:

Option 1:
water only

Option 2:
Water and mineral only

Option 3:
Water, mineral salts and some organic nitrogen only

Option 4:
Water, minerals salts and some organic nitrogen and hormones

Correct Answer:
Water, minerals salts and some organic nitrogen and hormones

Solution:

Ascent of sap -

The upward movement of water from the root to aerial parts of the plant body.

-

Xylem is the complex conducting tissue in the plants. It is responsible for the translocation of water, mineral salts and ions. Plants utilize nitrogen in the form of ammonia (NH_4^+) or nitrate

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(NO_3^-) ions. Those hormones that are transported with the transpiration stream have to be loaded into the xylem and unloaded at the target cells. Hence, the correct answer is option 4.

Q. 51 Persistent nucellus in the seed is known as:

Option 1:
Chalaza

Option 2:
Perisperm

Option 3:
Hilum

Option 4:
Tegmen

Correct Answer:
Perisperm

Solution:

Perisperm -

Nucellus is the central tissue of ovules that contains the embryo sac. In some seeds like black pepper, it presists after the double fertilization. The persistent nucellus is called perisperm. It

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performs nutritive function.

-

Nucellus is the central tissue of ovules that contains the embryo sac. In some seeds like black pepper, it presists after the double fertilization. The persistent nucellus is called perisperm. It performs nutritive function.

Q. 52 Match ColumnI with Column II

Column I	Column II
a. Saprophytes	i. Symbiotic association of fungi with plant roots
b. Parasite	ii. Decomposition of dead organic materials
c. Lichens	iii. Living on living plants or animals
d. Mycorrhiza	iv. Symbiotic association of algae and fungi

Choose the correct answer from the options given below

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Option 1:
a-i, b-ii , c-ii d-iv

Option 2:
a-iii, b-ii, c-i, d-iv

Option 3:
a-ii, b-i, c-iii, d-iv

Option 4:
a-ii, b-iii, c-iv. d-i

Correct Answer:
a-ii, b-iii, c-iv. d-i

Solution:

saprotrophs -

The detritus food chain (DFC) begins with dead organic matter. It is made up of decomposers which are heterotrophic organisms, mainly fungi and bacteria.

- wherein

They meet their energy and nutrient requirements by degrading dead organic matter or detritus. These are also known as saprotrophs (sapro: to decompose). Decomposers secrete digestive enzymes that breakdown dead and waste materials into simple, inorganic materials, which are subsequently absorbed by them.

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a.Saprophytes are the decomposers of dead organic material, e.g fungi.

b. Parasite derive their nutrition by living on the live host which can be plant or and animal, e.g. cuscuta, ticks, etc.

c. Lichens are the symbiotic associations of the algal partner called phycobiont and fungal parter called myconiont

d. Mycorrhiza is formed by the symbioyic association of fungi with the roots of higher plants. Fungi provides phosphorus to the rrots and rtrots provide nutritution to the rrots.

Q. 53 Which of the following can be used as a biocontrol agent in the treatment of plant disease?

Option 1:
Trichoderma

Option 2:
Chlorella

Option 3:
Anabaena

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Option 4:
Lactobacillus

Correct Answer:
Trichoderma

Solution:

Microbial Biocontrol agents -

fungus Trichoderma. developed for use in the treatment of plant disease

- wherein

Trichoderma species are free-living fungi that are very common in the root ecosystems. They are effective biocontrol agents of several plant pathogens.

Trichoderma is the genus of free living fungi that commonly occur in the root ecosystem. They are efficiently used as biocontrol agent aganist several plant phatogens. *Chorella* is a phathogenic bacterium. *Anabena* is nitrogen fixing bactria. *Lactobacillus* is used to obtain curd and yoghurt.

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Q. 54 What would be the heart rate of a person if the cardiac output is 5L, blood volume in the ventricls at the end of the diastole is 100mL and at the end of ventricular systole is 50mL?

Option 1:
50 beats per minute

Option 2:
75 beats per minute

Option 3:
100 beats per minute

Option 4:
125 beats per minute

Correct Answer:
100 beats per minute

Solution:

Cardiac Output -

The stroke volume multiplied by the heart rate (number of beats per min.) gives the cardiac output.

- wherein

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The cardiac output can be defined as the volume of blood pumped out by each ventricle per minute and averages 5000 mL or 5 litres in a healthy individual. The body has the ability to alter the stroke volume as well as the heart rate and thereby the cardiac output. For example, the cardiac output of an athlete will be much higher than that of an ordinary man.

Cardiac output is the product of heart rate (number of beats per minute) and te stroke volume.

Given

Cardiac output [CO] = 5L

Stroke volume [SV] = EDV - ESV = 100mL - 50mL = 50mL or 0.05L

Heart rate = $\frac{CO}{SV} = \frac{5}{0.05} = 100$ beats per minute

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Q. 55 Which of the follwoing protocols did aim for reducing emissions of chloroflurocarbons into the atmosphere?

Option 1:
Montreal protocol

Option 2:
Kyoto protocol

Option 3:
Gothenburg protocol

Option 4:
Geneva protocol

Correct Answer:
Montreal protocol

Solution:

Montreal Protocol -

Recognising the deleterious affects of ozone depletion, an international conference treaty, known as the Montreal Protocol, was signed at Montreal (Canada).

- wherein

In 1987 (effective in 1989) to control the emission of ozone depleting substances. Subsequently many more efforts have been made and protocols have laid down definite roadmaps, separately for developed and developing countries, for reducing the emission of CFCs and other ozone depleting chemicals.

- Monteral Protocol is an international treaty signed in 1987 to control the emission of ozone depleting substances, such as chloroflurocarbons (CFCs)
- Kyoto protocol commits the party states to reduce the emission of greenhouse gases.
- Gotherburg protocol was signed to reduce acidification , eutrophication and ground level ozone.

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- Geneva protocol prohibits the use of chemical and biological weapons as warfare methods.

Q. 56 Polyblend, a fine powder of recycled modified plastic, has proved to be a good material for:

Option 1:
making plastic sacks

Option 2:
use as a fertilizer

Option 3:
construction of roads

Option 4:
making tubes and pipes

Correct Answer:
construction of roads

Solution:

Polyblend -

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Polyblend is a mixture of polymers specifically recycled modified plastic. It is used in the construction of a roads as it produces shrink and crash resistant layers.

-

Polyblend is a mixture of polymers specifically recycled modified plastic. It is used in the construction of a roads as it produces shrink and crash resistant layers.

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Q. 57 Which of the following contracepative methods do involve a role of hormone?

Option 1:
Lactational amenorrhea, Pills, Emergency contraceptives

Option 2:
Barrier method, Lactional amenorrhea, Pills

Option 3:
CuT, Pills, Emergency contraceptive

Option 4:
Pills, Emergency contraceptives, Barrier method

Correct Answer:
Lactational amenorrhea, Pills, Emergency contraceptives

Solution:

Lactational amenorrhea (absence of menstruation) -

Is another natural method to avoid pregnancy.

- wherein

This method is based on the fact that ovulation and therefore the cycle do not occur during the period of intense lactation following parturition. Therefore, as long as the mother breast-feeds the child fully, chances of conception are almost nil.

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- Lactational amenorrhea is marked by the absence of menstrution in the absence of progestrone and estrogen
- pills are small doses of either progestrone or progestrone - estrogen combination
- Emergency contraceptives are also combination of synthetic hormones

Q. 58 Drug called 'Heroin' is synthesized by:

Option 1:
methylation of morphine

Option 2:
acetylation of morphine

Option 3:
glycosylation of morphine

Option 4:
nitration of morphine

Correct Answer:
acetylation of morphine

Solution:

Heroin -

Commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound.

- wherein

This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant Papaver somniferum

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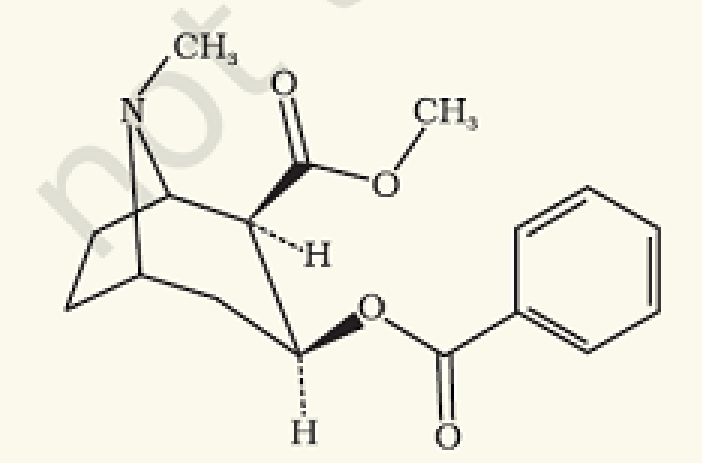
Generally taken by snorting and injection, heroin is a depressant and slows down body functions.



Herion is a chemically diacetylmorphine which is obtained by the acetylation of morphine extracted from latex of *Papaver somniferum*

Q. 59 Which of the following pairs of gases is mainly responsible for green house effect?

Option 1:
Ozone and Ammonia

Option 2:
Oxygen and Nitrogen

Option 3:
Nitrogen and Sulphur dioxide

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Option 4:
Carbon dioxide and Methane

Correct Answer:
Carbon dioxide and Methane

Solution:

As we have already discussed in Greenhouse Effects -

The term ‘Greenhouse effect’ has been derived from a phenomenon that occurs in a greenhouse. The greenhouse effect is a naturally occurring phenomenon that is responsible for heating of Earth’s surface and atmosphere.

- wherein

Green house effect is a natural phenomenon that heats up Earth's surface and atmosphere. Carbon dioxide and methane are common green house gases that have increased mainfold due to anthropogenic activities.

Hence, the correct option is (d).

Q. 60 Which of the following muscular disorder is inherited?

Option 1:
Tetany

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Option 2:
Muscular dystrophy

Option 3:
Myasthenia gravis

Option 4:
Botulism

Correct Answer:
Muscular dystrophy

Solution:

Myasthenia gravis -

Myasthenia gravis is a chronic autoimmune neuromuscular disease that causes weakness in the skeletal muscles, which are responsible for breathing and moving parts of the body, including the arms and legs. , myasthenia gravis is not inherited and occurs in people with no history of the disorder in their family.

-

Tetany is the condition of involuntary muscular contraction caused due to deficiency of calcium or infection of clostridium that interfers with neurons.

Muscular dystrophy is inherited as an X - linked recessive disorder.

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Myasthenia gravis is an autoimmune disorder

Botulism is food poisoning caused due to bacterial infecton.

Q. 61 Which of the following equipments is essentially required for growing microbes on a large scale, for industrial production of enzymes?

Option 1:
BOD incubator

Option 2:
sludge digestor

Option 3:
Industrial oven

Option 4:
Bioreactor

Correct Answer:
Bioreactor

Solution:

Bioreactors -

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Bioreactors are stirrer cylindrical tanks. The stirrer provides movement, even mixing and O_2 availability throughout the bioreaction.

- wherein

Bioreactors can be 100 to 1000 lt for large production.

In order to produce large quantities of the enzymes , bioreactors are used wherein large volumers of culture can be processed or provides optimal conditions for achieving the disered product.

Q. 62 The concept of " *Omnis cellula -e-cellula*" regarding cell division was first proposed by:

Option 1:
Rudolf Virchow

Option 2:
Theodore Schwann

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Option 3:
Schleiden

Option 4:
Aristotle

Correct Answer:
Rudolf Virchow

Solution:

Cell Theory -

Cell theory states that all living matter, from the simplest of unicellular organisms to every complex higher plants and animals, is composed of cells and each cell can act independently but functions as an inetgral part of complete organism

-

Rudolf Virchow proposed the concept of " *Omnis cellula -e-cellula*" which means that the new cells arise from the pre - existing cells. Thus, the first concept of the cell division was given by Rudolf Virchow.

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Q. 63 What is the fate of the male gametes discharged in the synergid?

Option 1:
One fuses with with the egg other(s) degentrate in the synergid

Option 2:
All fuse with the egg

Option 3:
One fuses with the egg, other(s) fuse with the synergid nucleus.

Option 4:
One fuses with with the egg and other fuses with central cell nuclei

Correct Answer:
One fuses with with the egg and other fuses with central cell nuclei

Solution:
Double Fertilisation -

After entering one of the synergids, the pollen tube releases the two male gametes into the cytoplasm of the synergid. One of the male gametes moves towards the egg cell and fuses with its nucleus thus completing the syngamy. This results in the formation of a diploid cell, the zygote.

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-Therefore, out of the two male gamete that are discharged in the synergid cell via pollen tube one of them fuses with the egg cell to form zygote (2n) and the other one fuses with polar nuclear (central cell nucleus) to form the primary endosperm nucleus (3n).

Hence, the correct answer is option d.

Q. 64 How does steroid hormone influence the cellular activities?

Option 1:
Changing the permeability of the cell membrane

Option 2:
Binding to DNA and forming a gene - hormone complex

Option 3:
Activating cyclic AMP located on the cell membrane.

Option 4:
Using aquaporin channels as secondary messenger

Correct Answer:
Binding to DNA and forming a gene - hormone complex

Solution:

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Steroid hormones, being lipid-soluble, can diffuse across the cell membrane and bind to specific receptors located within the cytoplasm or nucleus. This hormone-receptor complex then interacts with DNA, leading to the regulation of gene expression. The complex can act as a transcription factor, influencing the transcription of specific genes and ultimately affecting cellular activities. Hence, the correct answer is option 2.

Q. 65 Which of the following pair of organelles does not contain DNA?

Option 1:
Mitochondria and Lysosomes

Option 2:
Chloroplast and Vacuole

Option 3:
Lysosomes and Vacuoles

Option 4:
Nuclear envelope and Mitochondria

Correct Answer:
Lysosomes and Vacuoles

Solution:

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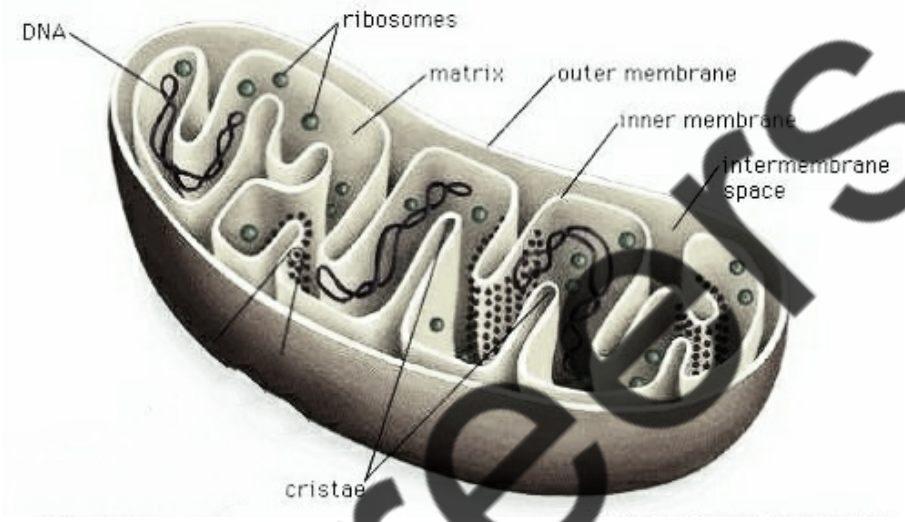
Mitochondria -

Each mitochondria is a double membrane bound structure with the outer membrane and the inner membrane dividing its lumen distinctly into two aqueous compartments.

- wherein

It is the site of aerobic respiration. It produces cellular energy in the form of ATP, hence called as 'power house' of the cell.



Nucleus, Mitochondria and Plastids contain DNA. Mitochondria and plastids are called semi autonomus because they can form some of the proteins using their own DNA . Vacuoles and lysosomes do not conatin DNA.

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Q. 66 A gene locus has two alleles A,a. If the frequency of dominant allele A is 0.4 ,then what will be the frequency of homozygous dominant, heterozygous and homozygous recessive individuals in the population ?

Option 1:
$0.36(AA); 0.48(Aa); 0.16(aa)$

Option 2:
$0.16(AA); 0.24(Aa); 0.36(aa)$

Option 3:
$0.16(AA); 0.48(Aa); 0.36(aa)$

Option 4:
$0.16(AA); 0.36(Aa); 0.48(aa)$

Correct Answer:
$0.16(AA); 0.48(Aa); 0.36(aa)$

Solution:

The Hardy-Weinberg formulas -

This formula allows detecting frequencies that change from generation thus allowing a simplified method of determining that evolution is occurring.

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- wherein

In a population at equilibrium for a locus of two alleles D and d having frequencies p and q respectively, the genotype frequencies are:

DD=p^2,

Dd= 2 pq,

dd= q^2

and p+q =1

p^2+2 pq +q^2 = 1

Given : $\rho = 0.4$

So, $q = 0.6$

Therefore,

- frequency of homozygous dominant $(AA) = p^2 = 0.16$
- frequency of heterozygous$(Aa) = 2pq = 0.48$
- frequency of homozygous recessive$(aa) = q^2 = 0.36$

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Q. 67 Match the following organisms with their respective characteristics:

(a) Pila	(i) Flame cells
(b) Bombyx	(ii) Comb plates
(c) Pleurobrachia	(iii) Radula
(d) Taenia	(iv) Malpighian tubules

Select the **correct** option from the following :

Option 1:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (ii) \quad (i) \quad (iv)$

Option 2:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (iv) \quad (ii) \quad (i)$

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Option 3:

$(a) \quad (b) \quad (c) \quad (d)$
$(ii) \quad (iv) \quad (iii) \quad (i)$

Option 4:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (ii) \quad (iv) \quad (i)$

Correct Answer:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (iv) \quad (ii) \quad (i)$

Solution:

Excretion in Platyhelminthes -

It is through single or paired photonephridium with flame cells.

-

Pila has radula (a rasping organ)

Bombyx is an arthropoda,it has malpighian tubules.

Pleurobranchia is a ctenophore, it has comb plates.

Taenia is a platyhelminth, it has flame cells.

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Q. 68 Which of the following is a commercial blood cholesterol lowering agent ?

Option 1:
Cyclosporin A

Option 2:
Statin

Option 3:
Streptokinase

Option 4:
Lipases

Correct Answer:
Statin

Solution:
Statins -

Statins produced by the yeast Monascus purpureus

- wherein

used as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Statins are produced by the yeast *Monoscus purpureus*. They are commercially used as blood cholesterol lowering agents.

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Q. 69 Variations caused by mutation, as proposed by Hugo de Vries, are:

Option 1:
random and directional

Option 2:
random and directionless

Option 3:
small and directional

Option 4:
small and directionless

Correct Answer:
random and directionless

Solution:

Hugo De Vries Mutation Theory -

This theory states that new species arise from the pre-existing one in single generation situation by sudden appearance of marked genetic variation called mutation

- wherein

According to De Vries, evolution in a discontinous and jerky process, rather than a continious and gradual one

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According to Hugo De Vries, mutuations are discontinuous, random and non-directional. These mutuations are responsible for the accumulation of variations that act as the raw material for evolution.

Q. 70 Select the **incorrect** statement:

Option 1:
Male fruit fly is heterogametic.

Option 2:
In male grasshopers,50% of sperms have no sex-chromosome.

Option 3:
In domesticated fowls, sex of progeny depends on the type of sperm rather than egg.

Option 4:
Human males have one of their sex-chromosome much shorter than the other.

Correct Answer:
In domesticated fowls, sex of progeny depends on the type of sperm rather than egg.

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Solution:

Basis of Sex Determination -

The cytological observations made in a number of insects lead to the development of the concept of genetic/chromosomal basis of sex.

- wherein

Experiments were performed by Henking (1891).

In fowls the sex is determined by the ovum and not sperm because in birds, the females are heterogametic (zw) and males are hemogametic (zz)

Q. 71 Which of the following immune responses is responsible for rejection of kidney graft ?

Option 1:
Auto-immune response

Option 2:
Humoral immune response

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Option 3:
Inflammatory immune response

Option 4:
Cell-mediated immune response

Correct Answer:
Cell-mediated immune response

Solution:

Cell-mediated immunity (CMI) -

The second type of immunity is called cell-mediated immune response or cell-mediated immunity (CMI).

- wherein

The T-lymphocytes mediate CMI

Rejection of organ transplants is performed by the T-cells based in the MHC (Major Histocompatibility Complex) Antigen.T-cells are involved in cell-mediated immune response.

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Q. 72 From evolutionary point of view, retention of thje female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in :

Option 1:
Liverworts

Option 2:
Mosses

Option 3:
Pteridophytes

Option 4:
Gymnosperms

Correct Answer:
Pteridophytes

Solution:

Reproduction of Pteridophyta -

Asexual by meiospores formed inside sporangia which is associated with sporophylls (sporophytes). Gamatophytes develop from spores and are haploid and independent and reproduce sexually. Zygote, thus produced from diploid sporophyte.

- wherein

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Two other ways of reproduction include: Apogamy and Apospory.

Rentention of the female gametophyte with developing young embryos on the parent sporophyte denotes the initiation of seed habit and it is observed in *Selaginella*, a heterosporous pteridophyte.

Q. 73 Select the correct sequence of organs in the alimentary canal of cockroach starting from mouth :

Option 1:
Pharynx ⟶ Oesophagus ⟶ Crop ⟶ Gizzard ⟶ Ileum ⟶ Colon ⟶ Rectum

Option 2:
Pharynx ⟶ Oesophagus ⟶ Gizzard ⟶ Crop ⟶ Ileum ⟶ Colon ⟶ Rectum

Option 3:
Pharynx ⟶ Oesophagus ⟶ Gizzard ⟶ Ileum ⟶ Crop ⟶ Colon ⟶ Rectum

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Option 4:

Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum

Correct Answer:

Pharynx → Oesophagus → Crop → Gizzard → Ileum → Colon → Rectum

Solution:

Alimentary canal of cockroach -

The alimentary canal of cockroach is divided into three parts, foregut, midgut, and hindgut.

- wherein

Foregut (stomodeum) is differentiated into five parts. Buccal chamber or mouth cavity, pharynx, oesophagus, crop and gizzard

The alimentary canal of cockroach has the following organs in order :-

Pharynx

Oesophogus

Gizzard

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Midgut

ileum

Colon

Rectum

Q. 74 Which of the following statements regarding mitochondria is **incorrect** ?

Option 1:
Outer membrane is permeable to monomers of carbohydrates, fats and proteins.

Option 2:
Enzymes of electron transport are embedded in outer membrane.

Option 3:
Inner membrane is convoluted with infoldings.

Option 4:
Mitochondrial matrix contains single circular DNA molecule and ribosomes.

Correct Answer:
Enzymes of electron transport are embedded in outer membrane.

Solution:

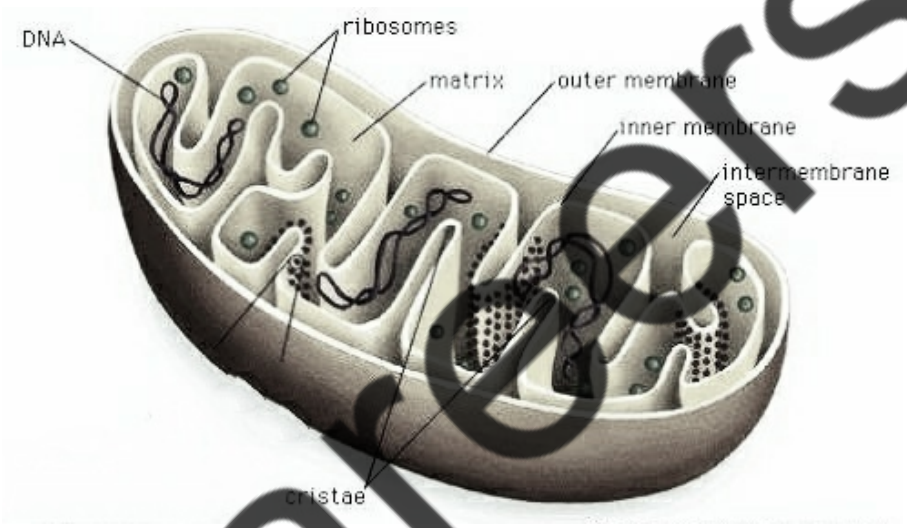
Careers360

Mitochondria -

Each mitochondria is a double membrane bound structure with the outer membrane and the inner membrane dividing its lumen distinctly into two aqueous compartments.

- wherein

It is the site of aerobic respiration. It produces cellular energy in the form of ATP, hence called as 'power house' of the cell.



Mitochondria are the ATP producing organelles. The electron transport chain is responsible for generation of ATPs through chemiosmosis.The enzymes for the electron transport chain are embedded in inner membrane.

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Q. 75 Which of the following statements is **not** correct ?

Option 1:
Lysosomes have numerous hydrolytic enzymes.

Option 2:
The hydrolytic enzymes of lysosomes are active under acidic pH
.

Option 3:
Lysosomes are membrane bound structures.

Option 4:
Lysosomes are formed by the process of packaging in the endoplasmic reticulum.

Correct Answer:
Lysosomes are formed by the process of packaging in the endoplasmic reticulum.

Solution:

Golgi Apparatus -

Principally performs the function of packaging materials, to be delivered either to intracellular targets or secreted outside of the cell.

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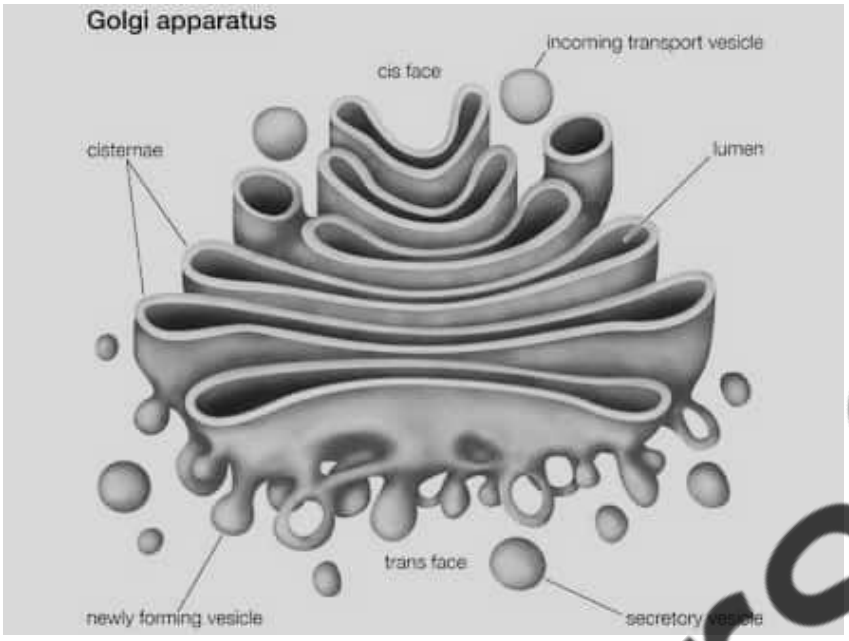
- wherein

It is the important site for the formation of glycoproteins and glycolipids.



Lysosomes are the membrane found hydrolytic enzymes containing vesicles that are given off through the golgi bodies.

Q. 76 Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology ?

Option 1:

Genetic code is not ambiguous

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Option 2:
Genetic code is redundant

Option 3:
Genetic code is nearly universal

Option 4:
Genetic code is specific

Correct Answer:
Genetic code is nearly universal

Solution:

Salient features of genetics code -

The code is nearly universe eg; from bactria it human, UUU would code for phenylalanine. Some exception are found in mitochondrial codons, and in some protozoans.

-

Genetic code is nearly universal. It means that the codons code for the same amino acid in all the organisms except for some minor deviations.This property of genetic code allows bacteria to produce human insulin through recombinant DNA technology.

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Q. 77 Tidal Volume and Expiratory Reserve Volume of an athlete is $500\ mL$ and $1000\ mL$ respectively, What will be his Expiratory Capacity if the Residual Volume is $1200\ mL$?

Option 1:
$1500\ mL$

Option 2:
$1700\ mL$

Option 3:
$2200\ mL$

Option 4:
$2700\ mL$

Correct Answer:
$1500\ mL$

Solution:

Expiratory Capacity (EC) -

Total volume of air a person can expire after a normal inspiration

- wherein

This includes tidal volume and expiratory reserve volume (TV+ERV)

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Expiratory Capacity $= TV + ERV$

$= 500mL + 1000mL$

$= 1500\ mL$

Q. 78 Match the following genes of the Lac operon with their respective products :

$(a)\ i\ gene$ $(i)\ \beta - galactoridase$

$(b)\ z\ gene$ $(ii)\ Permease$

$(c)\ a\ gene$ $(iii)\ Repressor$

$(d)\ y\ gene$ $(iv)\ Transacetylase$

Select the **correct** option.

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Option 1:

(a)	(b)	(c)	(d)
(i)	(iii)	(ii)	(iv)

Option 2:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (i) \quad (ii) \quad (iv)$

Option 3:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (i) \quad (iv) \quad (ii)$

Option 4:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (iv) \quad (i) \quad (ii)$

Correct Answer:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (i) \quad (iv) \quad (ii)$

Solution:

The lac operon -

In lac operon, a polycistronic structural gene is regulated by a common promoter and regulatory genes.

- wherein

Such arrangements are very common in bacteria and is referred to as operon.

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(a) $i\ gene$ codes for Repressor protein

(b) $z\ gene$ codes for β-galactosidase

(c) $a\ gene$ codes for Transacetylase

(d) $y\ gene$ codes for Permease

Q. 79 Use of an artificial kidney during haemodialysis may result in:

(a) Nitrogeneous waste build-up in the body

(b) Non-elimination of excess potassium ions

(c) Reduced absorption of calcium ions from gastro-intenstinal tract

(d) Reduced RBC production

Which of the following options is the most **appropriate** ?

Option 1:
(a) and (b) are correct

Option 2:
(b) and (c) are correct

Option 3:
(c) and (d) are correct

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Option 4:
(a) and (d) are correct

Correct Answer:
(c) and (d) are correct

Solution:

REGULATION OF KIDNEY FUNCTION -

The functioning of the kidneys is efficiently monitored and regulated by hormonal feedback mechanisms involving the hypothalamus, JGA and to a certain extent, the heart.

-

Kidneys are not only involved in the excertion of nitrogenous waste but also in the formation of RBCs and absorption of calcium from the GI tract through calcitriol hormones. When the artificial kidney are used, both of these functions will be compromised.

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Q. 80 Match the hominide with their **correct** brain size :

$(a)\ Homohabilis \qquad (i)\ 900cc$

$(b) Homoneanderthalensis \qquad (ii)\ 1350cc$

$(c) Homo\ erectus \qquad (iii)\ 650-800cc$

$(d) Homo\ sapiens \qquad (iv)\ 1400cc$

Select the **correct** option.

Option 1:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (i) \quad (iv) \quad (ii)$

Option 2:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (ii) \quad (i) \quad (iv)$

Option 3:

$(a) \quad (b) \quad (c) \quad (d)$
$(iii) \quad (iv) \quad (i) \quad (ii)$

Option 4:

$(a) \quad (b) \quad (c) \quad (d)$
$(iv) \quad (iii) \quad (i) \quad (ii)$

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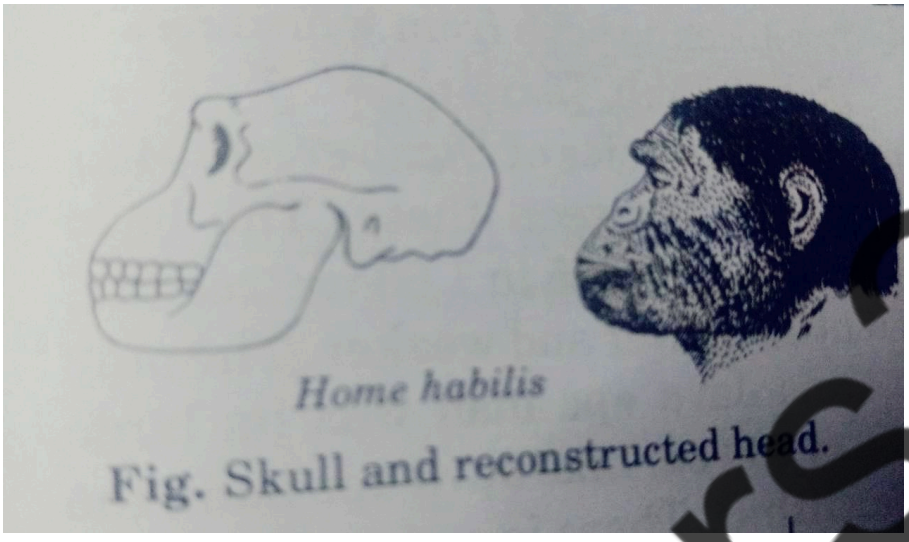
Correct Answer:

$(a) \quad (b) \quad (c) \quad (d)$

$(iii) \quad (iv) \quad (i) \quad (ii)$

Solution:



- wherein

Fossils were discovered from Pleistocene rocks of Olduvai Gorge in East Africa.

(a) $Homohabilies$ had brain capacities between $650 - 800cc$

(b) $Homo\ neanderthalensis$ had brain capacities of $1400cc$

(c) $Homo\ erectus$ had brain capacity of $900cc$

(d) $Homo\ sapiens$ had brain capacity of $1350cc$

Hence, the correct option is (c)

Q. 81 Select the hormone-releasing Intra-uterine Devices.

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Option 1:
$Vaults, LNG-20$

Option 2:
$Multiload\,375, Progestasert$

Option 3:
$Progestasert, LNG-20$

Option 4:
$Lippes\ Loop, Multiload\,375$

Correct Answer:
$Progestasert, LNG-20$

Solution:

contraceptive -

Contraceptive are methods which help prevent unwanted pregnancies.

- wherein

An ideal contraceptive should be user-friendly, easily available, effective and reversible with no or least side-effects. It also should in no way interfere with the sexual drive, desire and/or the sexual act of the user.

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The IUD's or Intra-Uterine Devices can act either through phagocytosis of sperms. suppressing sperm motility or through releasing hormones to make uterus unsuitable for implantation. Progestasert and $LNG-20$ are hormone releasing IUD's

---

Q. 82 Select the correct sequence for transport of sperm cells in male reproductive system.

Option 1:
Testis ⟶ Epididymis ⟶ Vasa efferentis ⟶ Rete testis ⟶ Inguinal canal ⟶ Urethra

Option 2:
Seminiferous tubules ⟶ Rete testis ⟶ Vasa efferentia ⟶ Epididymis⟶ Vas deferens ⟶ Ejaculatory duct ⟶ Urethra ⟶ Urethral meatus

Option 3:
Seminiferous tubules ⟶ Vasa efferentia ⟶ Epididymis ⟶ Inguinal canal ⟶ Urethra

Option 4:
Testis ⟶ Epididymis ⟶ Vasa efferentia ⟶ Vasa deferens ⟶ Ejaculatory duct ⟶ Inguinal canal ⟶ Urethra ⟶ Urethral meatus

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Correct Answer:

Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis→ Vas deferens → Ejaculatory duct → Urethra → Urethral meatus

Solution:

Testis -

In adults, each testis is oval in shape, with a length of about 4 to 5 cm and a width of about 2 to 3 cm. The testis is covered by a dense covering.

-

The pathway for the transport of sperms:

Seminiferous tubules

Rete testis

Vasa efferentia

Epididymis

Vas deferens

Ejaculatory duct

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Urethra

Uretral meatus

Q. 83 What is the direction of movement of sugars in phloem?

Option 1:
Non-multidirectional

Option 2:
Upward

Option 3:
Downward

Option 4:
Bi-directional

Correct Answer:
Bi-directional

Solution:

Translocation -

The transport of substances over long distances through the vascular system, xylem and phloem.

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-

Transportation of nutrients can occur from leaves to roots and other parts of plants or from roots to the new leaves. Hence,sugar transport is bidirectional.

Q. 84 Which of the following statements is **incorrect** ?

Option 1:
Viroids lack a protein coat.

Option 2:
Viruses are obligate parasites.

Option 3:
Infective constitiuent in viruses is the protein coat.

Option 4:
Prions consist of abnormally folded proteins.

Correct Answer:
Infective constitiuent in viruses is the protein coat.

Solution:

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Viruses -

Nucleoprotein entity which require living host to replicate themselves utilising its synthetic machinery.

-

Viruses are made up of protein coat and genetic material. The genetic material is inserted into the host for multiplication and growth of viruses. Hence, genetic material is the infectious constituent.

Q. 85 The ciliated epithelial cells are required to move particles or mucus in a specific direction. In human, these cells are mainly present in :

Option 1:
Bile duct and Bronchioles

Option 2:
Fallopian tubes and Pancreatic duct

Option 3:
Eustachian tube and Salivary duct

Option 4:
Bronchioles and Fallopiun tubes

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Correct Answer:
Bronchioles and Fallopiun tubes

Solution:

Ciliated epithelium -

Single layer of columnar or cuboidal cells bearing cilia on their surface.

- Contains goblet cells (secretory cells) in some locations.

- wherein

Locations - Upper respiratory tract, uterine (fallopian) tube, uterus, paranasal sinuses, ventricles of brain & spinal central canal (ependyma).

Functions - Moves mucous and other substances by ciliary action, movement of gametes.

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The ciliated epithelium lines bronchioles to move the air and trap particles and in fallopiun tubes to move the ovum and zygote after fertilization.

Q. 86 Which of the following statements is **incorrect** ?

Option 1:
Morels and truffles are edible delicacies.

Option 2:
$Claviceps$ is a source of many alkaloids and LSD.

Option 3:
Conidia are produced exogenously and ascospores endogenously.

Option 4:
Yeasts have filamentous bodies with long thread-like hyphase.

Correct Answer:
Yeasts have filamentous bodies with long thread-like hyphase.

Solution:

Ascomycetes / Sac fungi -

1. Mycelium is septate and branched.
2. Sexual reproduction is Isogamy or anisogamy.
3. Four to eight ascospores are formed in an ascus.
4. Fruiting body is Ascocarp.
5. Centrum is present.

- wherein

Eg. Saccharomyces, cerevisiae, Baker's yeast, Penicillium.

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Yeasts are unicellular fungi. They sometimes form pseudophypha by continuous budding, they are not filamentous.

Q. 87 Which of these following methods is the most suitable for disposal of nuclear waste ?

Option 1:
Shoot the waste into space.

Option 2:
Bury the waste under Antarctic ice-cover

Option 3:
Dump the waste within rocks under deep ocean

Option 4:
Bury the waste within rocks deep below the Earth's surface

Correct Answer:
Bury the waste within rocks deep below the Earth's surface

Solution:

As we have learnt in Radioactive waste -

It has been recommended that storage of nuclear waste, after sufficient pre-treatment, should be done in suitably shielded containers buried within the rocks, about 500 m deep below the

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earth's surface.

Hence, the correct option is (d).

Q. 88 Following statements describe the characteristics of the enzyme Restriction Endonuclease. Identify the **incorrect** statement.

Option 1:
The enzyme cuts DNA molecule at identified position within the DNA.

Option 2:
The enzyme binds DNA at specific sites and cuts only one of the two strands.

Option 3:
The enzyme cuts the sugar-phosphate backbone at specific sites on each strand.

Option 4:
The enzyme recognises a specific palindronic nucleotide sequence in the DNA.

Correct Answer:
The enzyme binds DNA at specific sites and cuts only one of the two strands.

Solution:

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Restriction Endonuclease -

These enzymes cleave DNA only within or near the specific base sequence. These sequences are called as recognition sites. They are of three types RE- I, RE-II and RE-III.

- wherein

1st Restriction enzyme (Hind II) was used in RDT.

Restriction endonuclease bind to the specific sites (palindromic sequences) the DNA and cut both the strands of the DNA.

Q. 89 In a species,the weight of newborn ranges from 2 to 5 kg. 97% of the newborn with an average reight between 3 to 3.3 kg survive whereas 99% of the infants born with weight from 2 to 2.5 kg or 4.5 to 5 kg die.Which type of selection process is taking place ?

Option 1:
Directional Selection

Option 2:
Stabilizing Selection

Option 3:
Disruptive Solution

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Option 4:
Cyclical Solution

Correct Answer:
Stabilizing Selection

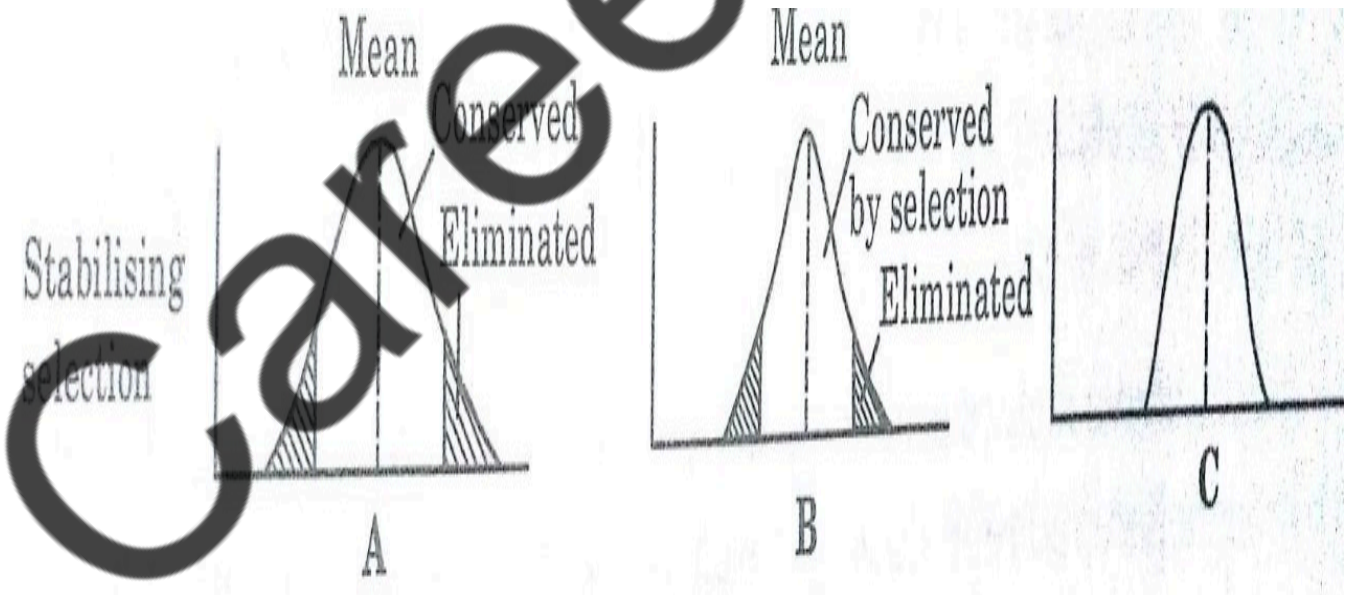
Solution:

Stabilizing or balancing selection -

A type of natural detection which leads to the elimination of organism having over specialised characters and maintains homogeneous population which is generally constant.

- wherein

It favours the average or normal phenotypes. Results in very low rate of evolution.



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Stabilizing selection favours the medium types $(3\ to\ 3.3\ kg)$ while the extreme phenotypes $(2\ to\ 2.5\ kg\ and\ 4.5\ to\ 5\ kg)$ are eliminated.

Q. 90 Select the **correctly** written scitific name of Mango which was first described by Carolus Linnaeus :

Option 1:
Mangifera indica Car. Linn.

Option 2:
Mangifera indica Linn.

Option 3:
Magniferra indica

Option 4:
Magnifera Indica

Correct Answer:
Mangifera indica Linn.

Solution:

Nomenclature -

Science of providing distinct and proper names to an organism.

- wherein

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There is a need to standardize the naming of an organism, so that it is known by same name across the world.

The scientific name along with the name of the author is written as :

Mangifera indica Linn.

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